OECD ECONOMIC SURVEYS

1999

SPECIAL FEATURE

Reforming social security

TURKEY

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1998-1999

TURKEY

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF TURKEY

THE LAND

Area (thousand sq. km)	779
Agricultural area (thousand sq. km)	275
Forests (thousand sq. km)	202

Major cities, 1990 (thousand inhabitants):	
Istanbul	7 309
Ankara	3 237
Izmir	2 695

THE PEOPLE

Population, 1997 (million)	63.7
Per sq. km, 1997	82
Annual rate of change of population, 1997	1.7

Civilian labour force, 1997 (million)	21.9
Civilian employment	20.5
Agriculture, forestry, fishing	8.6
Industry	3.5
Construction	1.3
Services	7.1

PRODUCTION

GDP, 1998 (TL billion)	51 625 143
Per head (US$)	3 111
Gross fixed investment, 1998 (TL billion)	12 794 101
Per cent of GDP	24.8
Per head (US$)	771

Origin of GDP, 1998 (per cent):	
Agriculture, forestry, fishing	17.4
Industry	29.1
Services	53.5

THE GOVERNMENT

Public consumption, 1998 (per cent of GNP)	12.3
Central government current revenue, 1998 (per cent of GNP)	20.4

Public debt, end-1998 (per cent of GNP)	37.6
Domestic	21.9
Foreign	15.7

FOREIGN TRADE

Commodity exports, 1998, f.o.b. (per cent of GDP)	13.6
Main exports (per cent of total exports):	
Machinery and transport equipment	15.1
Other manufactures	59.4
Food and live animals	14.0

Commodity imports, 1998, c.i.f. (per cent of GDP)	23.2
Main imports (per cent of total imports):	
Machinery and transport equipment	39.7
Other manufactures	24.2
Chemicals and related products	14.3

THE CURRENCY

Monetary unit: Turkish lira

Currency unit per US$, average of daily figures:	
1994	29 778
1995	45 738
1996	81 281
1997	151 595
1998	260 473

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Turkey by the Economic and Development Review Committee on 17 May 1999.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 9 June 1999.

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The previous Survey of Turkey was issued in June 1997.

Assessment and recommendations

Background to current economic policy issues

The Turkish economy has continued to be characterised by erratic bouts of rapid short-term growth and endemic high inflation, preventing the economy from fulfilling its longer-term growth potential. Above-trend growth between 1995 and mid-1998 has been followed by a recession, the economy having weathered the Asian crisis but proved vulnerable to the emerging-market crisis that followed the Russian default. The second half of 1998 was difficult in terms of declining activity and weakening international confidence due to the world financial crisis, but the policy response, building on the disinflation programme launched in early 1998, has been effective in stabilising the macroeconomic environment. Wholesale price inflation has fallen to around 50 per cent and there is a primary fiscal surplus. However, corrective action has needed to rely heavily on high real interest rates, which are not consistent with stable growth in the longer run. Explosive interest payments undermine longer-run efforts to reduce the budget deficit while sustaining high inflation expectations and exacerbating the country's vulnerability to global economic shocks. The macroeconomic adjustment so far achieved will only be lastingly successful if accompanied by a broad range of structural reforms bearing, in particular, on an overhaul of the social security system, a strengthening of the banking sector, accelerated privatisation, and the restructuring of state-owned enterprises. Full implementation of the official agenda for reform of these sectors would set the economy securely on a path of convergence with the rest of the OECD area, even if further reforms will be needed, notably in the domains of agricultural support, infrastructure development and an appropriate regulatory framework, to remain on that path. If implementation falls short, it would risk further

embedding high inflation expectations and associated severe resource misallocation.

Against this background, the *Survey* begins by analysing the response of the Turkish economy to the global financial crisis and describing the short-term outlook (Chapter I). Excessive budget deficits having been the main engine of high inflation, Chapter II discusses the process of fiscal consolidation needed to bring the inflation rate back to a sustainable level. Chapter III looks at ways of dealing with the financial pressure emanating from the social security system, one of the chief sources of the chronic budget deficit. The final section details the areas in which further structural reform is needed if Turkey is to achieve balanced longer-run growth.

The economy has been adversely affected by the world slowdown

Turkish exports were badly affected by the collapse of the Russian market and there has been increased competition in both foreign and domestic markets following the devaluations among the emerging market economies. Consequently, many hitherto fast-growing export industries have experienced a sharp slowdown, with a resultant loss in Turkey's aggregate export market share. Industrial output is estimated to have been almost stationary in the second half of last year. With imports declining due to the sharp deceleration in overall activity, the current account has shifted into moderate surplus. But as far as export prospects are concerned, the external trade environment does not favour a rapid pick up: export market growth is likely to remain quite sluggish, at less than 2 per cent in volume terms, before recovering to a still-modest 5 per cent in 2000. On the other hand, the policy of stabilising the real effective exchange rate (based on a dollar/DM basket) has been effective in preserving overall price competitiveness, and Turkey's competitive position remains far stronger than it was prior to the 1994 crisis. In these circumstances, exports could recover some of the momentum lost in 1998, although volume growth would still be slow by the standards of recent years.

With domestic demand also depressed, short-term growth prospects are subdued

In the aftermath of the Russian crisis there was an immediate and severe outflow of foreign capital, which imposed a major domestic credit squeeze on the already slowing economy. To ensure the continued financing of the government deficit, interest rates on treasury bills rose to 130-140 per cent. With wholesale price inflation declining to around 55 per cent by end-year, real rates exceeded 50 per cent. Domestic borrowing and lending have virtually ground to a halt as the public sector has "crowded out" the private sector. High interest rates are leading households to postpone spending on durables and a very high proportion of firms' revenues is now coming from the non-commercial activity of investing in government paper. With interest rates remaining in the 90 -100 per cent range, only slow growth is now expected for 1999, in contrast to growth rates in the 5-7 per cent range in recent years. Moreover, given Turkey's record of output volatility, any economic forecast is likely to have a high margin of error. Activity remains vulnerable to both external and domestic confidence factors. Added to the global economic uncertainty has been the political hiatus preceding the formation of a new government. With global economic prospects improving and the political situation becoming clearer, such uncertainties will diminish. But perhaps the most important uncertainty relates to whether the three-year stabilisation programme introduced in early 1998 will be fully implemented. If it were not to be completed, there might be some temporary resurgence in domestic demand, but foreign confidence would be adversely affected and the economy could not achieve a balance that would sustain growth in the longer term.

The stabilisation programme is likely to be partially successful...

The stabilisation programme is intended to bring inflation down to single digits by 2000, and makes a significant shift away from the former complacency about high inflation. A quarterly IMF review process, requested by the Turkish authorities in July 1998, has backed up the programme. The announcement of short-term targets for implementing a range of macroeconomic and structural policies is intended to increase transparency and continuity at a time of heightened political uncertainty. Despite the worsening economic and political environment, the interim IMF reports have

been relatively encouraging. Nevertheless, according to the OECD projections, the three-year disinflation profile is unlikely to be fully achieved. There is little evidence that the traditional inflation inertia present in public sector wage and price behaviour has been broken, and the assistance given by low commodity and energy prices is likely to be temporary. Inflation could be halved by 2000, but it will remain well above single digits.

... but the continuing high costs of inflation suggest the policy response is too gradual

Indeed, the gradual nature of the disinflation strategy betrays evidence of a policy dilemma, as to how rapid the adjustment process can and should be. On the one hand, since excessive public spending is behind the inflation process, rapid disinflation could cause severe adjustment problems for sectors such as agriculture and the state-owned enterprises, which are dependent on subsidies. Vested interests of this sort make for a policy of gradualism, based on a weak budget constraint. On the other hand, it is widely recognised that the costs of rapid inflation are very heavy. Extreme inflation severely distorts the level and composition of national saving and investment, undermining the economy's otherwise favourable supply-side performance. In a liberalised capital market, high and volatile inflation implies a very high risk premium in real interest rates. The consequences are extremely short planning horizons and a dearth of capital inflows, which account for only a disappointingly small proportion of capital investment. But at the same time, some segments of the population actually gain from this situation: especially net lenders (corporations, banks and high-income households), who receive higher unearned income receipts in the short term (though tax reform has begun to bite into these gains). Inflation thus works to the benefit of an urban *rentier* economy *vis-à-vis* other social groups, resulting in an extremely skewed income distribution. This has implications for social cohesion as well as economic performance. Both social and economic considerations point to the critical importance of putting in place a set of policies that would bring inflation down more rapidly.

Rapid reform requires a more consensual approach, involving institutional reform

It should also be noted that reducing inflation in an environment of high and inertial expectations can be very costly in terms of lost output. For the costs to be kept to a minimum, inflation inertia – based on endemic backward-looking indexation – needs to be attacked. As a first step, a forward-looking indexation has been adopted for the setting of wages in the public sector. However, the procedure has been overridden, and there is at present little confidence that the institutional framework is in place which will allow this system to work efficiently. Indeed, perhaps the biggest structural challenge is to build up a consensus for a forward-looking reform strategy that could be used as a nominal anchor for reducing inflation expectations, leading to the establishment of a fully independent central bank. There is a need – recognised in the setting up and recent reinvigoration of the Economic and Social Council – for an institution that can be used as a forum for building a consensus for change. Such an institution should be supported by an independent body with a wide-ranging mandate to undertake research into the costs and benefits of different policy options. This would help to strengthen policy-making by identifying more clearly whose interests are at stake and how resistance to reform could be overcome. Care would have to be taken to ensure that all stakeholders in the economy were properly represented and that the strategy of reform was enhanced to take income distribution aims into account. Experience elsewhere in the OECD shows that comprehensive reform, in the context of forward-looking, society-wide wage and macroeconomic policy commitments, can substantially reduce the adverse short-term output and employment costs of reducing inflation.

Monetary policy has played an essentially stabilising role, but could become more pro-active...

The conduct of monetary policy has undergone recent reform, and monetary restraint has begun to play a central, if essentially reactive, role in stabilisation. Chronic inflation has destroyed virtually all trust in the Turkish lira as a store of value as, until 1997, central bank control over money creation was limited by direct government access to credit. This has now been cut off, and the central bank's strategy with regard to the disinflation target has been to supply domestic liquidity consistent with a depreciation of the lira *vis-à-vis* the currency basket at a rate which maintains

international competitiveness. Such a strategy implies that liquidity creation is sufficient to prevent an unwanted short-term strengthening in the lira parity. An overvalued exchange rate, while it would be an effective deflationary instrument in the short run, would leave the financial system vulnerable to global financial-market volatility and speculative capital flows, leading at some point to another devaluation and a reversal of disinflation gains. Fundamentally, monetary policy alone cannot be expected to restore confidence in the lira in the face of high budget deficits so that monetary policy's role must necessarily be cautious and limited. Nevertheless, depending on permanent improvements on the fiscal side, the central bank could become more pro-active and use its policy instruments more forcefully, in line with disinflation objectives.

... but the financial sector situation remains weak, calling for tighter prudential regulation and reform

The financial sector remains a source of economic vulnerability, having come to rely on lax fiscal policy for its profits. This dependence inhibits macroeconomic flexibility and impedes structural reform. Banks in general have become highly dependent for their profits on intermediating between the foreign and the government sector. The resulting open foreign-exchange position is substantial, though smaller than prior to the 1994 crisis. Two banks have been transferred to the Saving Deposit Insurance Fund and the Treasury is continuing to scrutinise the financial soundness of individual banks under the authority of the Bank Act. The government has acted to strengthen the banks' foreign borrowing ratios through tighter regulation, but action in this direction has been limited because it has the effect of pushing interest rates up. The government and financial sector have thus become highly interdependent. The costs are to be seen in a high degree of dollarisation and large off-balance sheet positions which weaken monetary control. And financial intermediation, though well-developed in terms of infrastructure and human capital, is under-developed compared with the size of the economy. There is an unusually high degree of self-financing of business investment and few homes are mortgage-financed. Investment is thus probably not being directed to its most productive use. The larger private sector banks are part of industrial group holdings, channelling funds into sectors that might

not be able to compete for funds on an open market basis. In any reform programme, the banking sector will have to build up its financial intermediation skills, and could be vulnerable during a period of economic adjustment. This makes it particularly important that Turkey act to strengthen banking supervision. In recognition of this a draft banking law has been prepared, *inter alia* establishing an independent Banking Regulation and Supervision Board. This draft banking law would significantly strengthen supervision and accounting standards and passing it should be a priority. Another major challenge will be reforming the state banks. These banks are making subsidised loans according to government policy, diverting credit into particular sectors at the expense of others. As a precursor to eventual divestment, the quasi-fiscal operations with which these banks are entrusted should be gradually wound down.

Fiscal policy has become more balanced but is still the principal source of disequilibrium

The budget is the biggest single contributor to resource misallocation, monetisation risk and inflation inertia, through the chronic financial deficit. A fundamental element of the 1998-2000 stabilisation programme has been the commitment to a strong budgetary primary surplus. This objective has been backed up by reforms to the civil service wage indexation system, spending cuts and steps to improve the tax structure and revenue administration. After falling to zero in 1997, the primary surplus rose to 4½ per cent of GNP in 1998, despite weaker output growth than expected. For 1999, the primary surplus is scheduled to be only a little lower. But there is some concern that the primary surplus may not be sustainable insofar as expenditure control in 1998 was based heavily on real wage freezes, investment cuts, and deferral of certain spending into 1999. Moreover, political uncertainty has meant that only an interim half-yearly budget has been decided upon. The authorities will need to be extremely determined in their pursuit of the primary surplus target to maintain the credibility of the reform programme.

The primary surplus is not high enough

Despite the primary surplus, the consolidated central government deficit, at about 7 per cent of GNP in 1998, continues to exceed the corresponding OECD average by 6 percentage points. Mounting interest payments remain a major

source of spending pressure and potentially explosive debt in spite of the primary surplus. At 18 per cent of GNP, domestically-held debt is not exceptional in international terms, but a number of factors mean that there is still concern about its adverse impact on stability:

– Domestic debt is very short term, the bulk of the debt having to be rolled over within the year; the gross financing requirement is at times difficult for the market to absorb, increasing the risk of monetisation.
– Net foreign borrowing has been negative for five consecutive years, and this has put additional pressure on domestic financial markets and real interest rates.
– Real interest rates on marketable government debt are extremely high, forcing up debt interest payments to a degree that even the currently high primary surplus cannot contain and making the fiscal situation potentially explosive.
– On top of the consolidated budget, the government is heavily involved in "off-budget" operations. State-owned financial institutions hold a substantial amount of non-marketable, "non-cash" government debt that is not serviced at market rates; and state-owned banks also hold government debt as "receivables" in return for their quasi-fiscal operations of providing subsidised credit to agriculture and smaller firms.

As conventionally measured, and taken in aggregate, government debt has been kept under control partly because inflation has eroded its real value. The impact of inflation so measured has been substantial. However, even allowing for these factors, current interest rates now imply a substantial real burden. OECD calculations show that, at the current level of real interest rates, a primary surplus significantly higher than targeted under the 1999 budget would be needed to stabilise the domestic marketable debt ratio. Taking account of the liabilities incurred via state-owned banks, the primary surplus target would need to be even more ambitious.

Progress has been made towards a more efficient tax system and revenue administration

As an essential element of the fiscal consolidation process, the government introduced a major tax reform in 1998, aimed at widening the tax base, reducing tax rates, improving tax administration, and shortening lags in tax collection (which, in a high inflation environment, have proved very costly to the exchequer). The main measures included a tax amnesty geared to a new wider definition of taxable income (*i.e.* as an increase in wealth); the issuance of taxpayer identification numbers (including the use of third party information); a shift to quarterly pre-payments of corporate income taxes (as of 1999); criminalisation of collusive tax evasion and stiff penalties for late payment/evasion of taxes. At the same time, marginal tax rates have been reduced by up to 10 percentage points. Tax collection improved markedly in 1998. While this outcome may have included an announcement effect of the tougher enforcement measures, the government is confident that, together with the lowering of tax rates, sustained efforts in better auditing and enforcement will generate permanently higher tax revenue by making evasion less profitable.

Pension reform is a priority

The social security system is one of the chief sources of chronic deficit. Despite fairly minimal pensions and very narrow coverage of the population, low retirement ages and little linkage between benefits and contributions means that the pension outlays far exceed inflows. And despite a relatively young population, social security deficits are projected to worsen rapidly. Measures have been taken in recent years to address the causes of deficits, notably by increasing compliance in the payment of contribution premia, but these have been insufficient to arrest the rapid growth in such deficits. Re-establishing solvency will require reforms both to improve matters on the contribution side and to encourage people to remain in the formal work force for longer. This suggests the following strategy for reform, based on stabilising the public "pay-as-you-go" (PAYG) system:

– *Raise the retirement age* for existing workers to 60 and gradually to 65 for future workers and lengthen the contribution period required to obtain a full pension to 30 and 35 years, respectively. Benefit reforms, such as switching

to lifetime average income as the reference period, are also needed. The transition to the new system-solvent rules should be rapid.

– *Improve contribution collection rates*, while widening the coverage to include the current "unregistered" sector, through institutional reform of the funds. This will require political determination, substantial investments in information technology and skills, better governance mechanisms, and improved incentives of both individuals and firms to comply. As collection rates improve, statutory contribution rates can come down, in turn reducing the incentives to informal work.

At the same time, any future liabilities arising from an expansion of the system to include higher rates of registration and female participation should be pre-funded to prevent the growth of further unfunded debt. This would require tight controls to ensure that any funds thus built up are not wasted or spent.

Adding a private funded pillar could enhance old-age income security

Making adequate provision for income in old age necessitates an important and early policy choice. Either *i)* the PAYG system itself must be expanded by substantially raising currently low contribution ceilings, or *ii)* a second private funded pillar must be established, leaving the role of PAYG as one of minimum pension provision. The second, mixed approach could enhance pension income security by allowing for a broader diversification of risks, while generating positive economic externalities, such as lower labour market distortions and higher national savings. It is, thus, preferable. However, a modern and effective regulatory framework, and greater financial market transparency, are key preconditions for an expanded private sector arrangement. Indeed, it would appear difficult to create a second pillar in the current environment, and it is important that the authorities in this sense respect the proper sequencing of reforms. Another essential precondition is a sustained reduction in inflation such that investors have confidence that their savings will not be eroded by inflation. The development of markets for indexed debt would help to guard against inflation risk in the future, while allowing pension funds to invest abroad may help to diversify away domestic

inflation risk. Private schemes, furthermore, can be made mandatory or voluntary, occupational or individual, subsidised or not. A mandatory scheme may be preferable to a voluntary scheme as it covers the poor, especially if it includes refundable tax credits to top up their savings contributions, while individual schemes are likely to be more portable than occupational schemes and are less likely to be seen as a tax. But, as long as inflation remains high, mandatory schemes may continue to be seen as a "tax" if savings are thereby eroded.

Social assistance also needs to be strengthened

There will also be a need for a social safety net, financed out of general tax revenues, to pick up those who have never worked, those who have not worked long enough, or those who do not earn enough from their individual funded plans. To this end, the current very minimal old age assistance benefits need to be raised to closer to the subsistence level. The possibility for moral hazard implies that the minimum guarantee must remain modest and well targeted by rigorous means-testing. While a universal flat-rate benefit would be less distorting and easier to administer, it would be likely be too expensive for Turkey to consider at this time.

Efficient and equitable provision of health care is a long-standing issue...

The health expenditures of social security organisations, after stabilising in the first half of 1990s, have increased strongly in the past few years and are putting pressure on the social security fund balances without giving satisfactory health protection. Turkish health expenditures remain well below the OECD average both as a share of GDP and on a per capita basis, and the efficient and equitable provision of health care has been a long-standing issue. Turkey lags behind many countries with similar income levels in terms of certain basic health indicators, notably infant mortality rates. Symptoms of inefficiency are evident in the coexistence of congestion in hospital ambulatory care services, very low rates of hospital bed occupancy, and a sharp decline in expenditures for preventive care. There is also a very unequal distribution of health-care supply across the country, which leaves a significant part of the population without adequate health insurance coverage. Weaker public-sector supply and growing private sector involvement

has meant that access to quality care is now limited to the better-off who are able to make out-of-pocket payments. The system has steadily become more regressive.

... and reform proposals should be implemented

Serious health reform efforts have been under way since the end of the 1980s, culminating in a set of legislative proposals currently in the Parliament. The proposed reform would strengthen preventive and primary care by reinforcing the public health centres and establishing a system of family doctors in private practice, who would provide primary care and refer the patients to specialists for further treatment. Public hospitals would be made autonomous and would have to compete with each other, as well as with private hospitals, for block contracts from the Provincial Health Directorates. Such an arrangement could be expected to boost the efficiency of hospital services. These reforms would also limit the direct subsidisation of public hospitals and cross-subsidisation of those currently covered by social insurance, thereby eliminating one source of inequity. More importantly, the reform would extend insurance coverage to the whole population, via the General Health Insurance Scheme and by subsidising the insurance premium on a sliding scale according to the household's capacity to pay. Rapid implementation of these reforms is desirable. But the keys to success are the setting up and, above all, monitoring of contracts. Such a system would involve heavy budgetary costs, which raises questions of affordability. It would need to engineer a shift in resources towards problems and regions where health outcomes are currently the weakest and where high returns in improved population health can be expected.

The privatisation programme needs to be completed

An important message of the *Survey* is that wide-ranging structural reforms are needed both for restoring macroeconomic balance and improving long-run economic performance. Perhaps most critical in this respect is the reform of the state economic enterprises and the extensive system of subsidisation, especially for agriculture, which are both financially costly and resource-inefficient. The Turkish authorities embarked on an ambitious privatisation programme from 1984, with the aim of minimising state involvement in the industrial and commercial activities of the

economy. However, despite the completed sale of 130 companies, State Economic Enterprises (SEEs) still play a significant and even dominant role in some sectors of the economy, and privatisation – or "corporatisation" – still offers substantial scope for efficiency gains. The government's privatisation programme stalled in the second half of 1998, with the global financial crisis. While it is important that competition issues be resolved before monopolies or dominant players are privatised, it is also important for Turkey's future economic performance that these enterprises be privatised as rapidly as possible. And the new Competition Authority will need to play an ongoing monitoring role.

Reform of the agricultural sector is critical

The agricultural sector is extremely important in Turkey: it still contributes around 15 per cent to GDP and generates around 45 per cent of all employment, while markets are much more distorted that in other OECD countries. It is also the recipient of large-scale subsidies, which go to raise producer receipts (which are 65 per cent higher than they would have been at world prices) without reducing consumer prices (which are 50 per cent higher than if world prices prevailed). The OECD has extensively documented the damaging economic consequences of high levels of agricultural support and OECD Agricultural Ministers stressed the need to reduce market distortions again in 1998. Turkey is the only OECD country to have gone against the trend and increased its support to agriculture over the past ten years. Turkey's current agricultural policies are clearly not aligned with OECD goals and principles and impose a heavy burden on the economy. The most effective long-term reform would be to completely replace Turkey's current agricultural support arrangements with a single transparent direct income scheme targeted on small farmers. Current agricultural support arrangements could not be replaced overnight, but reforms would bring significant benefits in terms of lower prices for consumers and budget savings which could be used in priority spending areas such as health and education.

Fulfilling the country's growth potential requires better energy, transport and telecommunications infrastructure

Outside the public sector, Turkey's supply-side performance has been relatively dynamic, but needs to be underpinned by better infrastructure. The economy already benefits from a number of advantages. Import barriers are low and Turkey benefits from being in a customs union with the EU; it is also well placed geographically to take advantage of trade with emerging economies within the region. Its second strength is a young and rapidly growing work force. It also has a well-established tradition of dynamic performance in terms of entrepreneurship (albeit linked to a very large informal sector). The country thus has potential to grow very fast, if the obstacles to better resource allocation are removed, particularly with respect to infrastructure. Energy, transport and telecommunications are key sectors here.

– The *electricity sector* faces major challenges in improving the efficiency of its existing generation and distribution network and expanding supply. In both of these areas, the government's strategy has been to involve the private sector in an industry traditionally dominated by the state. The three crucial challenges now are: developing workable competition; getting consumers to share the benefits of liberalisation; and implementing regulatory mechanisms compatible with a market system. These regulatory issues have not yet been fully settled in Turkey, and need to be rapidly addressed. In addition, constitutional issues have emerged as a major impediment to privatisation in the energy sector. The constitution appears to prohibit the granting of concessions to foreign investors in certain infrastructure areas, and grants final authority over disputes to the Turkish Administrative Court, rather than to an international court of arbitration as foreign investors would require. Permanently solving these problems may require new legislation or constitutional change.

– *Transport:* Turkey also has an inefficient public railway system. The physical state of the rail network in Turkey is very poor and the Railways Company is the biggest loss maker of all state economic enterprises. A 10-year restructuring plan is being implemented, but restructuring is hampered because supporting legislation has not

yet been passed, limiting the capacity to make significant efficiency gains. Turkey faces a greater challenge than most OECD countries because it also has to deal with the poor physical state of its rail network. Pruning the network to eliminate low-density routes will be necessary and public service obligations should be reassessed within a broader transport framework to see if other ways of meeting them could be more economical.

– Liberalisation of the *telecommunications sector* is under way and it will be important to get the new legal and regulatory framework for telecommunications established. The Post Office remains under the direct control of the Treasury. As a chronic loss-maker and one of the larger state sector employers, "corporatising" the Post Office, as many OECD countries have done, should generate important improvements in efficiency and in turn also reduce the burden on public finances.

Raising human capital remains an important commitment

Turkey has made progress in addressing human capital development, most notably through rapidly raising compulsory schooling from 5 to 8 years. Inevitably, bottlenecks have occurred and it will be important to maintain the commitment to upgrading the quality of education through teacher training and upgrading school facilities. Ensuring that all children do attend school may require incentive programmes for the poorest families. Addressing the wide disparities in literacy rates amongst young adults, women in particular, will also contribute to raising human capital.

Summing up: reforms are needed to exploit Turkey's entrepreneurial strengths

To sum up, the medium-term stabilisation programme has helped Turkey to weather an adverse international economic environment. Inflation is falling and the current account is in surplus. But the projected GDP growth rate is slow and any shortfall from the coalition government's proposals for policy adjustment runs severe negative risks. Moreover, the programme itself seems inadequate to reduce the high fiscal deficits, high real interest rates and entrenched inflation expectations, which are the chief sources of domestic instability. Monetary policy thus needs to be supported by further fiscal restraint. There is a widespread acknowledgement in the domestic policy debate

that fundamental structural reforms are needed if macroeconomic adjustment is to be successful in the longer run. In particular, the roots of Turkey's current problems stem from the pervasive influence of the public sector. If reforms to rationalise and further reduce the involvement of the state in the economy were forthcoming, they would provide the basis for convergence founded on the evident supply-side strengths of the Turkish economy, which derive from an open and competitive economy and a strong entrepreneurial tradition.

I. Slow growth, instability and inflation

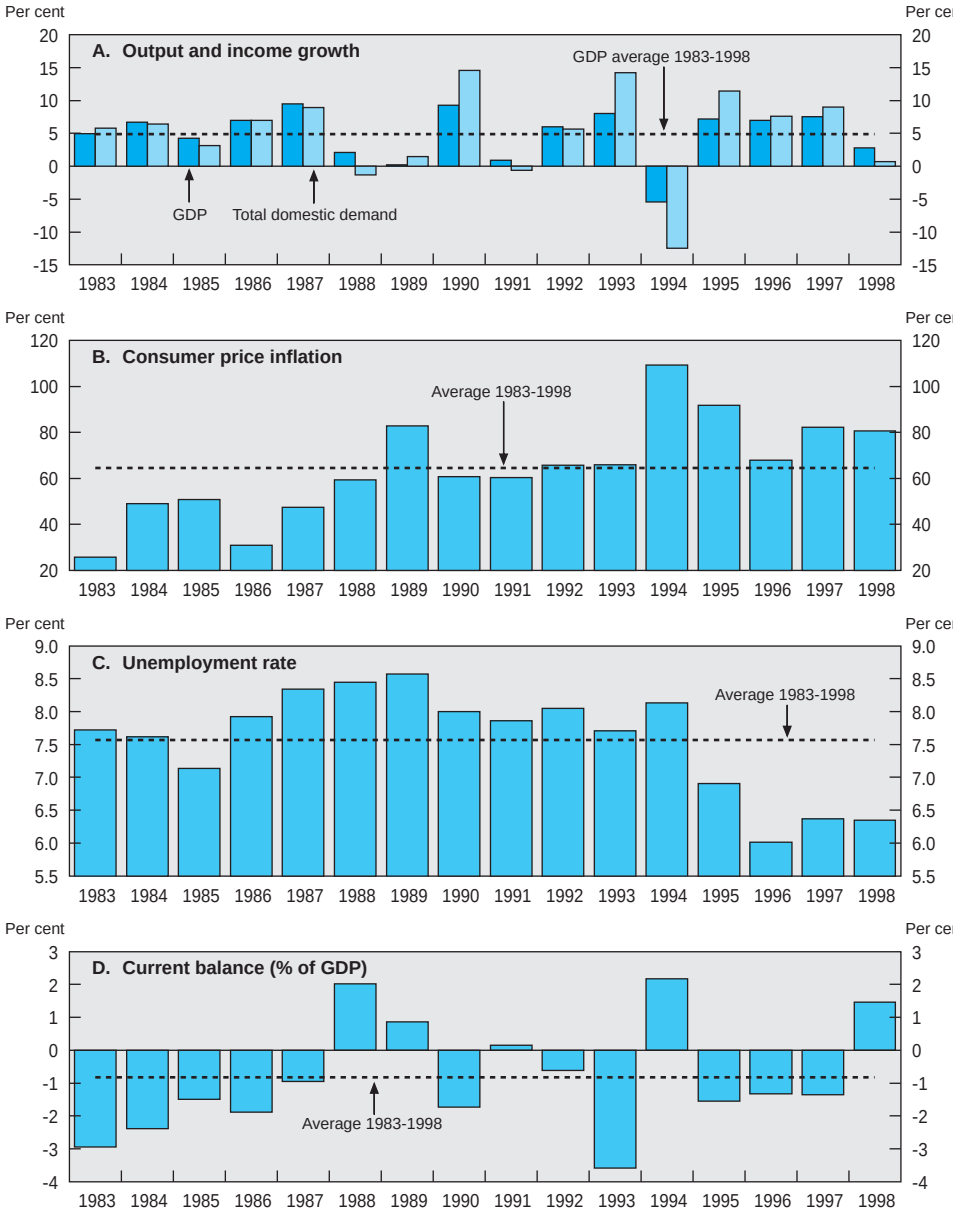
Turkish economic growth tends to be characterised by erratic bouts of rapid growth and persistent inflation, which severely diminishes the country's longer-run performance by reducing planning horizons and exacerbating the country's vulnerability to global economic shocks. Following a year of severe recession, the Turkish economy went through a boom period of above-trend growth between 1995 and mid 1998 (Figure 1). GDP growth averaged over 7.5 per cent in 1997 and peaked at 8.7 per cent in the first quarter of 1998. CPI inflation, which had remained steady at around 75 per cent in the first half of 1997, accelerated to more than 100 per cent by the beginning of 1998. Having shrugged off the turmoil in Asia – the stock market continued to boom and external financing of the current account deficit was no problem – the economy proved vulnerable to the emerging-market crisis which followed the Russian default. The second half of 1998 was particularly difficult in terms of massive capital outflows, rising real interest rates and declining activity. There is a general consensus that the policy response to the crisis has been effective and that the situation should gradually improve: CPI inflation has since fallen to around 65 per cent and the macroeconomic environment appears to have stabilised. However, the short-term prospects for the economy remain rather problematic in the light of a still-difficult global environment and the need to complete the stabilisation programme. Moreover, the record of past policy implementation being imperfect in Turkey, the risk that the stated goals of disinflation and stability will not be attained, with detrimental effects on Turkey's longer-run economic performance, remain quite considerable.

Cyclical background to the current instability

Rapid growth and weakening external balance, 1995-97

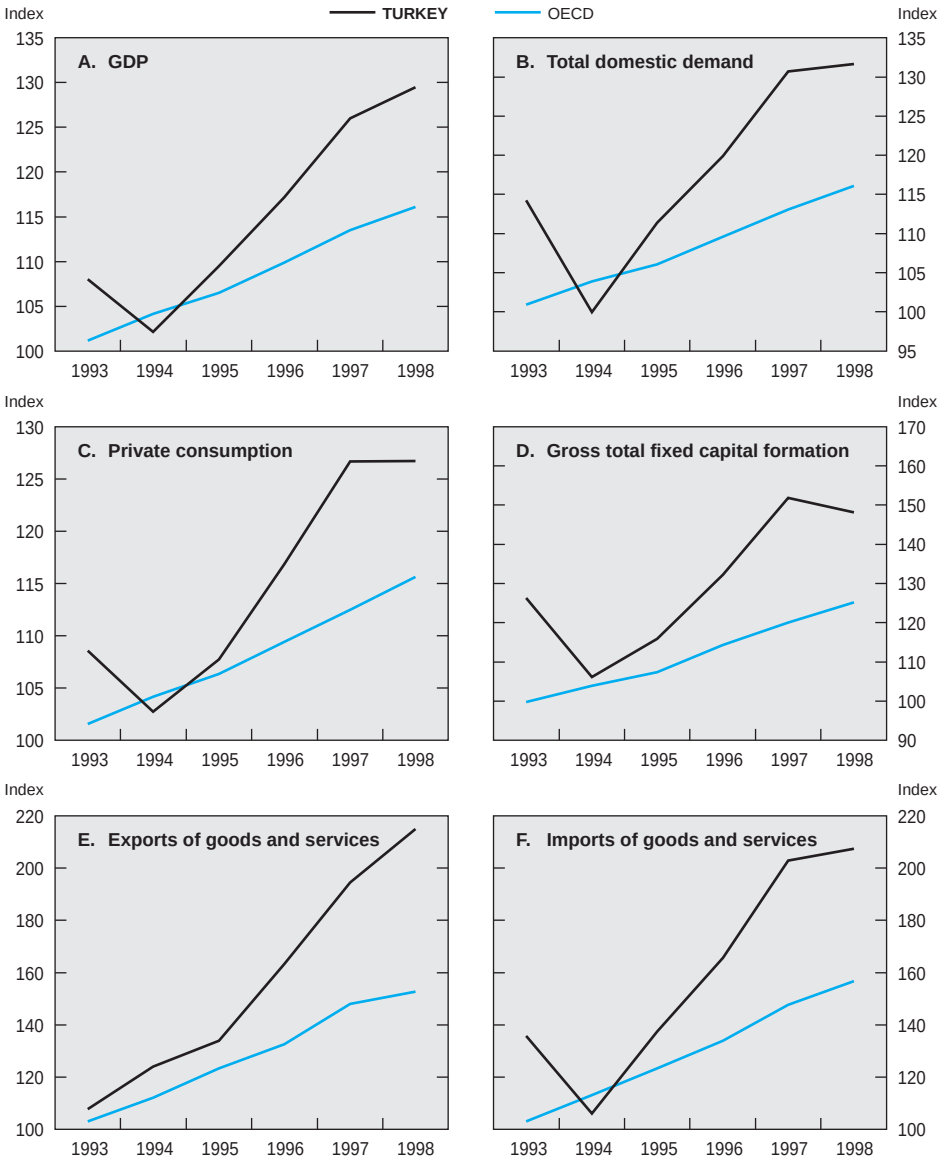
The upturn which began in 1995 was led by accelerating domestic demand (Figure 2) and particularly a recovery of consumption, under the combined impact of rising real incomes, the wealth effects of decelerating

Figure 1. **Macroeconomic performance**



Source: State Planning Organisation.

Figure 2. **Contributions to demand growth**
1992 = 100[1]



1. 1998 OECD data are estimates.
Source: State Planning Organisation and OECD.

Table 1. **Decomposition of output growth by sector**

	1995 (per cent of GDP)	1993	1994	1995	1996	1997	1995-97[1]
Agriculture	15.7	–1.3	–0.7	2.0	4.4	–2.3	4.0
Industry	26.3	8.2	–5.7	12.1	7.1	10.4	32.5
Manufacturing	22.6	9.3	–7.6	13.9	7.1	11.4	35.9
Services	54.3	8.8	–3.6	6.0	6.6	7.4	21.4
Construction	5.5	7.9	–2.0	–4.7	5.8	5.0	5.9
Trade	20.5	11.6	–7.6	11.5	8.9	11.7	35.6
Transport	12.6	10.8	–2.0	5.7	7.6	6.0	20.6
Bank services	3.7	6.9	–4.3	7.3	6.5	6.8	22.0
Government	8.0	1.8	0.8	2.5	–0.3	0.1	2.3
GDP		8.0	–5.5	7.2	7.0	7.5	23.3

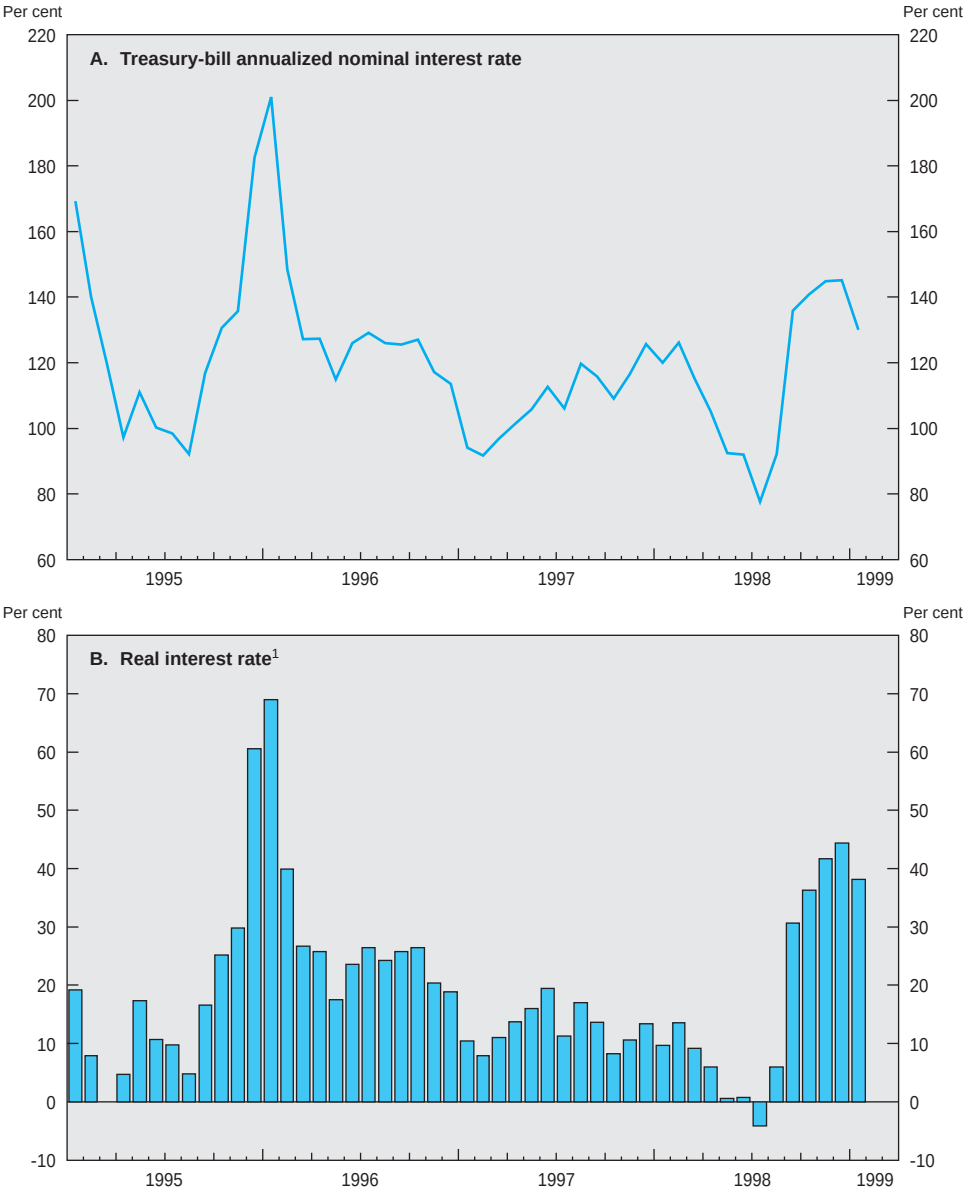
1. Cumulative percentage change.
Source: State Planning Organisation.

inflation and the positive income effects of higher interest rates. The recovery was assisted by a resumption of investment, as Turkish industry prepared for the commencement of a customs union with the European Union in 1996. The dynamism of activity was not evenly spread across sectors, as long-term sectoral shifts continued during the expansion, mirroring a more intense process of urbanisation. At just 4.3 per cent, the cumulative growth of agricultural output remained rather slow between 1995 and 1997, compared with a rise of 23 per cent for the economy as a whole (Table 1). By contrast, the economic upturn was driven by the output gains of the industrial sector, which rose, in cumulative terms, by 33 per cent over the period, mainly responding to buoyant purchases of consumption goods and capital equipment. Reflecting the strength of the tourism sector, output of services advanced by about 21 per cent.

With a lag, higher investment translated into a pickup of exports, which were boosted by a competitive exchange rate (the result of the 1994 crisis). As the level of demand directed towards the home market rose, the economy quickly approached its potential growth. At the same time, capacity utilisation rose almost to the level seen at the peak of the previous 1992-93 boom years, while labour market conditions, though improving slowly, also appeared to strengthen. Economic expansion, combined with a sizeable reduction of border protection and a re-appreciation of the real effective exchange rate, was mirrored in soaring imports and a weakening external position. The current account swung into a deficit and remained there until 1997.

High nominal and real interest rates continued to attract capital inflows to cover the growing current account deficit (Figure 3). Indeed, foreign capital

Figure 3. **Interest-rate developments**



1. Deflated using consumer price inflation.
Source: Central Bank of Turkey.

Table 2. **Official foreign exchange reserves**[1]

	1993	1994	1995	1996	1997	1998
In millions of US dollars	6 213	7 112	12 391	16 273	18 419	19 721
In months of import[2]	2.5	3.7	4.2	4.5	4.6	5.2

1. Excluding gold.
2. Official foreign exchange reserves expressed in terms of average monthly imports.
Source: Central Bank of Turkey.

recovered rather quickly from the 1994 crisis, and by 1995 foreign reserves were already equivalent to about four months of imports, a record level. Foreign reserves further rose to approximately five months of imports thereafter (Table 2). Nevertheless, despite the expectation of stronger inflows following entry into the EU Customs Union, foreign direct investment inflows picked up only temporarily (Figure 4), a development linked to political instability. Thus, owing to the short-term and volatile nature of the inflows the external position continued to attract scrutiny, and towards mid-1996 some international rating agencies placed Turkey on credit watch. Turkey's sovereign credit rating has remained at a historically low level since.

Policy response to high inflation: the 1998 stabilisation programme

Between 1995 and 1997 the rate of inflation systematically overshot the official projections set out in the annual budget. As the effects of real depreciation levelled off, there was a drop in inflation in 1995, but inflation subsequently hovered well above the levels seen at the beginning of the 1990s (Figure 5). Price pressures further escalated sharply in 1997, a consequence of rising agricultural support prices as well as soaring public sector prices and wages. (The combined weight of food and public sector goods in the consumer and wholesale price index is approximately 50 per cent.) Domestic supply bottlenecks, generated in connection with rising capacity utilisation, appear to have played a negligible role in mounting inflation.

A three-year stabilisation programme was launched in early 1998 (Box 1), based upon a commitment to tight fiscal policy, the abolition of backward indexation of public sector wages and a more active role in the disinflation strategy for monetary policy, which was having to cope with continued strong capital inflows. Measures to reduce overheating led to a deceleration in economic activity in the first half of 1998. The slowdown was also confirmed by the weakness of industrial output growth, although there is a presumption that Turkish exports were little affected by the impact of financial market turmoil in Asia until the second half of

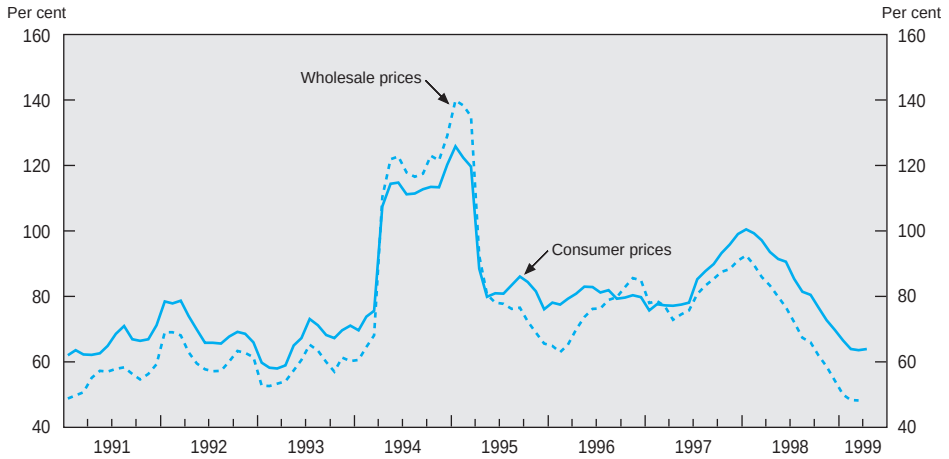
Figure 4. **Current account, direct investment and external debt**

Per cent of GDP

1. Methodological changes introduced at the beginning of 1999 include updating the external debt data in accordance with international standards, modernising the data code structure and enlarging the data scope. Treasury plans to complete the modification of the 1964-95 series by end 1999.

Source: Central Bank of Turkey, Undersecretariat of Treasury.

Figure 5. **Inflation developments**
Change over twelve months



Source: OECD, *Main Economic Indicators.*

Box 1. The 1998-2000 Stabilisation Programme

Recognising the close links between financing of large budget deficits, monetary expansion and chronic inflation, the Turkish authorities launched a *Three-year Stabilisation Programme* in January 1998. This programme provides a framework for concerted anti-inflation actions on a number of fronts, including stronger co-ordination between macroeconomic policies and structural reforms.

Key objectives of the programme

- To reduce wholesale price inflation from over 90 per cent at end-1997 to single digits by end-2000;
- To increase the primary budget surplus from near balance in 1997 to above 4 per cent of GDP in 1998 and to sustain that level over the process of disinflation.

Macroeconomic policy instruments

- A mix of higher tax revenues and stronger expenditure restraints was targeted to improve the primary fiscal position;
- A protocol between the Treasury and the Central Bank on short-term advances sanctioned the end of borrowing from the Central Bank;

(continued on next page)

(continued)

– Against a background of strong capital inflows, monetary policy was envisaged to limit the expansion of net domestic assets, while the exchange rate was to remain consistent with inflation targets.

Structural reform elements

To underpin credibility and to enhance private agents' confidence, the programme comprised a number of structural reform measures, including:

– The linkage of *public-sector wages* and *agricultural support prices* to targeted inflation, rather than past inflation;
– *Accelerated privatisation*, both to lower domestic borrowing requirements and to enhance economic efficiency;
– *Rapid elimination* of *social security* deficits. This effort involves appropriate increases in the minimum retirement age and minimum contribution period, an expansion of the reference period for calculating pension benefits and an increase in the ceiling on wages subject to contributions;
– *Reform of the tax system* by shortening the lag between the accrual of tax obligations and their payment, while also lowering the tax rates to assist the widening of the tax base;
– *Strengthening the banking sector* and banking supervision through more stringent enforcement of capital adequacy requirements and reducing the ceiling on banks' net open foreign exchange positions.

A *Memorandum of Economic Policies* of June 1998 clarifies the commitments set out in the original three-year plan. In addition, it takes the programme a step further by adopting the practice of announcing quarterly targets for the budget, domestic borrowing, privatisation and the expected behaviour of base money. Since June 1998 these quarterly targets have been monitored by the IMF staff and the staff findings have been made public.

Amendments in response to the Russian crisis

Several changes in the programme were necessary in the second half of 1998 to take account of the deteriorating external environment. First, the ceiling on the Central Bank's net domestic assets has been raised compared with the original plan. Second, the exchange rate has been managed mainly to preserve international competitiveness rather than as an anti-inflationary policy tool. On the fiscal side, the authorities' original commitments were confirmed. However, mounting unforeseen public spending in agriculture implied that alternative sources of savings had to be identified.

1998 (see below). Consumer price inflation, which peaked at more than 100 per cent in January (year-on-year), declined gradually to around 80 per cent by August. External forces also influenced price dynamics, primarily weak world primary oil and non-oil commodities. In addition, the declining trend of certain primary raw material prices intensively used by public enterprises (oil, aluminium, iron and steel, for example) helped to reduce the inflationary impact of the public sector. Administered price restraint also played a role.

Against a background of increasing political uncertainty, the commitment to disinflation was re-stated within the framework of the June 1998 *Memorandum of Economic Policies*, and was backed up by a self-imposed quarterly monitoring process by the IMF, as well as a stronger commitment to transparency in the disclosure of information.

Responses to the global financial crisis

Effects on trade

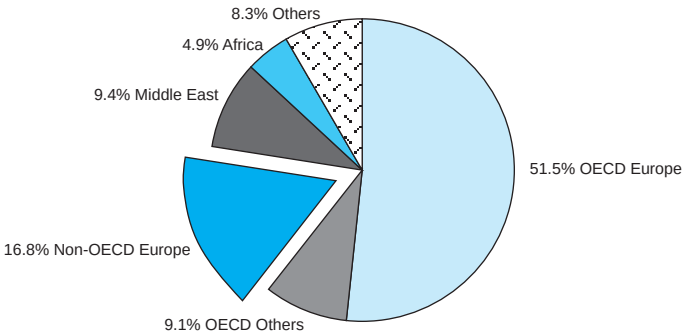
During 1998 a number of sectors for which Turkish exports tend to overlap with East Asian exports – notably foodstuffs, leather, and iron and steel – began to undergo a deterioration in their export performance. This was concentrated mainly in the second half of the year, however, as the improved competitiveness of East Asia seems to have reached Turkey with some lag. The Russian crisis, by contrast, had a more immediate impact, reflecting strengthening commercial links. Output developments in Turkey and the former Soviet Union have become increasingly interdependent in recent years. In 1997, Eastern Europe and the former Soviet Union accounted for about 16 per cent of total exports from Turkey (Figure 6). Assessing the full exposure to developments in Russia is difficult, however, reflecting the omission of a sizeable amount of the so called "shuttle" trade with Russia, as well as the neighbouring countries of the former Soviet Union, although estimates of such trade are made by the use of survey data. Also, there is a large volume of "unregistered" activities which are indirectly measured via foreign exchange receipts deposited with the commercial banks, and included in the "other goods and services" item of the invisible balance (Table 3).

Financial impact of global instability

The most immediate impact of the crisis in Russia was the sudden reversal of foreign investors' sentiment, reflected in capital outflows, which temporarily left the Treasury unable to borrow from international markets. The effect of financial instability was thus a strong contraction in foreign reserves, which had just recovered from the effects of the Asian crisis. Foreign reserves fell by some $ 2-3 billion. The large outflow of capital over the second half of 1998 ($ 7 billion, almost entirely reversing the inflows registered since the beginning of the year) took the form of a huge shift in the portfolio investment account (Table 4). In addition, an outflow of about $ 4 billion was attributed to residual errors and omissions in the balance of payments, probably resulting from hidden capital flight to offshore financial centres and under-reporting of export proceeds abroad in anticipation of further currency losses.

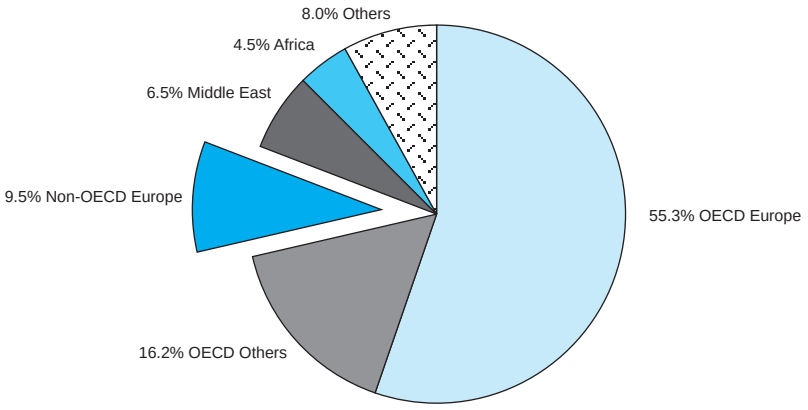
Figure 6. **Geographical distribution of foreign trade**
1997

Exports



Total exports = US$ 24.7 billions

Imports



Total imports = US$ 46.1 billions

Source: OECD, *Foreign Trade Statistics.*

Table 3. **Current account of the balance of payments**

Millions of US dollars

	1992	1993	1994	1995	1996	1997	1998		
							I	II	Year
Merchandise exports (f.o.b.)	14 891	15 611	18 390	21 975	32 446	32 647	15 765	15 363	31 128
Exports	14 715	15 345	18 106	21 636	23 225	26 261	13 332	13 549	26 881
Shuttle trade	..	..	..	..	8 842	5 849	2 182	1 507	3 689
Transit trade	176	266	284	339	379	537	251	307	558
Merchandise imports (f.o.b.)	23 081	29 771	22 606	35 187	43 028	48 005	22 891	22 649	45 540
Trade balance	**–8 190**	**–14 160**	**–4 216**	**–13 212**	**–10 582**	**–15 358**	**–7 126**	**–7 286**	**–14 412**
Other goods and services (net)	3 157	3 959	3 755	6 377	3 698	7 854	3 750	7 627	11 377
Private unrequited transfers (net)	3 147	3 035	2 709	3 425	3 892	4 552	2 415	3 153	5 568
Official unrequited transfers (net)	912	733	383	1 071	555	314	62	97	159
Invisible balance	**7 216**	**7 727**	**6 847**	**10 873**	**8 145**	**12 720**	**6 227**	**10 877**	**17 104**
Current balance	**–974**	**–6 433**	**2 631**	**–2 339**	**–2 437**	**–2 638**	**–899**	**3 591**	**2 692**

Source: Central Bank of Turkey.

Table 4. **Net capital movements**

Millions of US dollars

	1992	1993	1994	1995	1996	1997	1998		
							I	II	Year
Total net capital movements	**3 648**	**8 963**	**–4 194**	**4 643**	**8 763**	**8 737**	**7 648**	**–6 875**	**773**
Direct investment	779	622	559	772	612	554	220	353	573
Portfolio investment	2 411	3 917	1 158	237	570	1 634	1 420	–7 806	–6 386
Other long-term capital	–938	1 370	–784	–79	1 636	4 788	2 607	1 378	3 985
Short-term capital	1 396	3 054	–5 127	3 713	5 945	1 761	3 401	–800	2 601
Net errors and omissions	**–1 190**	**–2 222**	**1 769**	**2 354**	**–1 781**	**–2 755**	**1 421**	**–4 439**	**–3 018**
Current balance	**–974**	**–6 433**	**2 631**	**–2 339**	**–2 437**	**–2 638**	**–899**	**3 591**	**2 692**
Change in reserves[1]	**–1 484**	**–308**	**–206**	**–4 658**	**–4 545**	**–3 344**	**–8 170**	**7 723**	**–447**

1. A minus sign indicates an increase in reserves.

Source: Central Bank of Turkey.

Table 5. **Financial market developments: a comparison between three external shock episodes**

	1997	1998	1999
	Asian crisis (t = July)	Russian crisis (t = August)	Brazilian crisis (t= January)
One month prior to the shock (t – 1)			
Real effective exchange-rate appreciation[1]	6.9	13.3	17.8
Real interest rate[2]	–8.4	–22.0	20.3
Istanbul stock exchange[3]	90.2	25.3	–24.7
One month after the shock (t + 1)			
Nominal depreciation of the T-lira against the dollar	13.8	2.6	11.5
Real interest rate[2]	–4.0	–0.8	20.0
Istanbul stock exchange[3]	102.9	–34.3	12.8
Two months after the shock (t + 2)			
Nominal depreciation of the T-lira against the dollar	18.1	4.0	17.9
Real interest rate[2]	–9.8	4.6	9.9
Istanbul stock exchange[3]	165.7	–36.4	32.0

1. Percentage deviation of the (t – 1) index from average since the 1994 crisis (1994 QII). Figures have been deflated using the wholesale price index.
2. Simple deviation of monthly level from the 1994 crisis level (1994 QII).
3. Percentage change since the beginning of the year. Figures for the Brazilian crisis refer to rates of change relative to the beginning of 1998.

Source: OECD.

Average interest rates on Treasury bills jumped from 77 per cent in July to 136 per cent in September, but the Turkish-lira fluctuated against the dollar significantly less than at the time of the Asian crisis, suggesting that, at least initially, the Central Bank intervened quite strongly in defence of the currency (Table 5). The wider gap between interbank rates and the rate of nominal depreciation of the lira proved to be insufficient to arrest the massive outflow of capital, however, pointing to a serious confidence problem. Under the circumstances, fears that the liquidity squeeze could jeopardise the capacity to re-fund public sector debt hit the debt programme of the Treasury. Both the August and September auctions deviated strongly from the expected quantity and maturity of borrowing. Domestic factors also influenced the fall of confidence. One was the uncertainties brought about by the introduction of withholding taxes on interbank transactions in July. Another was the expansion of agricultural support funds which was required to compensate for the adverse impact of stronger agricultural output on farmers' revenues. This created a surge in demand for liquidity on the part of the state banks. A third factor was domestic political instability.

Monetary policy: a stabilising role

Monetary policy priorities

While the economic expansion was accelerating, up to the first half of 1998, the main monetary policy priorities remained those of ensuring the stability of the foreign-exchange and domestic money markets in the face of the high financing requirements of the public sector. Stability was achieved through a relatively predictable real exchange rate path, combined with high short- and long-term interest rates, which tended to attract large capital inflows and generated overall balance of payments surpluses up to the first half of 1998. This was a source of liquidity creation, and a high premium was placed on central bank liquidity management. Open market operations were required to sterilise episodes of excessive capital inflows or inject liquidity when faced with a reversal of foreign capital (as in late 1997 in the wake of the Asian crisis). Such an approach made room for financing the public sector deficit, while allowing private sector credit to grow rapidly. It was thus consistent with an overall neutral or accommodating monetary policy *vis-à-vis* inflation. Essentially, by focusing on quasi stability in the real exchange rate, the Central Bank continued to allow a significant element of indirect monetary financing by granting liquidity to the banking sector. This assisted financial stability in the domestic market, while sustaining Turkish export competitiveness, but was based on an implicit underwriting of chronically high inflation.

The *Memorandum of Economic Policies* launched in June 1998 envisaged a more active role for the Central Bank of Turkey in influencing aggregate monetary developments. It was announced that the exchange rate was to be managed in a way consistent with the annual inflation targets. Considering the difficulty in predicting aggregate monetary developments in an environment of decelerating inflation, greater emphasis was placed on control over the growth of domestic assets of the Central Bank, rather than reserve money, as a means of disinflation. Under the new approach, the expansion of net domestic assets would have been maintained under tighter control, while also allowing interest rates to move more freely. In implementing monetary policy, the central bank remained committed to taking all necessary action to maintain international competitiveness and confidence in the lira.

However, responding to the capital flight and liquidity crunch that followed the August 1998 crisis, the Turkish authorities had little choice but to alter radically the composition of money growth from that envisaged under the original plan. Thus the Central Bank allowed the volume of open market operations to expand, and net domestic assets surged in the later part of 1998. This expansion was also manifested in the broader monetary aggregates, with M2 accelerating well above inflation in the latter part of 1998, as the growth of bank savings

Table 6. **Monetary conditions**

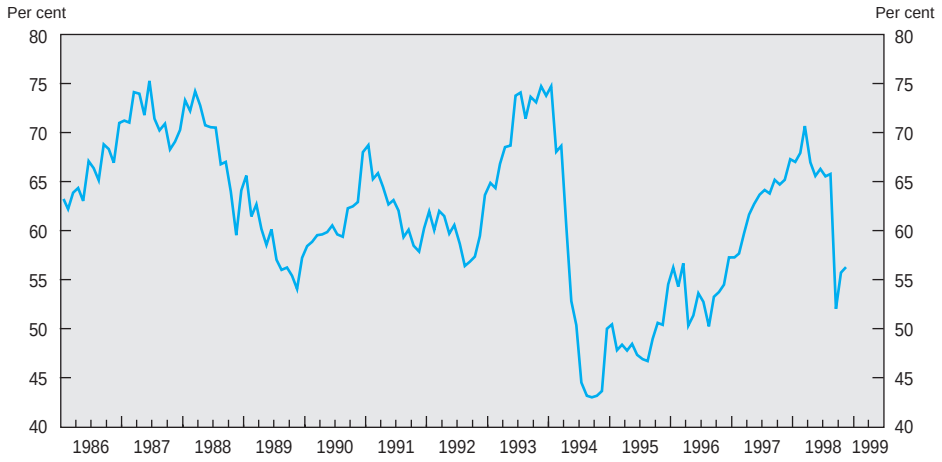
Annual percentage change

	1995	1996	1997	1998
Reserve money	84.9	80.0	84.7	78.7
Monetary base	92.3	72.8	–28.6	–56.1
Central Bank money	93.0	72.2	7.9	–14.6
Total domestic liabilities	113.1	94.5	74.3	13.0
Total balance sheet	149.2	101.6	128.1	69.9
M1	65.7	109.9	65.8	65.7
M2	97.7	120.5	87.9	106.2
M2Y	106.5	109.6	84.0	79.4
M3Y	102.3	124.8	98.7	87.0
Total deposits	100.9	132.7	97.6	101.4
Total credits	149.9	114.0	130.0	62.7
Memorandum item:				
Wholesale price index[1]	65.6	84.9	91.0	54.3

1. End-of-period figures.
Source: Central Bank of Turkey.

Figure 7. **Bank credit to the private sector**

Per cent of M2Y[1]



1. M2Y is M2 plus foreign currency deposits.
Source: OECD, *Main Economic Indicators.*

deposits strengthened (Table 6). Because of high deposit rates, foreign exchange deposits (in US dollar terms) and repo operations of the banks with their clients decreased. To balance the liquidity of the system, the central bank injected both the required lira and foreign exchange into the market. However, the proportion of deposits destined to fund credits to the private sector fell precipitously (Figure 7). Unlike the 1994 crisis, the government did not commit the error of trying to use administrative means to force financial institutions to absorb government paper at below-equilibrium interest rates.

Stresses in the financial system

The hike of lira interest rates also meant that a number of banks experienced distress. Since the release of the 1997 *Survey* two private banks have been transferred to the Saving Deposits Insurance Fund of the Central Bank. Increased arbitrage activity, involving the borrowing of less costly foreign funds and investing in lucrative domestic assets also played a role in increasing bank vulnerability (Table 7). At 31 per cent of capital by end 1998, the net foreign exchange position of the overall banking system was three points below the statutory ceiling. From January a 30 per cent ceiling has been in operation, and both investment and state banks were within this prudential ceiling, although there was a serious deterioration in the exposure of public banks. The large exposure of the private commercial banks was mainly the result of the high exchange rate risk taken by a number of small and medium-sized banks, which also carry a significant maturity mismatch between their short-term liabilities and long-term assets. Nonetheless,

Table 7. **Open positions of the banking system**

Millions of US dollars

	1992	1993	1994	1995	1996	1997	1998	
							Value	Per cent of the capital base
Investment and development banks	25	–404	–195	–80	–13	35	11	1.6
Deposit money banks	–2 616	–4 574	–823	–347	–1 216	–1 918	–2 530	–31.2
State banks	–660	–673	59	–213	136	287	–295	–17.2
Private banks	–1 944	–3 577	–720	–206	–1 324	–2 069	–2 112	–36.5
Foreign banks	–12	–324	–162	73	–29	–135	–123	–19.9
Total	–2 591	–4 978	–1 018	–427	–1 229	–1 883	–2 519	–28.7

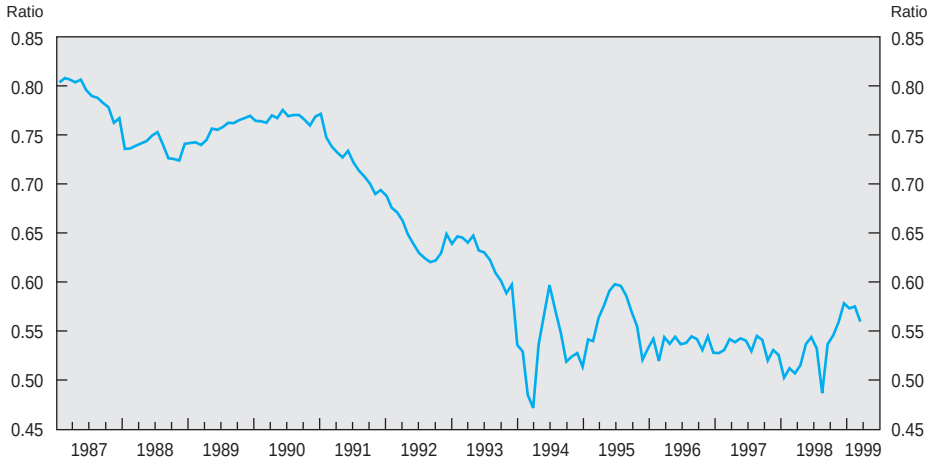
Source: Undersecretariat of Treasury.

the foreign-exchange open positions of the banking sector were much smaller than they were prior to the 1994 crisis (Table 7).[1]

The Turkish economy has gone through a period of extensive financial liberalisation over recent years (see Chapter IV); however, reflecting a difficult operating context, with high inflation, persistent fiscal imbalances and periodic episodes of exchange rate crises, allocational distortions increased. Economic agents have adapted by shortening the duration of their financial contracts, while banks have responded by focusing on government securities as a source of income rather than expanding their traditional lending business. With the largest private banks belonging to powerful family conglomerates, private bank lending is provided mainly to customers with known credit credentials. As a consequence, other firms, including many small and medium-sized firms, have been forced to turn to their own funds for the financing of their investment plans, limiting their borrowing to short-term working capital. Effectively, only exporting firms can borrow abroad. These developments have increased the dependence of the banking system on risk-free income while also making for a highly inefficient allocation of resources, and hence for sub-optimal growth.

As discussed in the 1997 *Survey of Turkey*, the high inflation environment has also led to a progressive switch from domestic to foreign currencies. Many

Figure 8. **Currency substitution**
Ratio M2 to M2Y[1]



1. M2Y is M2 plus foreign currency deposits.
Source: OECD, *Main Economic Indicators.*

Turkish firms, especially those that are more export-oriented, find that this can simplify their planning and minimise the adverse impact of unanticipated nominal exchange rate depreciation. Foreign currency deposits currently represent about one-half of broad money (M2Y) (Figure 8). However, widespread currency substitution raises the issue of whether domestic monetary aggregates can be expressed as a stable function of standard macroeconomic variables (like nominal GDP, for example) and can thus serve, as in other countries, as a useful guide for monetary policy.

One manifestation of financial distortions has recently been the rapid development of the short-term repo market, which was stimulated on the demand side by the former discriminatory tax framework and on the part of issuing banks by the lack of reserve requirements on this instrument. Repo transactions involve the sale of a government security – the highest yielding asset in Turkey – with a commitment to buy back the security at a given date and price. These transactions have been used widely by banks since 1995 to finance off-balance sheet portfolios of Treasury securities and by households as a highly remunerated short-term substitute for deposits. They increased by 13 per cent in dollar terms in 1997, decreasing somewhat in 1998 as the tax advantage compared with interest earnings on normal deposits was partly removed. Repo transactions are an off-balance sheet item, and to maintain the effectiveness of monetary policy operations, the Central Bank controls their amount indirectly by monitoring the liquidity position of the banks on a daily basis. This information is taken into account in the formulation of its monetary policy operations.

Another important source of banking sector weaknesses can be found in the problems of the public sector banks. These banks provide subsidised lending to specific favoured sectors, but lack adequate capital injection from the Treasury to cover associated losses, thereby creating a cash flow problem and a mounting contingent liability for the government as market interest rates have risen (see Chapter II). The distortionary effects of the financial operations of public banks materialise in upward pressures on market interest rates. Faced with mounting losses these banks need to attract an increasing amount of deposits to expand their cash-flow. The importance of these problems is magnified by the size of the operations of the public banks. The three major public banks accounted for some 32 per cent of the total assets of the banking system as of the end of 1998.

Economic prospects

Short-term outlook

The external trade environment does not favour a rapid pick up from the currently slow rate of economic growth. With a still-weak external environment

Table 8. **Short-term outlook**

Percentage changes, volume (1987 prices)

	1995		1998 Estimate	Projections	
	Current prices TL trillion	Per cent of GDP		1999	2000
Private consumption	5 457.9	70.3	0.1	2.0	3.8
Public consumption	837.2	10.8	5.0	5.0	5.0
Gross fixed investment	1 854.1	23.9	–2.4	0.0	5.2
Final domestic demand	8 149.2	104.9	–0.3	1.6	4.3
Stockbuilding[1]	127.1	1.6	1.1	0.0	0.0
Total domestic demand	8 276.4	106.6	0.7	1.6	4.3
Exports of goods and services	1 544.1	19.9	10.5	2.5	7.0
Imports of goods and services	1 890.2	24.3	2.2	3.0	8.0
Foreign balance[1]	–346.1	–4.4	2.2	–0.3	–0.6
Statistical discrepancy[1]	–163.9	–2.1	–0.2	0.0	0.0
GDP at market prices	7 766.3		2.8	1.4	3.9
GDP implicit price deflator			74.8	50.0	31.0
Memorandum items:					
Consumer prices[2]			80.7	54.0	32.0
Unemployment rate			6.3	6.5	6.4
Current balance ($ billion)			2.7	0.8	–1.4
Current balance (% of GDP)			1.5	0.4	–0.7

1. Change as a percentage of GDP in previous period.
2. Private consumption deflator.

Source: OECD.

this year, export market growth is likely to remain very sluggish, at around 2½ per cent in volume terms, before recovering to 7 per cent in 2000 (Table 8). While Turkish export competitiveness is assumed to remain constant in terms of relative prices, the real exchange rate could continue to appreciate in terms of relative unit labour costs, as has been the case under the new monetary policy regime. Turkey's world export share fell in 1998, due to the devaluations in Asia and Brazil, which have led to increased competition on third markets, and the impact of the Russian crisis.

On the international financing side, the situation is somewhat more propitious. The current account has shifted into small surplus and access to international capital markets is being gradually restored. In early 1999 foreign investors responded quite favourably to the issue of foreign exchange debt placements (see Chapter II). There has thus been some restoration in foreign confidence, which the projections assume will gather strength with the formation of a new government and a stabilisation of emerging markets globally. This should allow

real interest rates to decline steadily, although they are likely to remain relatively high in the absence of a more dramatic inflow of foreign capital, which would require a significantly strengthened structural reform programme.

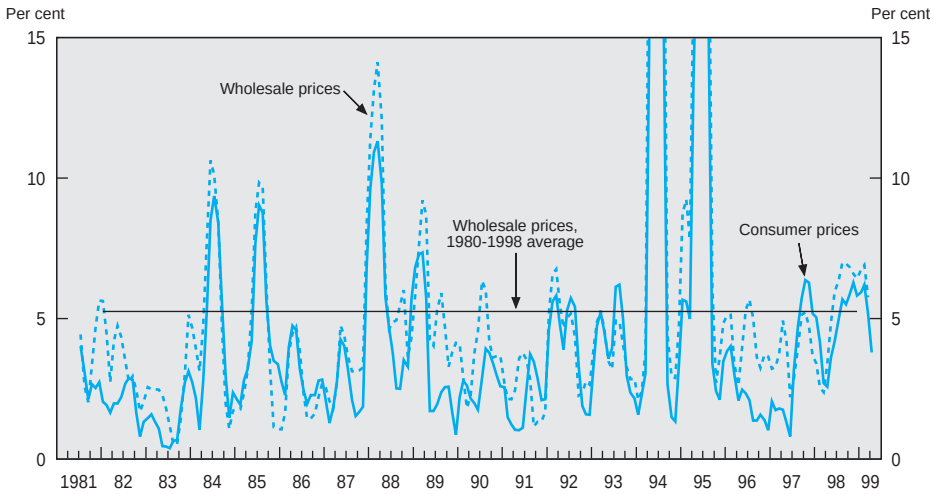
Domestically, it would be difficult to forecast a reduction in real interest rates to levels which would support a strong recovery. Even though a fall in nominal rates is projected to occur, the decline in consumer price inflation, from about 80 per cent (on average) in 1998 to 54 per cent in 1999 and lower in 2000, will have to be achieved against the background of still high inflation expectations. Indeed, inflation itself will be difficult to bring down to target levels because the trend in import prices is upwards and there is pressure on government wages and administered prices. The monetary stance will thus need to remain cautious. At the same time, the government's financing requirement is scheduled to remain high despite a continuing primary surplus. Indeed, domestic demand will be subject to the triple effects of credit restraint on private sector demand, constraints on public spending and higher tax pressure. Investment spending is not expected to recover very rapidly, although personal consumption, which is more responsive to lower real interest rates, will be supportive. Overall, total domestic demand is projected to pick up slowly this year, firming to a 4 to 4½ per cent rate of growth in 2000. With the foreign balance making a fairly neutral contribution, GDP growth would also recover to around 4 per cent, following modest growth this year.

The risks attaching to this projection are quite large. They relate principally to the developments of the external environment and the state of foreign confidence in the Turkish economy. In turn, these factors depend largely on fulfilment of the assumption that the government takes appropriate policy action to sustain the primary surplus as set out in the Stabilisation Programme, while also maintaining the momentum of structural reforms. If a fiscal policy is put in place which reduces inflation expectations to a degree consistent with current inflation targets, real interest rates could decline much more than projected, with important benefits for the strength and balance of the economy as it enters the new millennium. There would, however, be transition costs, which could further weaken domestic demand in the short term. If, on the other hand, implementation of the current stabilisation package is incomplete, there could be a temporary recovery of demand, without the economy achieving a balance which would sustain growth into the medium term, with the risk of repeating earlier cycles of growth and instability.

The longer-run costs of persistent inflation

Abstracting from 1994, when the rate of inflation rose to 140 per cent following the currency crisis, the variability of inflation, measured by the standard deviation of monthly inflation rates, has remained remarkably stable during the

Figure 9. **Inflation variability**[1]



1. Monthly average of the standard deviation of the inflation rates of the six preceding months.
Source: OECD, *Main Economic Indicators.*

1990s, pointing to a high degree of inflation inertia (Figure 9). However, high inflation, even if relatively predictable, has severe costs. Perhaps the most important are that planning horizons are arbitrarily shortened and in a non-neutral tax system there are resultant large distortions in rates of return, which exacerbate the income inequalities which are salient feature of the Turkish economy. Indeed, the burden of the inability to deal with chronic inflation tends to be concentrated on the lower income segments of the population. The consequences of inflation in reducing the value of money balances tends to be regressive ("inflation tax"). As inflation accelerates, there is a large unseen transfer of wealth from the savers to borrowers, via *ex post* negative real interest rates. Negative real rates of interest can encourage investment in assets which are preferentially taxed and discourage investment in assets which are taxed without indexation provision, encouraging tax evasion and erosion of the tax base. In this case, social programme spending may be crowded out. Negative real rates of interest will disappear as expectations adapt to sustained inflation, however, and on marketable debt savers will demand a high risk premium, as at present in Turkey. In this case, wealth is transferred from borrowers and taxpayers to the "rentier" class holding the domestic public debt. Scarce public resources are once again diverted from needed social expenditures to a rising public debt service. Ultimately, by worsen-

ing the inequality of incomes, inflation can exacerbate the risk of social instability, in addition to imposing significant economic costs.

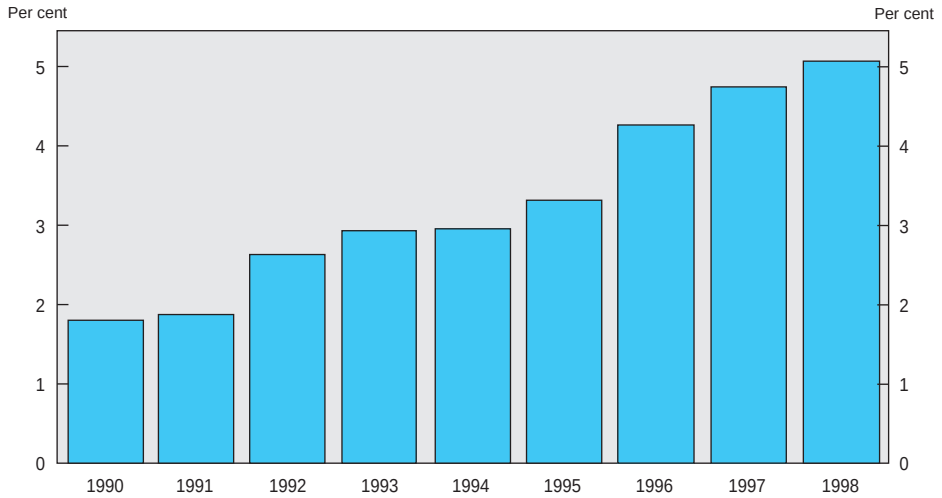
It is not clear how much productive investment in general has been crowded out by recently high real interest rates. As regards the export industries, these are able to borrow in foreign currency. Moreover, a salient feature of private investment in Turkey is that the bulk of these expenditures is not very sensitive to interest rates – a consequence of widespread lumpy, own-financed investment, particularly in the residential construction sector. (In this sense high inflation has severe costs in preventing the development of important market segments.) In addition, the major private conglomerates own their own banks and therefore have direct access to credit lines. These considerations mean that no precise measure of crowding out is possible. If the upper limit for the value of output losses caused by government borrowing were to be estimated using the assumptions that public debt crowds out private capital on a one-for-one basis (Figure 10), the impact is rather small. But this is because, with the extensive inflation erosion of the debt, the real value of the domestically-held debt is not very high in Turkey as a percentage of GDP (see next chapter). The burden of wealth erosion has not been borne directly by the business sector. Rather it has imposed an arbitrary and concealed tax on households, with a consequently high excess burden for the economy at large.

Of great significance is also the instability of capital imposed on the economy because of difficulties in attracting a steady flow of foreign capital in a high inflation environment. This does not just apply to foreign capital willing to risk investment in domestic currency, but also to foreign currency borrowing in a situation where the essential collateral – the tax base – is subject to erosion. These considerations imply that Turkey's longer-run growth potential is not likely to be fulfilled while public sector borrowing remains so high, and that tax reform has to be an integral part of the process of reduction. The problem is that, from the expenditure side, there would be quite difficult problems of resource reallocation, away from sectors which have so far been the recipients of inflation and subsidisation benefits, towards social programmes more generally. Disinflation would thus also entail costs. Such a process has already been put in train, but it is so far very gradual and incomplete. Perhaps the most important feature of the Turkish situation is that the economy is simultaneously suffering both from the costs of inflation and the costs of partial disinflation, as is the case with a gradual disinflation process. For the situation to change the fiscal situation is key. Developments and trends in fiscal policy are discussed in Chapter II.

Benefits of returning to low inflation

A recent development has been the widening consensus against inflation. At the beginning of 1999 the government and the social parties intensified their

Figure 10. **Cumulative simulated impact of public-sector debt on the level of GNP**[1]



1. Simulations assume that the government overall net domestic cash and non-cash debt is replaced by an identical amount of business-sector capital. Thus capital crowding-out is assumed to be one-for-one. The figure shows the effects of this increase in capital stock on aggregate output. The impact of the crowding-out is calculated as the percentage deviation of the hypothetical real GNP level (Y_H) from the observed level (Y_o), where:

$Y_H = A (K + D)^{1 - \alpha} L^{\alpha}$;

$Y_o = A K^{1 - \alpha} L^{\alpha}$ and

K = capital stock (assumed to depreciate by 5 per cent per annum for the machinery and equipment component and by 3 per cent for the building component);

L = employment;

D = total domestic debt;

α = share of wages in total factor income (assumed to be 2/3).

Source: OECD estimates.

dialogue on how this may be achieved. Several common goals have already been identified, including those mentioned above in respect of tightening the control of current public spending. This could set the stage for the launching after the election of several important structural reforms, which during 1998 were not implemented as planned. It is important that the process of reform be viewed as a consensual, integrated whole. This is not just because partial reforms suffer from credibility problems, but because experience in other OECD economies shows that consensus-based policies can create a virtuous circle of improving expectations and enhanced real performance, largely through the medium of financial market confidence. The institutional aspects behind the building of such an approach are discussed in Chapter IV.

In the context of the above debate, it is of interest to assess the possible longer-run gains for the Turkish economy of achieving a stable low inflation environment. Currently, Turkish financial markets are highly distorted, and business and investment behaviour are correspondingly skewed. As noted above, corporations have been largely insulated from the direct crowding-out effects of excessive government borrowing. However, the OECD has conducted some simulations of what the economy might look like if investment and consumption behaviour were to revert to the OECD norm in terms of interest rates behaviour as real rates themselves are normalised. (This would, of course, require a normalisation of capital markets and may take time.) OECD calculations (using the INTERLINK model: see Annex I) show that if the real interest rate were to fall gradually by approximately 20-25 percentage points compared with their central projection level of about 30 per cent, and the rate of inflation were to decline to one digit levels, total aggregate output could increase by a cumulative 15 per cent (relative to baseline) in six years. A particularly strong positive impact would be felt by total domestic demand, which would be 20 per cent above baseline. Stronger domestic demand would stem not only from investment but also consumption, via the job-creating consequences of surging business spending.

II. Budget deficits and debt: correcting structural imbalances

Policy background

The 1997 *Economic Survey of Turkey* stressed the importance of rapidly redressing the fiscal deficit and concluded that public sector reforms were needed to achieve this fundamental objective. Without broad-based fiscal consolidation the objectives of a durable decline in inflation and a sustainable improvement in confidence were considered unachievable, since the budget makes perhaps the most important contribution to inflation inertia. As noted in Chapter I, the minority coalition government that took office in mid-1997 adopted a gradual approach to disinflation, based on the immediate commitment to a strong budgetary primary surplus. It was intended that fiscal retrenchment be backed up by structural reforms, comprising specific privatisation targets, reforms to the civil service wage indexation system, steps to improve the tax administration and reforms to the social security system. This chapter looks first at the progress of the fiscal consolidation process so far, from the perspective of the primary balance and the sustainability issues that arise because of the effects of high real interest rates on government debt. In this context, the "consolidated" budget tells only part of the story about state involvement in the economy. Public sector banks provide subsidised credit to agriculture and smaller firms and state-owned enterprises run at a loss, so that government budgetary involvement is higher than the impression given by the narrower definition. The second section analyses the structural factors underpinning the commitment to a strong primary surplus (reform of the social security system being the subject of Chapter III). The final part assesses the current status of the consolidation process in terms of what needs to be done in order to ensure continued credibility.

Progress towards budget consolidation

Fiscal policy in 1998

Following a sharp positive adjustment as a result of the 1994-95 stabilisation package, the consolidated budget deficit relapsed to 8.3 per cent of GNP in 1996. 1997 saw a slight improvement, but the primary surplus fell to zero. A fundamental element of the three-year counter-inflation programme was the need to bring the primary surplus back to a level where debt accumulation could be brought under control. Consistent with the objectives of the three-year plan, the 1998 budget quantified targets for higher revenues and primary expenditure restraint which should have generated a primary surplus of 3.9 per cent of GNP in 1998 (Table 9). However, the authorities' commitment to disinflation also meant that a sharp rise of interest payments had to be expected. Consequently, the overall budget deficit of the consolidated government was to have increased by about half a percentage point of GNP, to 8.1 per cent.

The eighteen-month Staff Monitored Programme (SMP), requested from the IMF in June 1998, was meant to underpin the credibility of the authorities' policies.[2] The programme tightened the deficit target for 1998 to 7.6 per cent of GNP, envisaging a larger primary surplus (4.1 per cent of GNP), to be achieved mainly through additional cuts in expenditure. The main policies and objectives of the three-year programme were re-stated to comply with the new dialogue framework and were set out in a *Memorandum of Economic Policies*. A further improvement to the plan was announced in October 1998, targeting a higher primary surplus for 1998 at 4.6 per cent of GNP, despite slower than originally expected real output growth. Several emergency measures to alleviate the negative effects of the Russian crisis were also introduced on that occasion, mainly initiatives to increase the availability of public export support credits.

At about 7 per cent of GNP (Figure 11), the 1998 outturn for the consolidated central government deficit exceeded the OECD average by over 5 percentage points. The major source of spending pressure was mounting interest payments, reflecting not only high interest rates, but also a lengthening of maturities during 1997, which caused a significant shift of interest payments into 1998. By contrast, one positive fiscal development compared with 1997 was the sizeable increase in the primary budget surplus to 4.6 per cent, high enough to meet the October official target. To a large extent, this was induced by a rise in total revenues, which at 22 per cent of GNP were 2.6 percentage points higher than in 1997. Stronger tax collection, as part of a wider tax reform effort to reverse the effects of tax collection lags,[3] also helped the performance of total revenues, as did one-off privatisation revenues, notably the sale of mobile phone licences. For the rest, the improvement of the primary balance was due to a decline in non-interest expenditures, which fell to 17.4 per cent of GNP – 2.2 percentage points

Table 9. **Consolidated budget statement**

Per cent of GNP

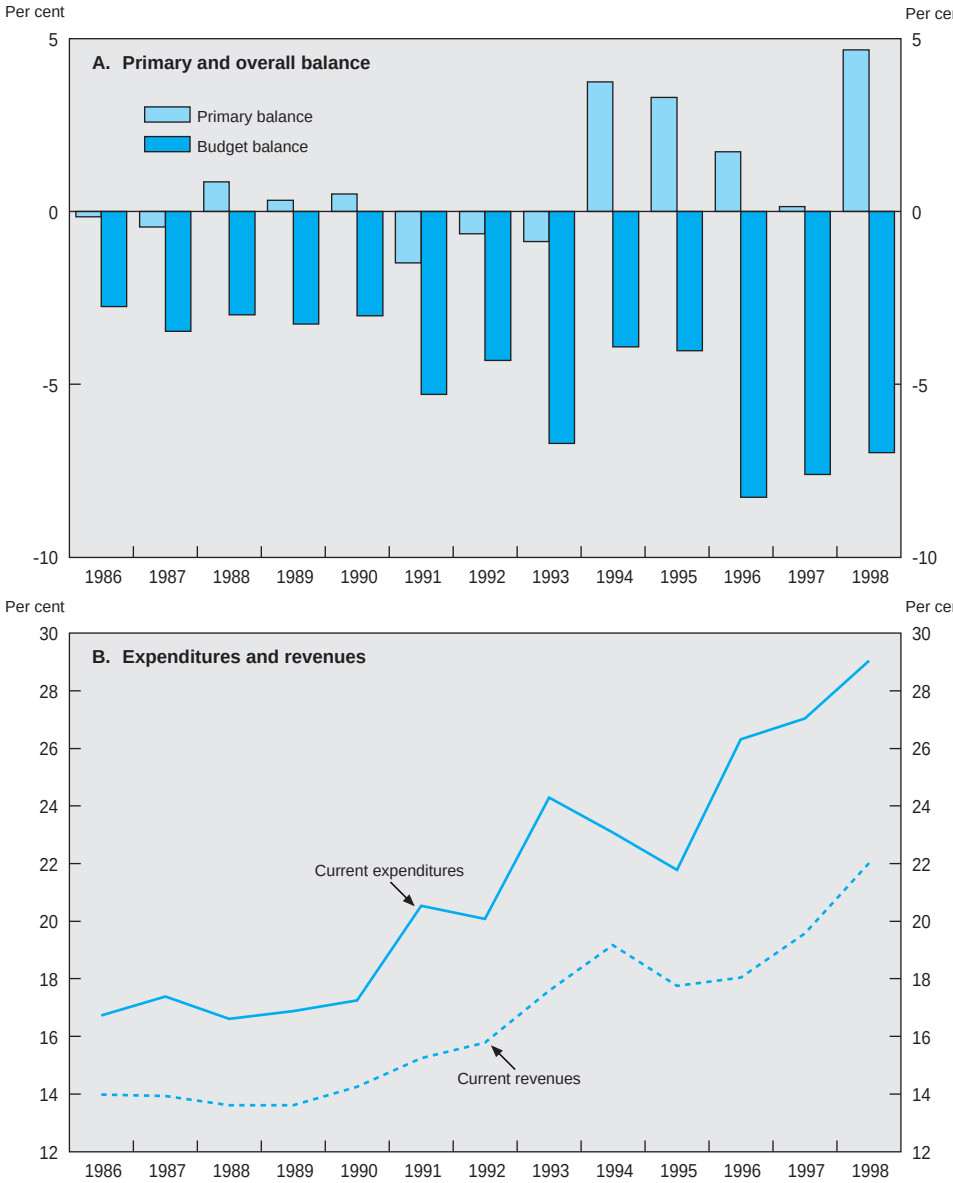
	1993	1994	1995	1996	1997	1998			
						Budget	SMP[1]	October revision	Outturn
Revenues									
Tax revenues	13.2	15.1	13.8	15.0	16.0	18.1	17.7	17.7	17.4
Non-tax revenues	4.1	3.8	3.7	2.9	3.4	3.9	3.9	4.5	4.6
Total receipts	17.6	19.2	17.7	18.0	19.4	22.0	21.6	22.2	22.0
Expenditures									
Wages	8.5	7.0	6.4	6.5	7.0	7.1	7.1	7.3	7.3
Other current	1.8	1.9	1.8	2.1	2.4	2.7	2.5	2.5	2.4
Investment	2.7	1.9	1.2	1.6	2.0	2.0	1.9	1.8	1.7
Transfers to SEEs	1.3	0.5	0.6	0.3	0.4	0.3	0.3	0.3	0.3
Other transfers	4.3	4.1	4.5	5.8	7.5	5.9	5.7	5.6	5.7
Total non-interest disbursements	18.5	15.4	14.4	16.3	19.2	18.1	17.5	17.6	17.4
Interest on public debt	5.8	7.7	7.3	10.0	7.7	12.0	11.7	11.7	11.7
Foreign borrowing	1.2	1.7	1.3	1.1	1.0	1.2	1.2	1.1	1.0
Domestic borrowing	4.6	6.0	6.1	8.9	6.7	10.9	10.5	10.6	10.6
Total disbursements	24.3	23.1	21.8	26.3	26.9	30.1	29.2	29.2	29.1
Primary balance	–0.9	3.8	3.3	1.7	0.1	3.9	4.1	4.6	4.6
Overall balance	–6.7	–3.9	–4.0	–8.3	–7.5	–8.1	–7.6	–7.1	–7.0
Deferred payments	0.5	0.5	0.7	0.1	0.5	0.0	0.0	1.1	0.4
Advances	–0.2	–0.5	–0.4	–0.3	–0.4	0.0	0.0	0.0	–0.6
Cash balance	–6.3	–3.9	–3.7	–8.5	–7.5	–8.1	–7.6	–6.0	–7.2

1. IMF Staff Monitored Program.

Source: Undersecretariat of Treasury.

Figure 11. **Consolidated budget and primary budget balances**

Per cent of GNP



Source: Undersecretariat of Treasury.

lower than in 1997. Restrictions on some types of budgetary transfers, and the blocking of certain cash outlays authorised in 1997 to recapitalise public banks, helped to offset the impact of higher public wages and spending overruns in the social security system (see Chapter III).

The 1999 Budget situation

For 1999, the original budget proposal envisaged targets for the consolidated budget deficit and the primary surplus of 7 and 4.3 per cent of GNP, respectively, to be achieved mainly through revenue increases in the form of higher tax receipts. These objectives were based on an expected annual rate of real GNP growth of 3 per cent, an optimistic assumption, perhaps, considering subsequent trends. Parliamentary debate on the new draft was brought to a halt by an early election, which prompted the approval of an interim budget, covering only the first half of the year. Under this budget, expenditure items have been subjected to ceilings, limiting all state agencies' spending entitlements to their share of total budget as allocated in 1998.

The fiscal programme agreed with the IMF on the occasion of the quarterly monitoring visit in February 1999 envisaged a continued improvement in revenue collection as well as non-interest expenditure restraints. For the year as a whole, under current policies, the authorities are still committed to achieving a primary fiscal surplus of at least 3 per cent of GNP. However, due to slower growth of industrial output and imports, tax revenues undershot in the first months of the year, particularly VAT receipts. The target surplus thus looks extremely difficult to achieve.

Off-budget developments and the PSBR

The consolidated budget is defined as the net borrowing requirement of the central government and autonomous government agencies. It includes transfers to cover the social security deficit, so that it does not differ greatly from the general government balance (Table 10). The borrowing requirement of the non-financial state economic enterprises (SEEs) amounts to around 1 per cent of GNP, implying that the total public sector deficit (PSBR) is that much larger. It amounted to 11½ per cent of GNP in 1996 and 1997, when it failed to decline despite the better performance of both central and general government. An improvement was registered in 1998, when the PSBR fell to 10½ per cent. The Turkish public sector, including SEEs and civil servants, is estimated to employ about 2¼ million employees or close to 11 per cent of total officially registered employment.[4]

The engagement of the public sector in extra-budgetary and off-budget operations complicates any attempt to measure the government's overall

Table 10. **Fiscal accounts at different levels of government**

Per cent of GNP

	1993	1994	1995	1996	1997	1998
Consolidated budget						
Primary balance	–0.9	3.8	3.3	1.7	0.1	4.6
Interest payments	5.8	7.7	7.3	10.0	7.7	11.7
Cash balance (Treasury definition)	–6.3	–3.9	–3.7	–8.5	–7.5	–7.2
Off-budget transfers	–3.3	–2.6	–2.7	–3.5	–2.5	–2.5
Adjusted cash balance[1]	–9.9	–6.5	–6.4	–12.0	–10.0	–9.7
General government						
Primary balance	0.7	4.8	5.4	4.8	2.0	–
Interest payments	9.4	10.4	9.6	12.9	10.0	–
Cash balance (Treasury definition)	–8.7	–5.6	–4.2	–8.1	–9.0	–7.6
Adjusted cash balance[1]	–11.7	–8.2	–6.9	–11.6	–10.5	–10.1
Public-sector balance						
Cash balance (Treasury definition)	–11.3	–7.2	–3.3	–8.0	–9.1	–7.9
PSBR[1]	–14.6	–9.8	–6.0	–11.5	–11.6	–10.4

1. Including off-budget transfers.
Source: Undersecretariat of Treasury and OECD estimates.

fiscal activity. Extra-budgetary funds include subsidies to and finance for the investments of public industries which are funded through earmarked duties and taxes. The most outstanding examples comprise the defence industry fund and the social solidarity and aid fund. Extra-budgetary funds were used extensively in the past because they do not require preliminary submission to the ordinary budget decision process and therefore can be implemented more quickly. Since 1993, however, recognising that they undermine the effectiveness of budgetary control, while also posing risks of an arbitrary diversion of public spending, the government has taken the majority of extra-budgetary funds into budget.

Quasi-fiscal, or off-budget operations, are now the major source of weakness in the process of government financial reporting in Turkey, largely because of the cash based reporting system. They consist of three items:

– The foreign borrowing of the public institutions (for example, local governments and SEEs) are typically assumed under full *government guarantees*. If and when default occurs, these borrowings become part of government debt, although these institutions are expected to repay loans to the Treasury in instalments.
– The Treasury meets its financial obligations to the state banks, central bank and SEEs by issuing special non-marketable bonds if budget laws

permit. These *non-cash debt payments* are not recorded as an expenditure item in the budget, but are treated as below-the-line items. Government spending is thus underestimated.

– The third off-budget item is the *unpaid duty losses* of the state banks. Some operations of Treasury, mainly related to agricultural support policy, are effected through the mediation of state banks (Ziraat Bank for individual farmers and Halk Bank for small businesses). Such mediation turns into a "receivable" on the part of the state banks, which includes interest charged but not being paid by the Treasury due to inadequate financial resources.[5] The accumulated receivables of state banks are shown as an asset item in their balance sheet, but not as a liability item in the budget accounts.

In recent years, some progress has been made towards controlling some of these off-budget operations in Turkey. Guarantees for the local authorities have been limited by the budget law of 1998 and the temporary budget law of 1999, which also authorised the Treasury to liquidate unpaid duty losses by issuing special bonds to an amount not exceeding 15 per cent of the existing stock. Issuing special or non-cash bonds has also been limited gradually. Under the 1998 budget law, this type of borrowing has been included in the general borrowing limit defined by the parliament. Nevertheless, off-budget operations remain sizeable and are distorting the financial statements of the government, while undermining their credibility. They also break the link between the financial accounts and the debt stock of the government.

As mentioned, quasi-fiscal operations represent an important source of compensation for the duty losses generated by years of generous agricultural support policies. Recent OECD estimates suggest that in the last few years the direct transfers to agriculture resulting from price-support schemes exceeded 4 per cent of GDP per year, peaking at around 5.5 per cent of GDP in 1998.[6] Not all of the losses have been compensated annually, so that the counterpart has been the rapid growth of receivables owed by the Treasury to the agricultural state-owned bank (Ziraat). These effectively compensate Ziraat for the costs of subsidised lending and for the writing off of the debt owed to it by other public entities obliged to purchase agricultural output.[7] For 1998, mounting receivables were a reflection of the larger than expected agricultural harvest and the corresponding increase in the costs of agricultural support that was needed to offset the associated downward pressures on agricultural prices. Overall, the amount of off-budget operations are estimated to have remained at approximately the same level in 1998 as in the previous year, at 2.5 per cent of GDP. If off-budget losses were taken into account, they would thus halve the effective size of the consolidated primary surplus for 1998.

The debt burden

Any judgements about the adequacy of the consolidation process depend on the extent to which debt accumulation has been stabilised by the primary surplus. Turkish government debt is very short-term, which means that interest rate changes are very quickly incorporate into debt service charges, making the budget deficit very vulnerable to "debt dynamics" where the real interest-rate burden rises too fast for the primary surplus to prevent a debt explosion. Indeed, the current danger is that very high real interest rates could trigger a situation where debt continues to rise, creating incentives to monetisation. This risk in turn leads market participants to demand both higher real rates and shorter maturities (see Annex II). The authorities, on the other hand, have been quite active in attempting to extend the maturity of the debt. But while such debt management can help budget financing in the short term, the credibility of the consolidation process depends on more fundamental factors.

Debt management

Since 1994 debt management has been geared towards increasing reliance on domestic financing to cover the consolidated deficit (Table 11). Net foreign borrowing has been negative for several years (Figure 12, Panel A), so that domestic borrowing has exceeded the total financing requirement by a significant margin. For 1998, this tendency was accentuated by lower than expected privatisation receipts and a squeeze on foreign borrowing induced by the Russian financial crisis.

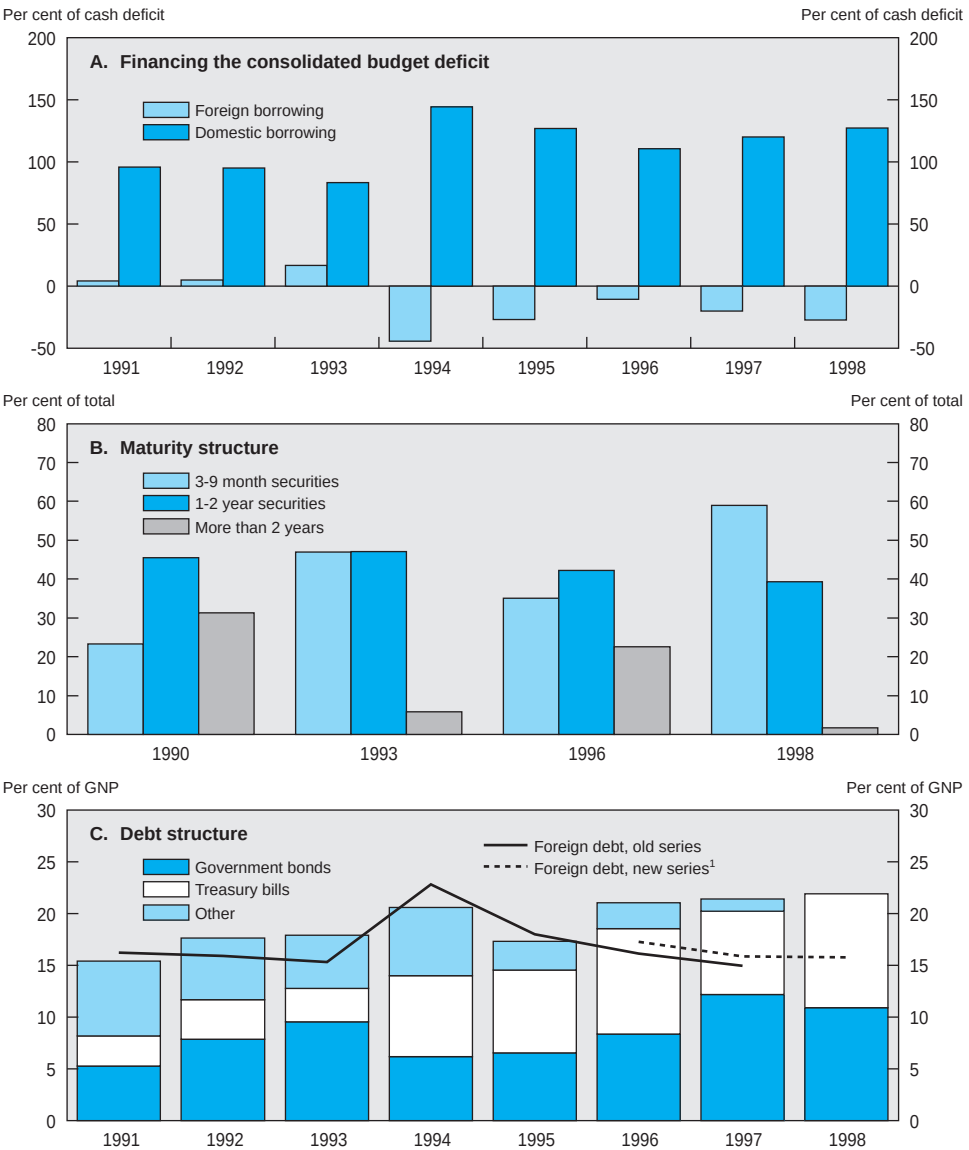
Table 11. **Consolidated budget financing**

Per cent of GNP

	1993	1994	1995	1996	1997	1998
Foreign borrowing	**1.1**	**–1.7**	**–1.0**	**–0.9**	**–1.5**	**–2.0**
Domestic borrowing	**2.6**	**4.5**	**3.6**	**7.1**	**8.4**	**8.7**
Government bonds, net	1.5	–1.8	1.1	1.8	5.0	2.5
Receipts	3.2	0.6	2.8	3.9	7.0	5.3
Payments	–1.7	–2.4	–1.7	–2.1	–2.0	–2.9
Treasury Bills, net	1.1	6.3	2.5	5.3	3.4	6.2
Receipts	8.6	16.1	14.6	21.8	10.0	17.3
Payments	–7.5	–9.8	–12.1	–16.5	–6.6	–11.1
Central Bank advances	**2.7**	**1.3**	**1.2**	**1.5**	**0.0**	**0.0**
Other	**0.0**	**–0.2**	**0.0**	**0.7**	**0.5**	**0.5**
Total financing	**6.3**	**3.9**	**3.7**	**8.5**	**7.5**	**7.2**

Source: Undersecretariat of Treasury.

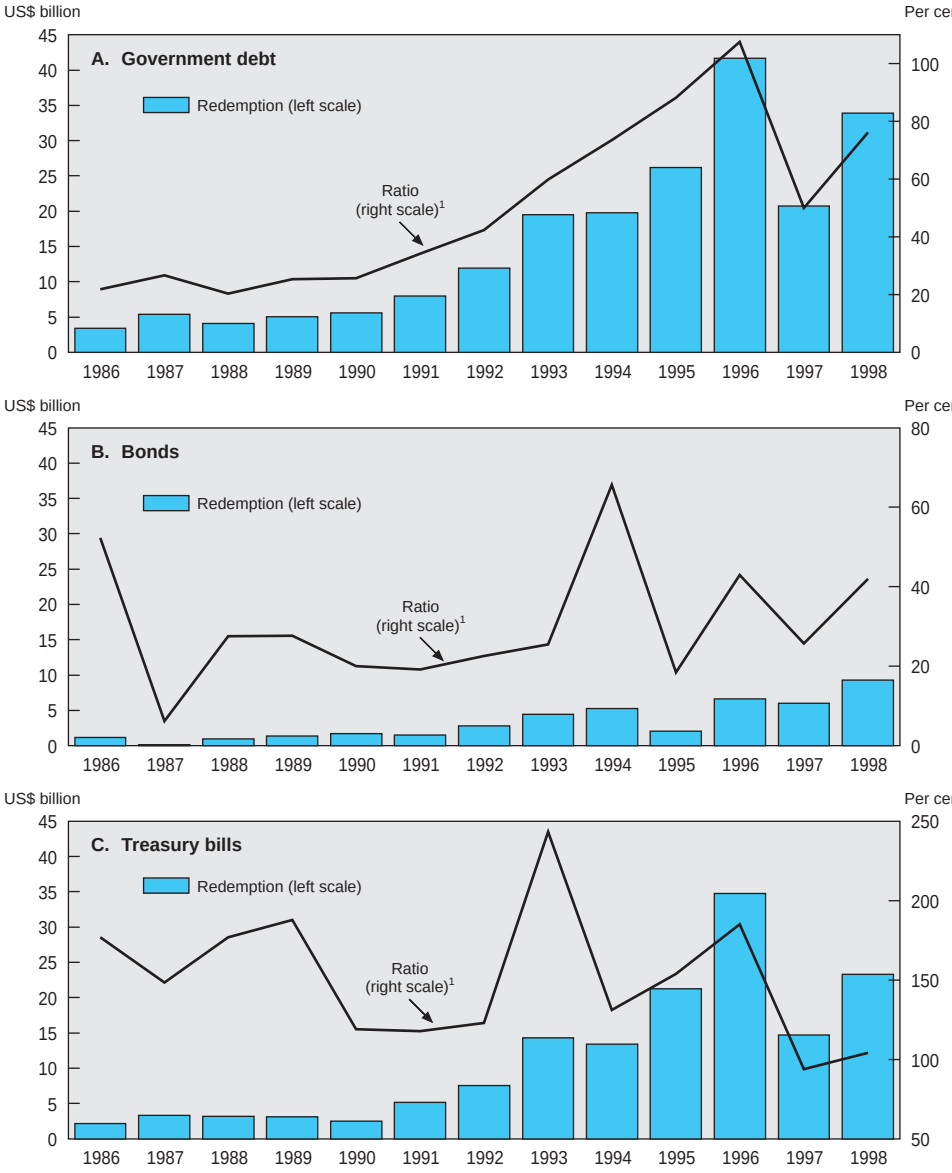
Figure 12. **Public sector debt indicators**



1. Methodological changes introduced at the beginning of 1999 include updating the external debt data in accordance with international standards, modernising the data code structure and enlarging the data scope. Treasury plans to complete the modification of the 1964-95 series by end 1999.

Source: Central Bank of Turkey, Undersecretariat of Treasury.

Figure 13. **Debt redemption and redemption ratios**



1. Defined as principal repayment as a proportion of the corresponding debt stock. For government debt, the total includes bank advances and consolidated debt.

Source: Undersecretariat of Treasury.

Over the last few years, the composition of the debt stock has changed, mirroring the pattern seen in the financing items of the budget deficit. New measures have been taken to streamline debt instruments and enhance transparency, including giving priority to a limited number of liquid instruments, intended to encourage the expansion of secondary trading. Nevertheless chronic inflation, combined with domestic and external uncertainties, meant that during 1998 the main securities issues were six- and nine-month Treasury bills, and one-year government bonds (Figure 12, Panel B). Debt placements did not always meet with buoyant demand, an impediment that also reflected the adverse reaction of financial markets to the Russian crisis. Nominal interest rates at three digit levels were required to prevent the cover ratio from declining sharply. With the average maturity of the debt contracting significantly over the year (Figure 12, Panel C), falling to 8 months, compared with 11.9 months in 1997, the conditions were created for a sudden rise in debt redemption equivalent to nearly 80 per cent of the stock (Figure 13). Such difficulties greatly compromised the Treasury's debt strategy, which was intended to lengthen the maturity structure of its borrowing.

For 1999, the government had some major rollovers to finance in the first quarter ($ 17 billion), which the Treasury would have liked to finance partly with funds raised in foreign markets. Foreign borrowing is much cheaper than domestic, the spreads over LIBOR ranging around 600-800 basis points. Recently, the Treasury was able to sell two Eurobond issues on the market relatively easily. The authorities have also started to issue foreign exchange-rate indexed domestic debt to domestic residents. Since 1997 they have also been selling 1- and 2-year inflation-indexed bonds, while a new 2-year instrument with coupons was successfully placed at the beginning of 1999. The government subsequently suspended the issue of indexed instruments.

The credibility of the fiscal consolidation process

The rise in the primary surplus over the last few years has been consistent with a decline in the overall (foreign and domestic) government debt-to-GNP ratio. It fell to an estimated 43 per cent by 1998, approximately the same level as in the early 1990s. Domestically-held debt, however, has continued to rise, and Table 12 examines the dynamics behind this. Using domestic marketable cash debt as the measure of debt, it would appear that since the financial crisis of 1994 the high positive real interest rate acted to increase the debt ratio by close to 4½ percentage points of GNP. In the face of this pressure the conventionally-defined primary surplus would have needed to be 2 percentage points of GNP higher in order to achieve a stable domestic cash debt to GNP ratio for 1998 (Table 12).

Table 12. **Contribution to changes in the central government "cash" debt ratio**

Per cent of GNP

	1994	1995	1996	1997	1998
Level of domestic "cash" debt[1]	8.7	9.3	12.8	15.8	17.9
Change in debt	1.8	0.6	3.5	3.0	2.1
Contributions to change in debt from:					
Primary balance[2]	2.1	2.0	0.6	–0.9	3.6
Debt dynamics	3.6	1.8	2.6	1.9	3.9
Other factors[3]	0.3	0.8	1.5	0.1	1.8
Total	1.8	0.6	3.5	2.9	2.1
Stabilisation conditions					
Required primary surplus to stabilise the debt[2]	3.9	2.6	4.1	2.1	5.7
Memorandum items:					
Nominal GNP growth (g_t)	94.7	102.0	90.7	96.2	80.4
Nominal interest rate on government debt (r_t)[4]	196.0	144.0	143.0	126.0	125.0
$r_t - g_t$	101.3	42.0	52.3	29.8	44.6
GNP inflation	107.2	87.1	77.9	81.2	73.9
Real interest rate on government debt	42.9	30.4	36.6	24.7	29.4

1. Domestic "cash" debt is marketable debt. See Annex II, "The dynamics of government debt accumulation" for technical details concerning the decomposition of debt changes into different contributing factors.
2. Defined as the overall primary balance less external interest payments.
3. Changes in net debt for which no corresponding flows are recorded, including statistical discrepancies.
4. Average across maturities.

Source: OECD.

This is only part of the story, however. The non-marketable (non-cash) debt, issued to other public entities to compensate them for losses incurred in their off-budget operations, is issued at below market rates and is partially inflated away. Indeed, looking at the aggregate (cash and non-cash) government debt ratio, interest payments (which can also be affected by timing factors) have had, until recently, a somewhat smaller effect in pushing up the debt burden than nominal income growth (which is mainly inflation-driven) has had in reducing it (Table 13). "Debt dynamics" have, in this sense, acted to alleviate the debt burden, at least until 1997. The problem is that if net off-budget bond-issuance is added to the consolidated budget balance, the primary surplus is effectively turned into a deficit. Debt dynamics in Turkey have thus exhibited an important dichotomy: explosive interest payments matched by discretionary restraint in the marketable sector, accompanied by inflation erosion and primary budget indiscipline in the non-marketable government debt sector.

Since 1998, and looking forward, high real interest rates, increasing recourse to more expensive domestic cash debt, and decelerating nominal income growth can be expected to exert an increasing upward pressure on the

Table 13. **Evolution of total domestic government debt**

Per cent of GNP

	1994	1995	1996	1997	1998
Total "cash" and "non-cash" domestic debt[1]	14.0	14.6	18.5	20.2	21.9
Change in debt ratio	1.2	0.6	3.9	1.7	1.7
Contributions to change in debt ratio from:					
Primary balance[2]	2.1	2.0	0.6	–0.9	3.6
Interest paid[3]	6.0	6.1	8.9	6.7	10.6
Debt dynamics	–6.2	–7.1	–6.9	–9.1	–9.0
Other factors[4]	3.5	3.6	2.6	3.1	3.7
Memorandum item:					
Nominal GNP growth	94.7	102.0	90.7	96.2	80.4

1. "Cash" domestic debt is marketable, whereas "non-cash" domestic debt is a rough approximation for non-marketable debt. See Annex II, "The dynamics of government debt accumulation" for technical details concerning the decomposition of debt changes into different contributing factors.
2. Defined as the overall primary balance less external interest payments.
3. Defined as total interest paid less external interest payments.
4. Changes in net debt for which no corresponding flows are recorded, including statistical discrepancies.

Source: OECD.

overall government debt ratio. Table 14 presents two forward-looking scenarios for domestic cash debt, beginning from a *status quo* scenario where inflation-adjusted interest rates are set at 25 per cent.[8] Excluding external interest payments, a targeted primary surplus of 3.5 per cent of GNP is assumed over the period, with an average annual inflation rate that remains above 40 per cent.[9] In this case, the primary balance would be insufficient to prevent a rise in the cash debt ratio. The "debt dynamic" term would add increasing pressure on actual debt, implying that the additional primary surplus required to achieve a stable cash debt-to-GNP ratio would be close to 2 per cent of GNP. To prevent the total (cash and non-cash) debt ratio from increasing would require significantly more than this unless off-budget spending is brought under control.

The variant to this scenario assumes a more aggressive consolidation policy, the counterpart of which would be a higher initial level of the real interest rates. More specifically, the OECD calculations indicate that under the assumption of close to the current 30 per cent inflation-adjusted rate of interest on government debt in 1999, the domestic debt would increase from about 17.5 per cent of GNP to more than 20 per cent. The implication is that the primary surplus required to achieve a stable debt to GNP ratio would amount to approximately 6 percentage points of GNP, or around 7 per cent of GNP if the primary surplus is correctly defined to include external interest payments. Nevertheless, as inflation starts to fall, the "debt dynamics" term would begin to exert a smaller overall adverse contribution to debt reflecting both falling real interest rates

Table 14. **Central government debt scenarios**

Per cent of GNP

	1999	2000	2001
Status quo scenario			
Central government domestic "cash" debt	19.5	21.3	23.3
Implied debt dynamics[1]	4.0	4.3	4.5
Contribution to changes in domestic debt due to "other factors"[2]	1.0	1.0	1.0
Required primary balance to stabilise the debt[3]	5.0	5.3	5.5
Targeted primary balance[3]	3.5	3.5	3.5
Gap between required and assumed primary balance	1.5	1.8	2.0
Underlying assumptions:			
Real GNP growth (per cent)	1.7	2.5	3.0
GNP inflation (per cent)	60.0	50.0	45.0
Nominal interest rate paid on government debt (per cent)	100.0	87.5	81.2
Stronger adjustment scenario			
Central government domestic "cash" debt	20.4	20.2	18.9
Implied debt dynamics[1]	5.0	2.3	1.2
Contribution to changes in domestic debt due to "other factors"[2]	1.0	1.0	1.0
Required primary balance to stabilise the debt[3]	6.0	3.3	2.2
Targeted primary balance[3]	3.5	3.5	3.5
Gap between required and assumed primary balance	2.5	–0.2	–1.3
Underlying assumptions:			
Real GNP growth (per cent)	1.0	3.2	5.7
GNP inflation (per cent)	45.0	20.0	15.0
Nominal interest rate paid on government debt (per cent)	87.1	38.0	28.8

1. Reflects the contribution to the increase in the debt ratio due to the gap between the interest rate on government debt and the nominal GNP growth rate.
2. Changes in net debt for which no corresponding flows are recorded, including statistical discrepancies.
3. Defined as the overall primary balance less external interest payments.

Source: OECD.

– a consequence of sizeable credibility gains – and stronger output growth. By 2001, an overall properly defined primary surplus of about 3¼ per cent of GNP, which would be broadly in line with the official target, would suffice to stabilise the debt. Again, it should be remembered that the above calculations take no account of the prospective acceleration in the real burden of off-budget debt as the effect of inflationary erosion weakens. Controlling this element of debt will demand additional discretionary restraint.

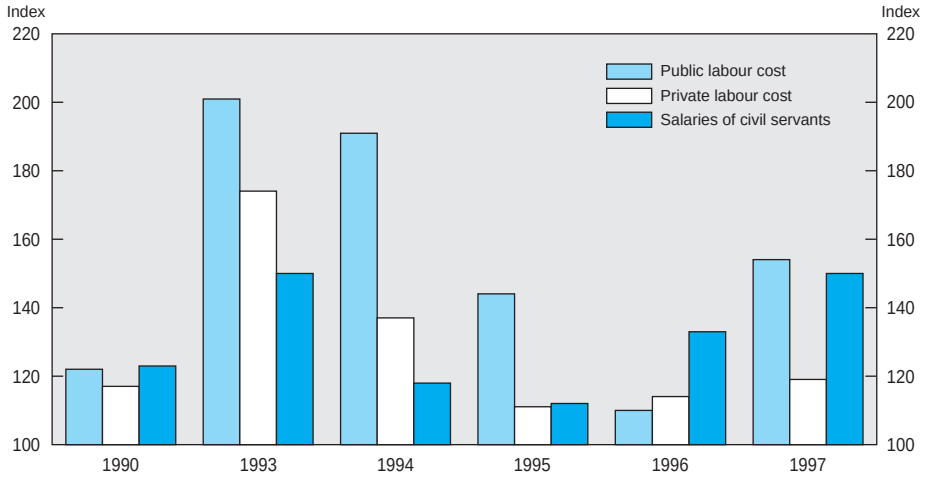
Correcting structural budget deficiencies

Controlling public sector wage and price developments

Making fiscal consolidation credible requires not just that the primary balance adjusts further, but also tangible evidence of the government commit-

ment to deliver moderate public sector wages and agricultural support prices in line with world market trends. Years of indexed adjustments have complicated macroeconomic management directly by raising government spending for wages and agriculture, and indirectly, through their adverse effects on inflation inertia, expectations and ultimately interest rate premia (Figure 14). Recognising these difficulties and in a move that was presented as a sea change in policy, the June 1998 *Memorandum of Economic Policies* announced that public sector salaries and agricultural support prices were to be adjusted in line with the targeted reduction in inflation, rather than backward indexed. Accordingly, the salaries of civil servants, which had been adjusted by 30 per cent at the beginning of 1998, were scheduled to increase by no more than 20 per cent in July and then kept at that level for the remainder of the year. Adjustments of between 60 and 70 per cent were announced for wheat, tea and tobacco, consistent with the inflation objective. However, in the context of a rapidly deteriorating political environment since August 1998, under mounting political pressure from inside the governing coalition, the government had to agree an additional 10 per cent salary increase in October. Another rise – worth 5 per cent – was caused by the decision to bring forward the reduction of the tax brackets approved in the tax reform (see below). As a result, the year-on-year civil service wage bill increased by about 80 per cent.

Figure 14. **Real wage dynamics**
1981 = 100



Source: State Planning Organisation.

Agricultural prices rose broadly in line with the inflation rate objective, although the existing positive gap with agricultural international prices continued to widen. In December 1998, the outgoing government announced a further salary increase for civil servants of 30 per cent for the first half of the year, significantly higher than the officially expected inflation rate of 25 per cent over the same period.

Correcting the defects in the tax system

The 1997 *Economic Survey of Turkey* reviewed the major issues involved in improving the tax system. In that context, lags in tax collection, limited coverage and failure to bring the unregistered economy into the tax net were identified among the major factors underlying the slow rise in tax revenues, one of the causes of systematic deficit overruns. In its conclusion, the *Survey* stressed the urgent need for Turkey to reinforce and speed up the process of tax reform, including the design of a transparent income measurement framework and the introduction of effective sanctions. The 1998-99 tax reform, one of the cornerstones of the government three-year stabilisation package, allows for progress in a number of areas covered by these recommendations, notably personal and capital income taxation, corporate taxation and tax administration. The two pillars of the reform are expanding the tax base and reducing marginal tax rates. The originally planned special consumption tax was not implemented.

Broadening the tax base

For many years, a major factor underlying the implementation of tax measures in Turkey has been the need to protect taxpayers from the distortions and inequities generated by chronic inflation.[10] An inflation indexation mechanism was put into force in the early 1980s and has since been used regularly to allow the gradual adjustment of statutory income tax brackets, as well as to modify requirements for exemptions. Thus taxpayers were prevented from being mechanically pushed by inflation into higher tax brackets. Little progress was achieved, however, with respect to widening the tax base and substantially reducing the top income rates, which are two areas of reforms where a number of OECD countries have concentrated their recent initiatives. Thus the top income tax rate was still 55 per cent in 1997, while a complex tax structure was the other major factor discouraging tax compliance, seven tax brackets being abnormally high by OECD standards. At 6½ per cent of GDP, revenues from personal income taxes are significantly lower than the international average.

Recognising this large scope for improvement, one salient feature of the 1998-99 tax reform has been the reduction of tax rates on personal incomes within a period of two years, in expectation that greater compliance will widen the tax base. The first of such reductions took effect in July 1998, by which time the number of tax brackets was limited to six, while the minimum tax rate set by the

central government declined from 25 to 20 per cent and the top rate was lowered to 45 per cent. The bottom and top household income tax rates were lowered again at the beginning of 1999, to 15 and 40 per cent, respectively. As a result, the new top rate is some points below many European countries, although the new starting rate remains relatively high by international comparison. When the reform is fully implemented by 2000, Turkey will move from the high-rate group to an average position among OECD countries, and this should result in greater compliance. The corporate income tax regime has also been modified, the income tax rate having been lowered from 44 to 33 per cent for retained income and 40.3 per cent for distributions. Furthermore, the inheritance tax, the real estate tax and charges on property transactions have been reduced.

In addition to amending tax rates, the new law has introduced important changes through the revision of the payments regime. Taxation based on payments for the previous year's declared income has been replaced by a "pay-as-you-earn" system based on quarterly financial statements. The expectation is that this new process will lead to considerably shortened collection lags, reducing inflation-related revenue losses. For 1999, the new payment system should increase corporate taxes by about 1 percentage point of GDP, reflecting the fact that in the first year of operation income taxes will be paid on both the income accrued in the previous year and on that earned in the current year. This will be a one-time double payment effect, although government revenues should be protected against the impact of inflation on a more permanent basis.

Another important objective of the reform is to cover all different forms of personal income, including incomes accruing from financial assets and business activities. Until the reform, dividends were subject to an imputation system, while capital gains realised via financial intermediaries were exempt. Since undistributed corporate profits are exempt from taxation under the new tax regime, this could reinforce the bias in favour of self-financing which, as discussed in Chapter I, is already widespread in Turkey.

Regarding capital gains taxation, the tax base has been broadened by making the gains on securities trading taxable. Despite this, taxation of interest income, which accounts for the bulk of capital income, has been fraught with problems. A controversial withholding tax on the nominal income from government securities was introduced in November 1996, and subsequently raised from 10 to 12 per cent in 1997. However, by year-end it was reduced to 6 per cent, while a 6 per cent withholding tax was imposed on revenues from repo transactions. The tax bill of July 1998 envisaged the introduction of a 12 per cent withholding tax on interest from interbank operations, the same as already existed on savings accounts. It also contained a clause establishing that the tax-deferral advantage previously accorded to financial investors would expire one year ahead of the original proposal. By September 1998, however, a number of

corrective amendments had to be urgently introduced, as the financial-sector responded to the law, and to global financial market instability, by demanding sharply higher interest rates on government securities and shorter maturities. In order to restore the attractiveness of government debt the 6 per cent withholding tax on Treasury bills and bonds was cancelled, a move accompanied by the elimination of the withholding tax on interbank operations.

The need for a more efficient tax administration

Widespread tax evasion is revealed by the low number of individuals filing tax forms, and by a tax collection rate of only about 89.5 per cent of total tax assessments, a modest level by international standards. Penalties are severe but not implemented, two important factors underpinning the unwillingness to pay assessed taxes being a very permissive Reconciliation Board and the expectation of regular tax amnesties. Until the reform the incentive to apply for reconciliation was very strong, reflecting the fact that admittance to the Board was totally unrestricted. This has resulted in lengthy administrative procedures, mounting unprocessed paper and consequently low efficiency of the tax administration.

The new tax law is based upon two basic principles, simplicity and effectiveness. An important step in this direction was achieved by the introduction of rules to limit the possibility of appeal, which is no longer possible when tax evasion is evident. Illegal charges have been better defined, including the criminalisation of collusive tax evasion which can be sentenced with prison, a penalty no longer convertible into monetary terms. Payment delays for outstanding monetary penalties have been shortened. A second element of simplification is contained in the abolition of lump-sum taxation. This is a flat tax typically paid by individuals owning small businesses or the self-employed. Under the new norms, professionals, of whom there are more than 1 million, have been brought into the general system of progressive taxation.

The launch of the so-called *Financial Millennium* was also intended as step towards simplification. The *Millennium* was a temporary programme created to encourage holders of undeclared money to bring their cash into the banking sector. The opportunity remained open between April and September 1998 and its sole officially reported objective was broadening the tax base, without secondary implications for criminal administration. During the period, a total of 8 419 new corporations emerged and many others increased their capital. Foreign exchange deposits by residents increased by $ 5 billion. Some individuals may have used the programme for purposes of money-laundering. Insofar as the *Financial Millennium* contained the elements of a tax amnesty, it could have introduced a longer-run disincentive to declare income because of the expectation of regular windows in the future. The last of a long series of tax amnesties was accorded in 1997.

Overall, the success of the reform is intimately linked to the ability of the tax administration to enhance enforcement. The size of the tax administration is set to expand, 2000 new graduates having been already taken on as tax officials in 1998. Expanded automation capacity is also being used to induce taxpayers to respect their obligations. A new project entailing the full automation of 153 tax offices and the establishment of a modern information support system is due to be completed by 2000. The year 1995 marked the introduction of tax identification (TI) numbers for all registered taxpayers. The TI programme was further expanded in 1998 with the establishment of a system that permits a wider circulation of identification numbers (for example, real estate and motor vehicle transactions and major shareholders of companies). By 1998, nearly 11 million TI numbers had been assigned, almost doubling the corresponding level for 1997. TI numbers for all bank transactions and accounts are intended to be introduced towards the end of 1999, in line with the completion of the project for full automation of the tax administration and tax offices. Overall, this progress can be expected to enhance the auditing effectiveness of the tax administration.

At the same time, making tax collection more efficient also requires better co-ordination between different levels of the tax administration. The Ministry of Finance is responsible for tax policy and tax administration at the national level through the General Directorate of Revenues (GDR). However at present, there is no direct link or chain of responsibility between the central level tax administration (GDR) and local tax offices in provinces. The GDR uses its authority through the Financial Directorates (*Defterdarlik*) in the provinces. And since the Financial Directorates represent the Ministry of Finance, they are responsible not only for taxation and inspection activities in the provinces but also for public expenditure, state property and government accounting matters. Dispersed responsibilities of Financial Directorates lead to a less effective tax administration in the provinces in terms of enforcement of tax laws, tax collection and auditing activities. However, the establishment in 1998 of six regional auditing centres which report directly to the GDR was a significant step to improve tax auditing at the provincial level.

The privatisation programme

Structural aspects of the privatisation programme are discussed in Chapter IV. Privatisation of state assets earmarked for debt redemption plays a key role in the IMF-backed stabilisation programme of the government. While important progress was made in the first half of the year, key privatisation initiatives were either suspended or postponed in the second half, as state sell-offs were influenced by the volatility of international capital markets, legal and institutional obstacles, and mounting internal political instability. By the end of 1998 the state had sold assets totalling $ 1 billion, corresponding to about 0.5 per cent of GDP.

Revenues thus fell far short of the $ 2 billion target set in the *Memorandum of Economic Policies*. Nevertheless, the year 1998 should be regarded as a relative success, taking account of the fact that in dollar terms total receipts were at a record. The bulk of privatisation proceeds was realised through the sale of two 25-year leases for mobile-phone operation rights, and the public offering of 12.3 per cent of *Is-Bank*.

Improving budget planning and enhancing transparency

The 1997 *Economic Survey of Turkey* suggested that an excessively short-term approach had made the implementation of sensible tax, spending and management policies very problematic and that this in turn makes it difficult to generate a consensus for reforms. The *Survey* concluded that, in these circumstances, introducing a rules-based system would provide the basis for establishing a more stable planning horizon. Not only could expectations be adjusted by such an approach, but damaging "quick fixes" would also be avoided. A number of recent planning and control initiatives to discourage public sector entities from undertaking certain types of activities can be seen as a starting point towards meeting these recommendations.

Of significance is the new relationship with the central bank. Since 1997 the Treasury has used short-term advances predominantly as an "overdraft" facility to accommodate intra-month cash flow purposes rather than as a financing item.[11] For 1998, the amount of the overdraft for each month was stated within the monthly borrowing programme, together with the respective pay-back deadline. Such deadlines have always been met in a timely way and have never exceeded the four weeks. Passive monetary financing of the deficit is thus ruled out. Credibility and transparency are recognised as playing an important role in breaking

Table 15. **Seignorage and inflation tax**

Per cent

	1995	1996	1997	1998
Seignorage[1]	2.0	1.9	1.8	1.7
Inflation tax[2]	3.7	3.5	4.1	2.8

1. Changes in reserve money relative to GDP. Seignorage measures the revenues accrued by a government from monetary expansion.
2. Consumer price inflation at time (t) multiplied by the ratio of reserve money relative to GDP at time (t – 1). As inflation increases it reduces the real value of Central Bank liabilities (currency plus reserves) to the extent that these are interest-free, thus acting as an "inflation tax" on depositors and currency-holders. Banks operating in Turkey are obliged to hold reserve requirements which are 8 per cent for TL deposits and 11 per cent for FX deposits. No interest is paid by the Central Bank on these reserves at present.

Source: Central Bank of Turkey.

inflation expectations. For this reason, along with stronger efforts towards fiscal consolidation, quarterly fiscal financing and monetary programmes have been implemented since 1998 and the public has been kept informed of the results achieved. Treasury cash management and borrowing requirements are announced each month in accordance with the *Memorandum of Economic Policies* of June 1998.

The Central Bank, nevertheless, remains an important source of seignorage revenue (Table 15). Indeed, the combination of monetary expansion and inflation in Turkey has provided a revenue substitute (an "inflation tax") amounting on average to 3½ per cent of GDP in recent years. The spending counterpart of this "tax" is to be found in the subsidised credit allocated to preferred sectors of the economy.[12]

Assessment of the consolidation process

Notwithstanding the significant slowdown in economic activity, Turkey was successful in meeting the targets set for the primary budget balance and the public sector deficit during 1998. Tax reform has triggered a sizeable increase in revenues, essentially reflecting the wider diffusion of TI numbers, which led to stricter enforcement. With the tax administration set to expand in terms of personnel, and an automation project being implemented, the auditing effectiveness of the tax system should be enhanced and tax collection continue to improve in performance. Significant progress has also been made towards achieving better overall fiscal policy planning and control. In particular, the initiative taken in 1998 to publish quarterly fiscal financing and monetary programmes has increased transparency, thus contributing to a better understanding of Turkish macroeconomic policies.

Nevertheless, and perhaps most crucially, the primary surplus objective for 1999 appears very unlikely to be achieved on current spending and revenue projections, and is in any case inadequate to stabilise the government debt ratio. Difficult decisions will thus still need to be taken to prevent debt from exploding, given current high real interest rates. There is no systematic calculation of tax expenditures, and off-budget subsidisation activities remain a source of chronic budgetary weakness which require immediate reform. Reinvigorating the stalled privatisation process would be an important step in this direction. The benefits that authorities can hope to obtain from the measures adopted thus far in terms of credibility are modest in relation to those which could be expected from the elimination of off-budget losses, or from tightening the conventional budget by implementing a long awaited far reaching reform of the social security system. The following chapter discusses the need for such reform.

III. Reforming the social security system

Introduction

The scope, quality and cost-effectiveness of the Turkish system of social protection all have serious shortcomings. The share of social expenditures in GDP is around 7 per cent against 25 per cent for OECD Europe (Figure 15),[13] leaving large segments of the population without an adequate social safety net. Yet the system raises a serious problem for the budget balance. The social security deficit reached 2.7 per cent of GDP and accounted for over a third of the total budget deficit in 1998. The "informal/unregistered" sector is estimated to have exceeded 50 per cent of total civilian employment (20 per cent of total salaried workers), indicating that the design and enforcement of contribution and benefit rules is resulting in serious imbalances in the distribution of costs and benefits across the economy and society. Since the mid-1990s successive governments have initiated reviews and discussions on reform. Draft laws were prepared by the government in 1995 on health reform and in 1997 on pension reform, but have not been adopted. In the context of the need to tackle the budgetary problem, reform of social security has taken on an added urgency and is expected to be at the top of the legislative agenda of the new government.

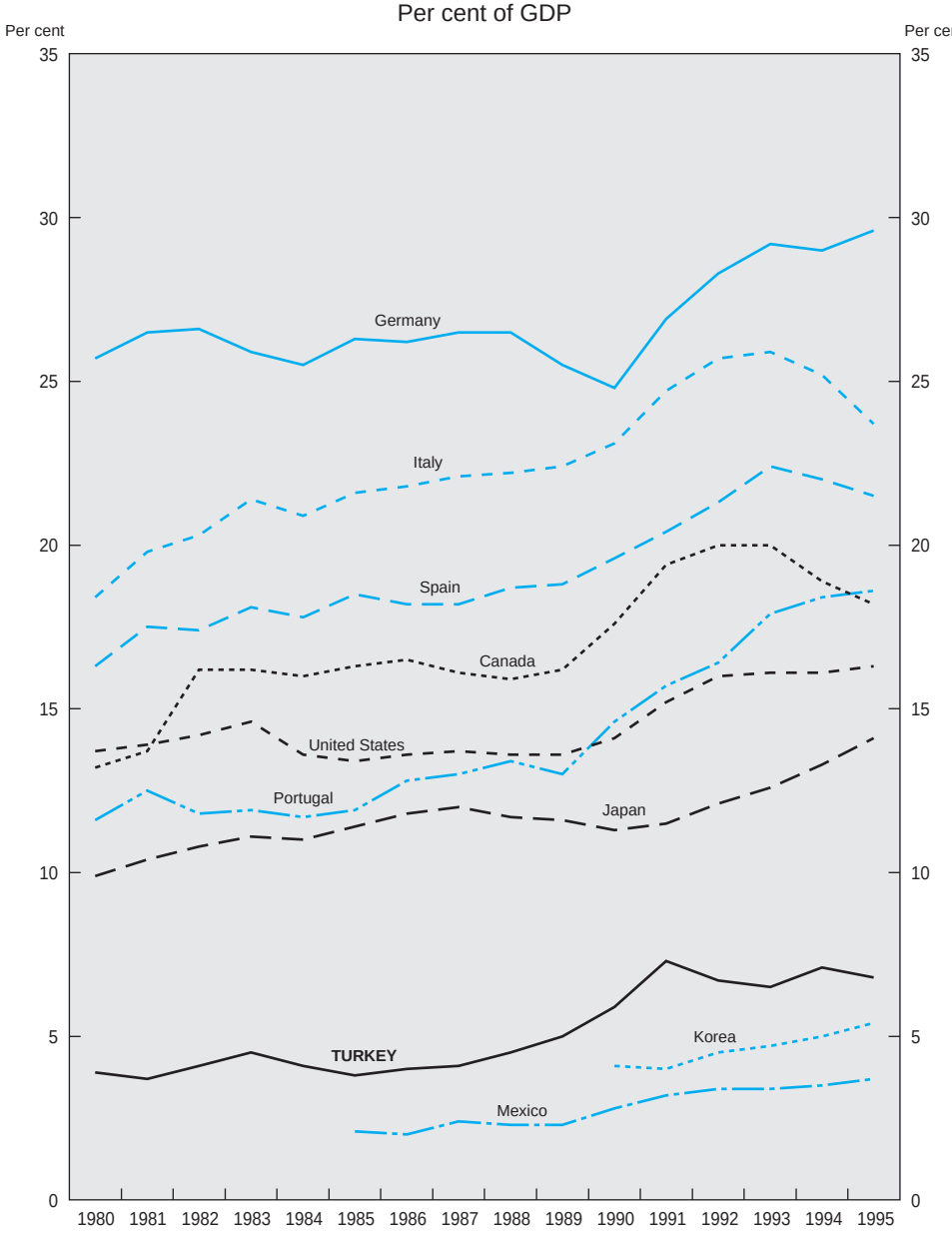
The aim of this Chapter is to set out the issues and propose an agenda for reform. It begins by describing the institutional background and broad trends behind the Turkish system of social protection, of which pensions and health insurance are the main components, though unemployment insurance and social assistance may assume greater importance under a reformed system. It then provides an overview of the detrimental economic and social impacts of the system. The final section discusses possibilities for reform.

Organisation and development of social protection

The Turkish constitution establishes the right of all citizens to social security, but subjects this objective to fiscal constraints, specifying that "the State fulfils its economic and social obligations by taking into account the economic

Figure 15. **Social expenditures in selected OECD countries**

Per cent of GDP



Source: OECD, Directorate of Employment, Education and Social Affairs.

stability and sufficiency of financial resources". The present status of social protection reflects this tension between generous objectives and limited means.

Social protection has a half-century history, reflecting an early policy ambition to reach the social protection standards of advanced countries stated in the Universal Declaration of Human Rights of 1948, and in other international declarations. Among the nine social protection areas targeted by the ILO Convention of 1952 are: *i)* old-age, *ii)* invalidity, *iii)* death, *iv)* health, *v)* sickness, *vi)* maternity, *vii)* work injury and occupational diseases, *viii)* family assistance, *ix)* unemployment. All except family assistance and unemployment insurance – which are major vehicles of social transfers in other OECD countries – have been incorporated in Turkish legislation. However, invalidity, death and maternity insurance are allocated only limited resources. Social expenditures concentrate on old-age and health benefits, which respectively account for 41 and 36 per cent of total social expenditures.

The social insurance system in Turkey

The social protection system has two components, which are social insurance and social assistance. Insurance is compulsory for all salaried workers and self-employed, and covers their pension and health needs, as well as the needs of their dependants (families); assistance covers the remaining non-insured in need, via two separately managed health and old-age components.

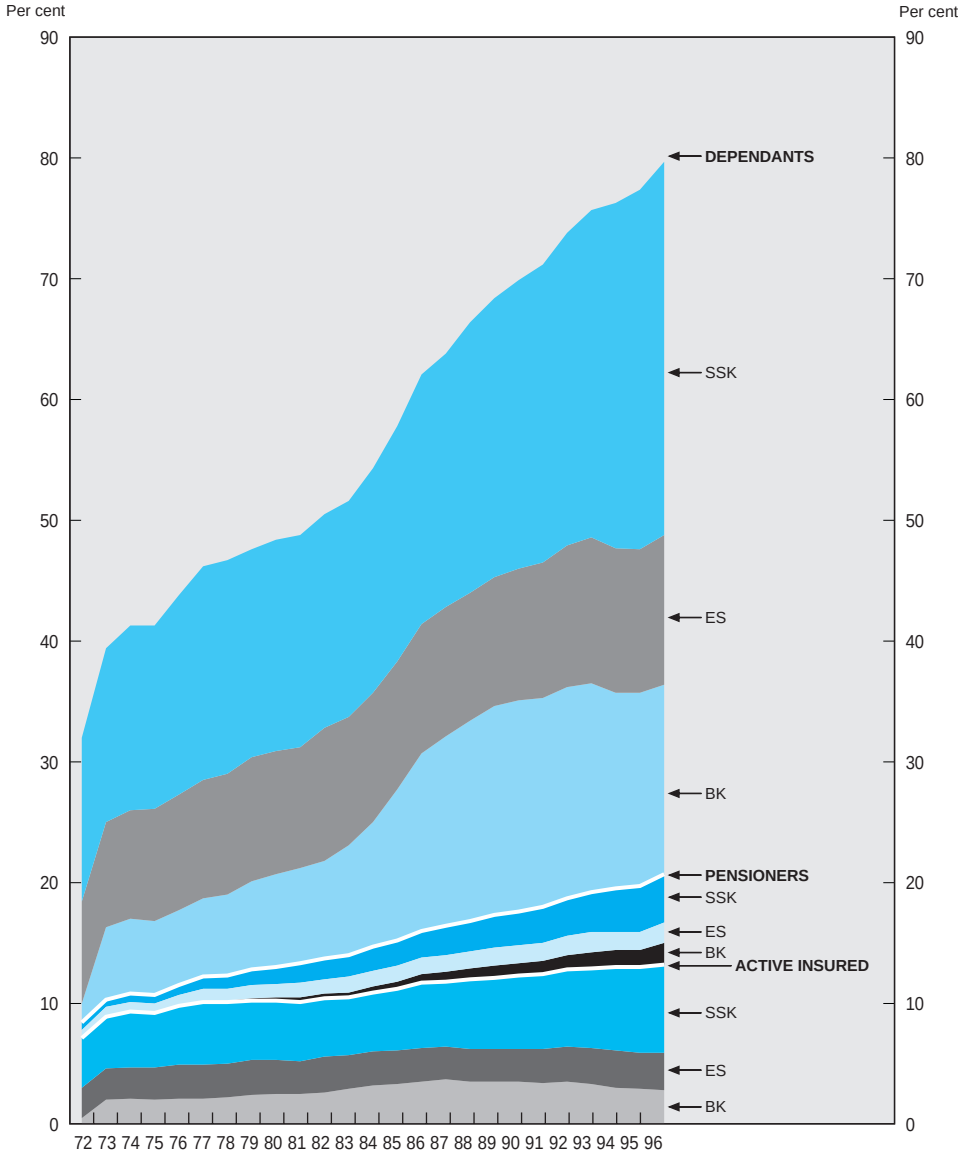
Social insurance is the principal pillar of the system. As of 1997 it covered 85 per cent of the population, either as active insured and pensioners or their dependants. The definition of "dependence" in social protection is generous and all-encompassing: for example, unmarried women with no age limitation are admitted at the charge of their parents or children. In practice, however, access to health benefits is uneven and pension benefits are limited to the directly insured and their survivors. Three employment-based social security organisations (SSOs) are responsible for the coverage: *SSK* is the main fund which covers wage earners in the business sector, while *ES* insures civil servants and military personnel and *BK* covers the self-employed including farmers (Figure 16).

Basic architecture

The social insurance system was initiated as an earmarked, self-financing scheme, with an objective to set up a trust account to accumulate savings from contributions of a value actuarially equivalent to the present value of future liabilities. It has degenerated progressively into a tax-funded system. Excess funds were built up in the system until the late 1970s due to the advantageous age structure of the members, but the funds accumulated were subsequently eroded under the impact of inflation and negative real returns on investments (see below). SSOs have ended up as *de facto* "pay-as-you-go" operations (Box 2).

Figure 16. **Three main social security organisations: active insured, pensioners and dependants**

Per cent of total population



Source: State Planning Organisation.

Box 2. Turkish social security: social insurance or tax-funded?

Social protection can be financed either by separate, earmarked contributions via *social insurance* schemes, or from the *general budget* using tax income. Turkish social insurance was initially designed as a funded social insurance system but has ended up as a PAYG system partly financed by the government. The system made payments to retirees from the inception of the system even before full rights had been established, thus building up unfunded pension liabilities. Contributions to the insurance funds (social security organisations – SSOs) by the *active insured* are used to finance the benefits for *pensioners*, the *sick*, and *dependants*. *Retirement* and *health* accounts are in principle separate, and the SSOs are supposed to keep them in actuarial balance by setting appropriate premia. In practice, the funds have faced increasing deficits, requiring substantial transfers from government for both components. There are three SSOs – the *Social Security Institution* (SSK), *Bag-Kur* (BK) and *Emekli Sandigi* (ES).[1] SSK accounts for around 60 per cent of the insured population, covering wage earners as well as a voluntary branch for unemployed former members. BK and ES each account for around 20 per cent of the insured population, the former covering the self-employed including farmers, as well as unemployed persons and housewives who can join its voluntary branch, and the latter covering the military and civil servants. Two additional, though relatively small, components of the social insurance system: are the *occupational funds*,[2] and *private life insurance* institutions.[3]

The main characteristics of the system are as follows:

Contributions

The contribution system varies with the fund. For the SSK, a contribution rate of 33.5 to 39 per cent (14 per cent employee and 19.5 to 25 per cent employer[4]) is applied to earnings up to a ceiling (*i.e.* insurable earnings). This ceiling is fixed by the government at low levels relative to average earnings but can vary substantially over time. It is currently equal to 2.3 times the minimum wage. For the BK, a contribution rate of 32 per cent is applied to a range of "revenue brackets", the first one based on declared income on entry. Insurees are then placed in progressively higher income brackets over a six year period, until they reach an intermediate "floor". They can then choose whether to move up further in the revenue scale until reaching a "ceiling", thereby increasing their insurable earnings for calculating the pension. ES contributions are 35 per cent (15 per cent employees and 20 per cent employer) of wages without a ceiling. In SSK and BK, roughly two-thirds of the overall contribution rate goes to fund pensions, the other one-third to health. In ES, members pay a single, consolidated premium for pensions and health.

Contribution collection and coverage are only weakly enforced for the first two schemes, particularly for state economic enterprises' contributions to SSK, as firms, in the past, faced negative real interest rates on contribution arrears. Interest rates on delayed payments were recently increased to punitive levels in the SSK. Current contribution collections rates for the SSK are 85 per cent, and for the BK, 50 per cent for self-employed and 32 per cent for farmers. Contributions to the ES are enforced but, nonetheless, its deficit represents almost 40 per cent of the total national social security deficit.

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Pension benefits

Pension benefits are based on an adjusted level of insurable income but the formulae for calculation vary across the three schemes. Key features include the following:

– *There is no minimum retirement age:* in the SSK, individuals can retire after a contribution period of 5 000 days, or less than 14 years, conditional upon either *a)* having been insured (*i.e.* registered with the system)[5] for at least 20 years for women and 25 years for men,[6] or *b)* having attained the age of 50 for women and 55 for men. Retirement after a contribution period of only 10 years is allowed in case workers have attained the age of 50 for women and 55 for men and have been insured for at least 15 years. In the BK and ES, individuals can retire after a contribution period of 20 years for women and 25 years for men, but for men reaching 55 and women 50 the contribution period is reduced to 14 years. The average retirement age in 1998 was low: 48 for women and 51 for men.[7]

– *Benefits are based on insurable earnings over a short reference period:* the last 5 years of insurable income (low paid) to 10 years (higher paid) for the SSK; the previous year for the BK; and the previous month's salary for the ES.

– *Pension rights accrue at very generous rates:* All schemes multiply insurable income by a replacement rate (defining a "full" retirement income) of 60, 70 and 75 per cent respectively for the SSK, the BK, and the ES. For SSK, the replacement rate is increased by 1 per cent for each additional 240 days (8 months) of contribution beyond 5 000 days. For BK and ES, it is increased by 1 per cent for each additional year beyond 20/25 years for men/women. In all three schemes, this rate is further increased by 1 per cent a year for those having worked beyond the "normal retirement age" of 50 for women and 55 for men, up to a ceiling (85, 90, and 100 per cent for SSK, BK, and ES respectively). Thus, individuals accumulate from between 3 to as much as 5 per cent per cent of insurable income as pension rights per year of contributions, which is very high by international standards. Accrual rates for shorter work histories can be even higher.[8]

– *Because insurable income is low, benefits tend to be low:* in 1998 average benefits ranged from around $ 188 per month for the SSK to $ 121 for the BK and $ 237 for the ES (36, 23 and 45 per cent respectively of the average net wage). In principle, benefits are indexed to civil servant wages, but the government also intervenes. In the case of the SSK and BK, pensions include a top-up in the form of a flat-rate element, the SYZ (paid from the SSO's budget): this is not related to contributions and has been used to partly compensate for real declines in civil servant salaries. The value of this supplement varies substantially from year to year and has been recently frozen in nominal terms.

Fluctuations in the contribution base and in the value of the SYZ mean that the value of pension rights are difficult to predict from one year to another (particularly for the SSK), making the calculation of past and future returns on pension contributions difficult. But, in general, the short contribution and reference periods, the low average retirement (and hence the long period of pension receipt, which stretches to 30-35 years in many cases) ensures that participants receive generous lifetime pension earnings in comparison to their working-life contributions.

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Health benefits

The SSK offers free health care to members in its own hospitals. SSK facilities also contract with other SSOs (ES and BK) for service provision. Health expenditures by active civil servants are directly paid by government ministries employing them, while retirees and their dependants remain at the charge of the ES. In the BK, health insurance expenditures rose sharply in 1997 and 1998. Health coverage was extended to rural sector participants of the BK only in November 1998, and the impacts of this latest extension are yet unknown. The high number of dependants and "pseudo-dependants" in the SSOs widens the coverage of health services substantially: many unregistered sector workers gain coverage as pensioners or dependants or by fraudulent use of other individuals' cards.

1. A more detailed description of the specifics of the pension and health rules, "Funding and benefit rules of social security organisations" (1999 mimeo), is available from the OECD Economics Department (Turkish Desk) on request.
2. There are 22 employer-sponsored pension funds, mostly run by banks and companies for their own staff, governed by SSK law. The members of these occupational funds are exempt from being registered with the SSK scheme, but their employer has to provide the same benefits as the SSK, and contribution rates and benefit regulation are identical to the SSK. Many employers insured under SSK have also established supplementary benefit provisions and occupational pension arrangements for their senior employees.
3. In 1996 there were 34 Treasury-regulated companies providing life insurance policies. The dominant policy type is "with profit endowment", generally written over 10-year periods. Tax privileges apply to these policies. However, high inflation and inadequate management of services have led to extensive mis-selling and deceptive performance. The market is characterised by heavy sales of new policies and a high level of lapses and cancellations of existing policies.
4. The employer's share varies with the premium paid for work injuries and occupational diseases, which in turn depends on the riskiness of the work.
5. The main reason for the distinction between contribution period and insured (working) period is that in SSK casual workers may not have the chance to work through the whole year due to seasonal features of their job.
6. With a minimum working age of 18, this implies a minimum retirement age of 38 for women and 43 for men.
7. Weighted averages for SSK, BK and ES. In SSK alone, the average retirement age is 50 and the average contribution period is 20 years.
8. For the SSK, after a contribution period of 10 years (and women having reached age 50 and men 55), the replacement rate is diminished by 1 per cent for each 240 days (2/3 of a year) missing from the normal minimum contribution period of 14 years. Thus, for a 10-year contribution period, the replacement rate would be reduced by merely 6 percentage points, *i.e.* to 54 per cent. This amounts to a roughly 5½ per cent accrual rate, compared with 4¼ for the normal 14-year contribution history earning a 60 per cent replacement rate.

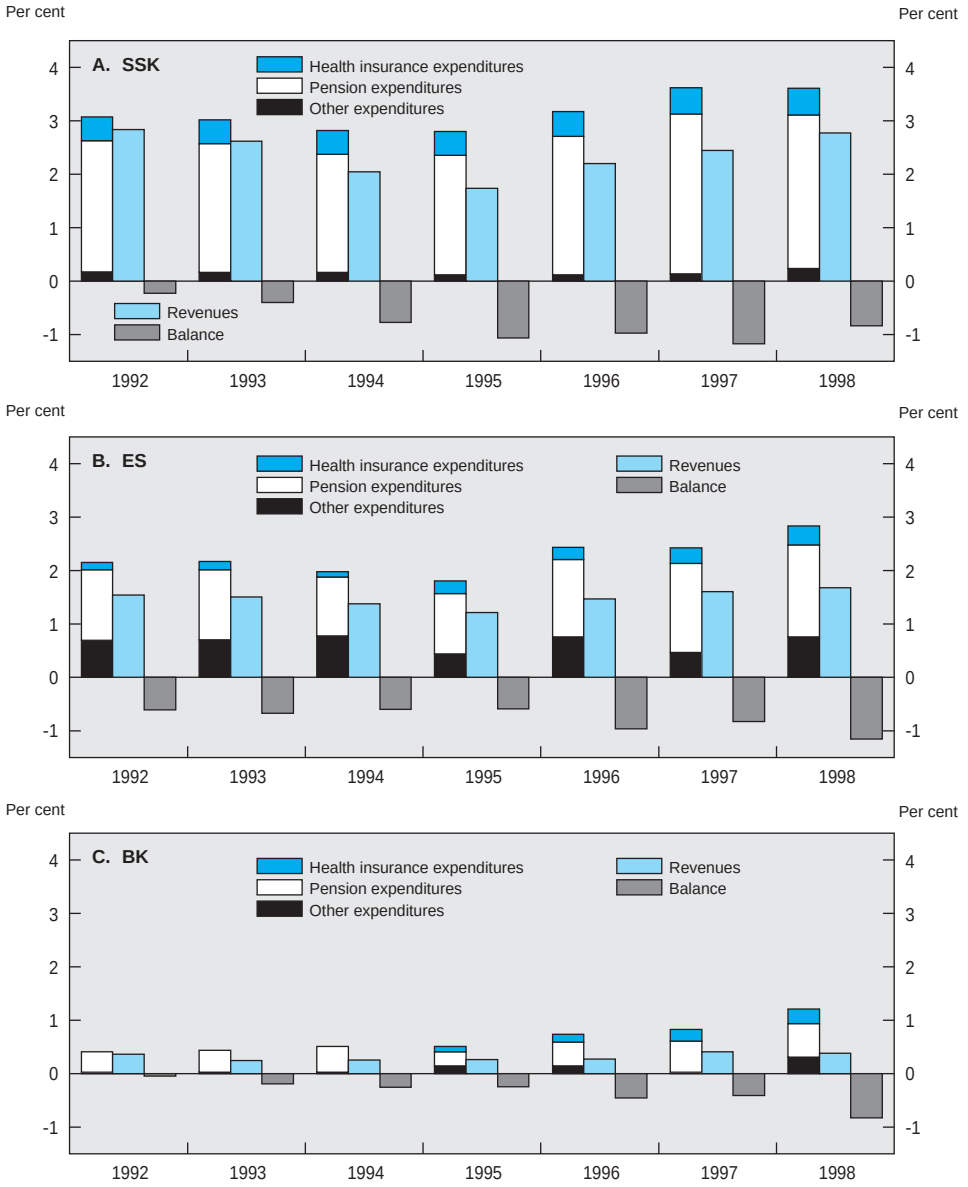
Social security organisations are in principle independent on a wide range of matters, including the right to propose benefit and contribution rules/ rates according to actuarial considerations, and to manage their asset portfolios. In practice however, the government guides the "structural" decisions of the funds, concerning their premia and benefit rates and their investment policies. The administrative costs, managerial continuity and investment performance of the funds have all suffered as a result of such political interference. Even day-to-day management may come under policy influence, as evidenced by the rapid turnover of general managers. The sizeable job creation and employment potential of SSOs (SSK alone has 57 000 workers) may be one reason for this interference. Another is the scope and magnitude of their investment and asset management powers: SSOs have been frequently forced to invest in placements bearing negative interest rates. Indeed, rates of return on invested assets have been excessively low by international standards.[14] This left a large unfunded liability.

The SSOs moved into deficit in the early 1990s, even before attaining their demographic maturity (Figure 17). Deficits first affected the pension accounts, and then in the second half of 1990s the health accounts, despite indirect government subsidies to SSO's hospital costs (see Box 3 below). Rising general budget transfers have funded the SSO deficits, as the SSOs have become progressively more dependent on direct tax-funding. This transformation of formally self-funding "insurance" schemes into "tax-funded" systems is the distinctive characteristic of the current Turkish social security system. Three main factors underlie this recent drift towards deficit: *i)* the narrowing of the contribution base by shifts to the unregistered sector; *ii)* the increasing decoupling of pension contributions and benefits; *iii)* and the growth of health expenditures.

Narrowing of the contribution base

Undeclared wage and self-employed income deprives the SSOs of a significant portion of their potential revenues. Workers (and their employers) not declaring their activities and not paying any premia find their way into the system as beneficiaries in case of need. This may be done by registering during either the early or the latter part of their working life, just enough to document the minimum years of contributions (14 years in SSK and 20/25 years in BK and ES). Registration may be avoided altogether by getting indirect pension and health benefits as "dependants" of formally insured relatives. Registration avoidance is particularly blatant in the case of health services. The use of health services by dependants is extensive, while workers from the unregistered sector with no insured relatives register only when they face health problems. When these means are not available, access to coverage by borrowing SSO membership (identification documents) is widely practised. Health authorities do not resist these practices actively, in order not to have to deny vital services.

Figure 17. **Current balances of the social security organisations**
Per cent of GNP



Source: State Planning Organisation.

The unregistered sector has expanded further since the mid 1980s, as industrialisation and urbanisation have increased the opportunities for sub-contracting, construction, and various types of services. In welcoming greater entrepreneurial activity, the authorities may have been unwilling fully to enforce the social security contribution obligations of newly created firms, thus providing an implicit "labour subsidy". The perceived benefits to competitiveness from the non-payment of social security payroll taxes by the unregistered part of the exporting sector may also have been behind this policy deficiency. Established enterprises have benefited indirectly from these labour subsidies, via sub-contracting and procurement of goods and services in the unregistered sector. But because they are placed at a competitive disadvantage, particularly in the low wage sector, there is pressure on them to under-report earnings and wage costs or not to register new employees at all.

In the registered sector, a similar failure of government policies to enforce the accrual and collection of social security contributions from firms with registered labour has effectively helped to decrease SSO revenues. At the same time incentives to delay payment or to under-report income have arisen from the negative real interest rate penalty for contribution arrears,[15] and from successive "contribution amnesties".[16] A disproportionately large number of workers in their early working life are reported to be earning minimum wages, whereas reported wages in the reference period before retirement (last five years) show abrupt increases.

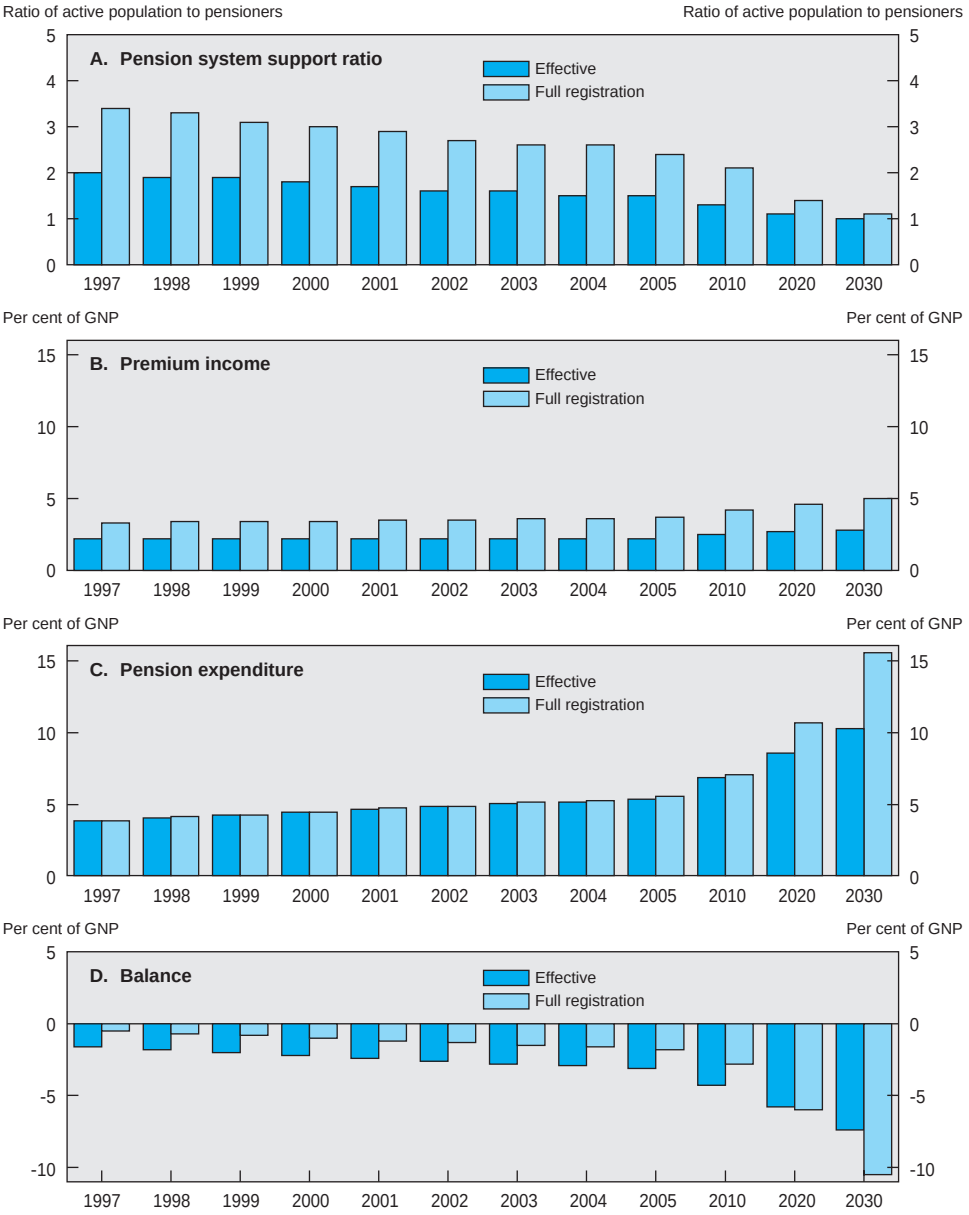
Problems with the pension system

Extensive non-registration and the poor collection record with respect to the registered population reduce the resources available for current pension payments and limit the scope of the public pension scheme. However, in addition to problems of contribution collection and system coverage, the pension leg of the SSOs faces a serious imbalance between contributions and benefits received over an individual's lifetime. Hence, widening registration would increase resources for current pension payments, and temporarily improve the apparent balances of the funds, but it would also increase future pension liabilities, exacerbating deficits later on (Figure 18). Furthermore, given the relatively youthful structure of the funds themselves, they should be in large surplus at the present time in order to be able to match the future liabilities of all current contributors.

The imbalance between contributions and benefits reflects a combination of three main factors (see Box 2 for more detail):

i) Contribution periods which are short compared with the length of pension-drawing periods. Retirement ages can be as low as 38 for

Figure 18. **The effect of non-registration on pension fund balances**



Source: Undersecretariat of Treasury.

Table 16. **Pay-as-you-go pension disequilibrium**

	Per cent of GNP			Ratio of contributors to beneficiaries	Ratio of per capita benefits to per capita contributions
	Benefits[1]	Contributions	Pension balance		
SSK					
1992	2.2	1.5	–0.7	2.3	3.3
1993	2.2	1.3	–0.9	2.3	3.8
1994	2.1	1.0	–1.1	2.4	5.1
1995	2.1	0.8	–1.3	2.4	6.2
1996	2.4	1.1	–1.3	2.4	5.1
1997	2.7	1.3	–1.3	2.3	4.6
1998	2.9	1.5	–1.5	2.2	4.5
BK					
1992	0.4	0.2	–0.2	4.1	6.8
1993	0.4	0.2	–0.2	3.7	8.7
1994	0.3	0.1	–0.2	3.3	8.0
1995	0.3	0.1	–0.2	3.0	6.6
1996	0.5	0.2	–0.3	2.8	8.7
1997	0.6	0.2	–0.3	2.7	6.5
1998	0.6	0.2	–0.4	2.8	7.5
ES[2]					
1992	1.3	1.3	–0.0	1.8	1.8
1993	1.3	1.3	–0.0	1.8	1.9
1994	1.1	1.1	–0.0	2.1	2.1
1995	1.2	1.1	–0.1	2.0	2.1
1996	1.5	1.2	–0.2	1.9	2.2
1997	1.7	1.3	–0.4	1.8	2.4
1998	1.7	1.3	–0.4	1.9	2.6
Total					
1992	3.9	3.1	–0.8	2.5	3.2
1993	3.9	2.8	–1.1	2.5	3.5
1994	3.6	2.3	–1.3	2.5	4.0
1995	3.5	2.0	–1.5	2.4	4.2
1996	4.3	2.5	–1.8	2.3	4.0
1997	4.9	2.8	–2.1	2.3	4.0
1998	5.3	3.0	–2.3	2.2	3.9

1. Including administrative costs, which amounted to 0.09 per cent of GNP for the total of the three funds in 1998.
2. Assumes allocation of the full 35 per cent of insurable wage contribution to the pension accounts.

Source: Undersecretariat of Treasury and OECD estimates.

women and 43 for men, and contribution periods can be as short as 14 years. The rates of accrual of pension rights are correspondingly high.

ii) Use of income levels at the end of the contribution period as the reference for calculating pension benefits.

iii) The presence of a flat-rate top-up of pensions benefits (SYZ) which is not related to contributions.

The outcome of weak contributions and inappropriate system parameters has been reflected in a structure of pension payments and contributions which is significantly at odds with the overall age structure of the population. Due to early retirement ages and short contribution periods, the system support ratio is already down to just over 2 workers per retiree – *i.e.*, at the level predicted when the effects of ageing are strongest in the average of OECD countries (in 2030).[17] The current levels of benefits and contributions, by contrast, requires that each beneficiary receive the contributions of 4 workers to make the system balance (Table 16).

Expansion of health spending

Health expenditures, after stabilising in the first half of the 1990s, have increased strongly in the past two years and currently approach total pension outlays. Turkish health expenditures remain, nevertheless, well below the OECD average both as a share of GDP and on a per capita basis.[18] The expected growth of health spending and its fiscal and social implications have become important issues. Health expenditures are putting pressure on SSO balances despite the fact that the SSOs take responsibility for only around 30 per cent of these expenditures. Government hospital services are heavily subsidised from the general budget, so that all insurance spending induces an indirect budget cost (Box 3). This induced cost is lower in the case of SSK, which has its own hospitals, whereas ES and BK purchase most of their service needs from government hospitals.

A significant factor in the recent increase in health spending is the rising importance of pharmaceutical costs (Figure 19). This may be related to the fact that increases in other types of health services have been restrained by physical rationing. Indeed, long queues for ambulatory and hospital care are indications that rationing is a major problem. This is especially true for the two business sector funds which are more budget constrained than the civil servants' fund. Budgetary constraints limit the supply capacity of health care providers which contract with SSOs for the supply of services. Each SSO tries to limit its own spending by means of these contracts, by transacting with a limited number of hospitals (mostly government and SSO hospitals), physicians and pharmacists, and, in the case of the SSK, by restraining expenditures in its own hospitals.

Unemployment protection

The ambitious scope of pension and health coverage contrasts with the lack of social insurance to protect against loss of income due to unemployment.

Box 3. Supply of health services and their rationing

Hospital services

Hospital facilities are in limited supply, distributed unequally across the country and inefficiently managed. The number of beds per capita is low by OECD standards (2.5 beds per thousand) and is below the WHO target for developing countries (3 per thousand). Beds are concentrated in the three major cities, which had 36 per cent of the capacity in 1998 for only 25 per cent of the population. Hospital beds are in limited supply in outlying areas and 10 per cent of the population lives in provinces where there is only 1 bed per thousand.* In addition, because of inefficient utilisation of hospital resources and doctor shortages, bed-occupancy ratios average 58 per cent in many hospitals leading to pent-up demand for services and delays in treatment, particularly in rural areas. Public confidence in the quality of public hospital services appears to have deteriorated.

Partly reflecting this supply situation, there has been an increase in the supply of private hospital services. These are purchased partly by the emerging private health insurance schemes, but most importantly, by the out-of-pocket payments of higher-income groups. A few private hospitals now offer international quality services at fees well above standard insurance rates and provide quality care for the wealthy who would formerly have gone abroad.

Despite rising private supply, the bulk of hospital services is supplied by the SSK, government, university and military hospitals, all of which are directly subsidised to varying degrees. 60 per cent of the costs of government hospitals are directly funded from the general budget. The other 40 per cent is billed to customers, of which the bulk is reimbursed by social security. Remaining out-of-pocket payments include services that are not reimbursed by social insurance, as government and SSK hospitals have attempted to increase their income by offering such fee-based services and care (*e.g.* more comfortable surroundings). Largely as a result of rapid growth of such non-reimbursable services and amenities, the share of "billed" revenues increased from 13 per cent in 1992 to 25 per cent of total financial resources in 1996 for public hospitals, and from 37 per cent in 1992 to 66 per cent in 1996 for university hospitals.

Ambulatory and physician services

Turkey has 12 doctors per 10 000 persons, considerably below the OECD average (27 per 10 000). Roughly half are found in the three main cities (and the ratio is higher for specialists). As a consequence, rural areas are under-supplied. In addition, rationing and quality problems in the public sector have encouraged increased supply of private-sector ambulatory health-care services. This development may reflect low salaries in the public sector, which encourage doctors to channel patients towards their private practice and may explain part of the decrease in public sector supply. Most private physicians are also part-time practitioners in SSK and government hospitals (hospitals have important out-patient services as well), where they develop part of their customer base: in 1996, only 15 per cent of physicians were engaged in purely private practice. Prescriptions and pharmaceutical costs from these private consultations are probably charged to SSK and other insurance funds. In this environment SSO members often forego their free consultations in public facilities, opting for private physician services paid for out of their own pocket. This has meant that the poor receive increasingly limited services and the quality of care is said to be declining.

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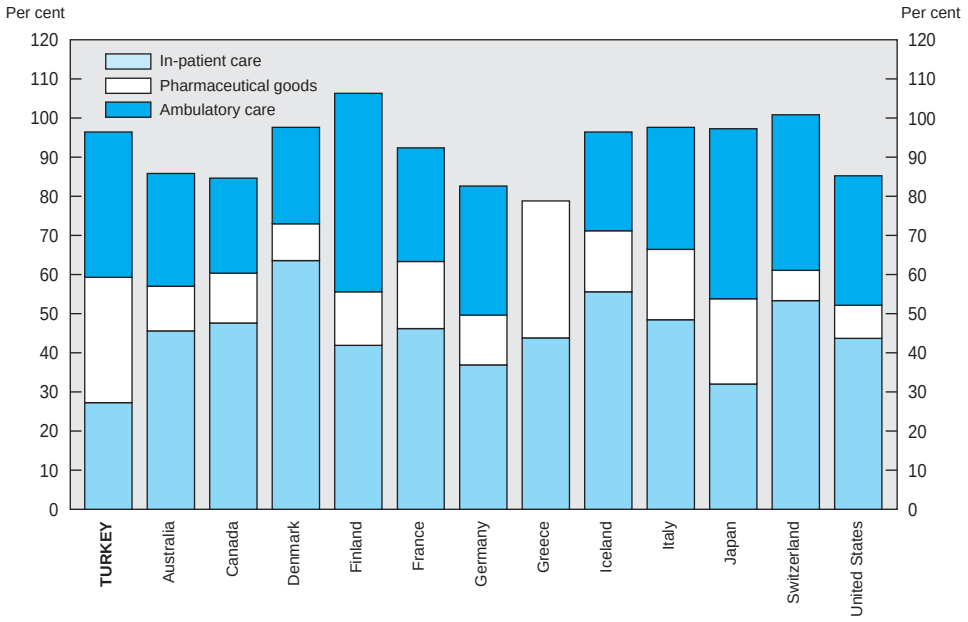
Preventive care and public health centres

Many rural areas are serviced by public health centres, which also provide a range of basic health services and preventive care. Even though such services have had a large impact on overall health outcomes in these areas, they remain poorly financed and they have been the first to suffer under public budget constraints. Their share in total spending has fallen steadily in the 1990s, from 7 per cent of the Ministry of Health budget in 1992 to 3 per cent in 1996. The implications of this trend for public health remains a source of concern, as illustrated by the recent outbreak of malaria in 1994-1996 (which, however, was subsequently brought under control).

* See TESEV (Turkish Economic and Social Studies Foundation)/UNDP (1998), *Turkey Human Development Report*, Ankara.

Figure 19. **Structure of health care spending in selected OECD countries**

Per cent of total national health expenditures



Note: Data for individual spending components do not always sum to estimates for total health care spending.
Source: OECD, Health Data 98.

Unemployed individuals receive only severance payments, a minimum level of which is mandated by labour law but is often more generous under collective agreements. If they qualify, employees can also take early retirement.

Labour regulations and collective agreements prescribe that employee dismissals should give rise to a lump sum benefit proportional to tenure. Most collective agreements grant additional compensation to fired workers.[19] According to the Labour Act, these liabilities are at the charge of individual employers, as there is no collective insurance fund or general budget contribution. This increases the cost of laying off workers. To limit these liabilities, some employers fire workers, especially unqualified/unskilled ones, before one year's tenure. Further, only 35 per cent of registered workers are covered by collective agreements and half of the working population is not registered and therefore receives no severance benefits at all. Thus, this form of unemployment protection is very unequally distributed across the working population.

Early availability of retirement rights allows registered employees facing unemployment risks to qualify for early retirement and secure a minimum income benefit, although the extent to which those taking up pensions on dismissal is not known. This would help explain why unions are unwilling to discuss the introduction of a minimum retirement age and longer contribution periods, unless such reforms are accompanied by the creation of a formal unemployment insurance scheme.

Social assistance

The social assistance system in Turkey is not well developed. It plays the role of provider of "last-resort" in the face of old-age and health distress situations, but its benefits are limited:

Old age income support

Old-age income support consists of means-tested social assistance to disabled and elderly people (over 65), which is paid for and managed by ES, although the total payment is refunded to ES by Treasury. They are meagre in value – 14 per cent of the minimum wage ($ 25 per month) – and, in practice, they leave the uninsured poor dependent on family and charity support. In early 1999, a little under a million persons out of total population in this age group of 6.5 million were receiving these benefits. Means and asset tests determine eligibility for support. However, the selection process may not be transparent. Such administrative problems may become a more serious source of concern if the level of support is raised in the future.

Access to health care by the poor

"Health green cards" are issued to Turkish citizens unable to pay for health services, even at highly subsidised government rates. A survey prior to the establishment of the "green card" system revealed that the access of uninsured low-income groups to hospital care was severely income-constrained.[20] "Green cards" give access to free hospital care (and pharmaceutical reimbursements) and a complex means-testing system has been instituted for their delivery. Implementation problems have been accompanied by a large number of entitlements: 6.6 million Turkish citizens had "green cards" in 1997. Nonetheless, it is reasonable to assume that "green card" holders face even more serious health-care rationing problems than those mentioned above for SSO members.

Economic and social distortions implied by the current social security system

Effect on industrial organisation

Social security payroll tax wedges are substantial, and introduce considerable distortions into the competition between "registered" and "non-registered" firms. Full compliance may mean labour cost increases amounting to up to 35 per cent of the wage bill for registered companies employing low-income workers. On the other hand, being outside the formal system tends to impede the growth and innovative capacity of successful informal sector firms, because of their "obligation" to avoid higher visibility. This could argue for more funding through sources other than a payroll tax, but there is also probably considerable scope for reducing contribution rates by improved collection, provided this was accompanied by a reform of the parameters underlying the pension system.

Labour market impact

The current social security system distorts the labour market in three main ways:

- The work/retirement decision for formal sector workers is biased in favour of early retirement, since there would be a drop in the net pension wealth of low-wage workers who continue to work. In the SSK system, the rate of replacement of insurable wages in retirement increases by 4 per cent per year up to the end of the minimum contribution period but falls to 1 per cent beyond that if a worker elects to continue to work. If eligible workers can find work in the unregistered sector, which seems to be the case for many workers, it is rational for them to retire at the earliest opportunity.
- Retirees who enter the unregistered sector further accentuate labour-cost differentials with the formal sector. The presence of early retirees

– who already have pension income – puts downward pressure on wages in the unregistered sector and further undercuts labour costs in the formal sector.
– The system creates a vicious circle against registration. To cover existing pension benefits and non-payment of contributions, contribution rates need to be kept high and this tax wedge leads firms to limit registration. At the same time, because young workers can save and earn a higher (market) rate of return by working in the informal sector, they also have an incentive to escape the social security tax.

Income adequacy

The social protection system, in its present form, provides very unequal coverage and benefits. Pensions are disproportionately large relative to contributions but are nonetheless small relative to the average wage (around 35 per cent).[21] This partly reflects a situation where many registered workers take their pensions as soon as possible and then go to work in the informal sector. However, the system creates inequities for people unable to work after retirement. The situation is particularly critical for those on social assistance.

Quality of health care

In the area of health coverage, two distinct issues deserve attention. These are:

– *Increasing disparities in access*: Access to health care is becoming increasingly income dependent. Middle-to-high income groups consume more health services per capita than low-income groups,[22] and as these services are subsidised by both social security funds and the general government budget, these groups are likely to receive, on a per capita basis, a higher share of public subsidies. The effects of public subsidies on health service provision has therefore become regressive; however, this may be partly compensated by the fact that a growing share of higher income groups are foregoing the use of public health facilities, using private physicians and hospitals. With higher income groups deserting the public health care system, pressures to maintain quality are weakened and a dual system of health care is developing.
– *Regional disparities*: Both health status and health care supply remain unequally distributed between urban and rural areas and between districts. Patterns of poor health and low access to care reinforce each other. Infant and maternal mortality rates remain extremely high in rural areas and, despite this fact, rural health services have been cut back (see Box 3 above).

Directions for reform

The above analysis has demonstrated serious shortcomings in the present system, which distort the organisation of business activities and the labour market, and make for an unequal distribution of social protection. It has also put pressure on the general government deficit: social security currently accounts for well over a third of the consolidated budget deficit in 1998 and, if no corrective action is taken, the share of social security deficits in GDP could reach at least 5½ per cent in 2010 and 7½ per cent in 2020.[23] With the share of social expenditures in GDP so much lower than for OECD-Europe, the limited scope and quality of Turkish social protection relates partly to the limited resources allocated to it. But reform requires a complete overhaul of the institutions and incentive structures underlying the system.

Pension reform

Reform objectives and strategy

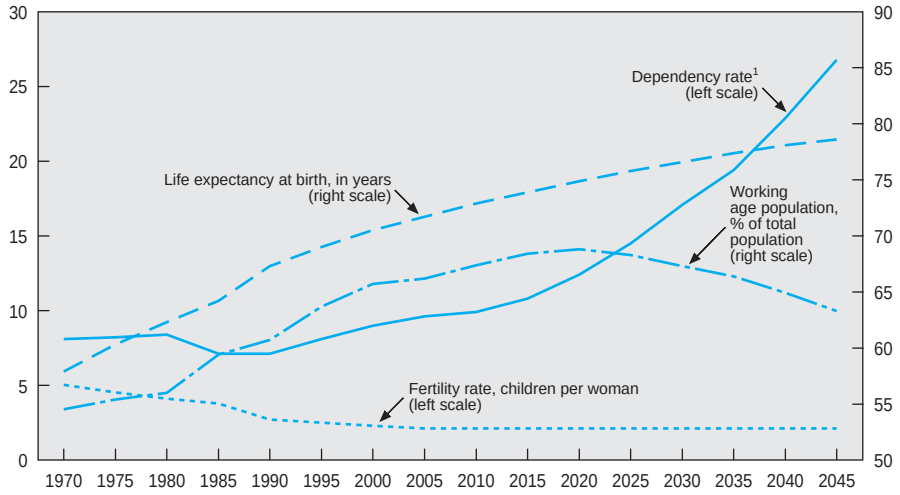
The first priority for reform will be to re-establish solvency of the three social security funds. This will require correcting glaring deficiencies in public pension arrangements, notably unreasonably low retirement ages, short contribution periods, poor collection and enforcement, and perverse incentives in benefit formulas. Beyond this immediate objective, however, there needs to be a reconsideration of the broader and longer run goals of the system. Issues such as old age income sufficiency and poverty alleviation, reduction of microeconomic distortions, intergenerational fairness and risk sharing will need to be addressed by further reforms of the pension system. Health reforms, to be discussed in the next section, will also contribute to a more effective social security system.

Turkey is in one sense ideally placed to undertake fundamental pension reforms in that it will not face an ageing problem for about another generation (Figure 20). Strong population growth will lighten the fiscal costs of transition, allowing the development of a partially pre-funded public scheme to reduce the requirements for pay-as-you-go funding,[24] and of an additional private funded pillar. Such arrangements would leave the pension system well prepared for the demographic shift later on, while inducing positive economic spillover effects, *e.g.* through increased national savings. It is for these reasons that the principles for reform agreed to by OECD Ministers last year remain relevant for a youthful country such as Turkey.[25] This suggests a reform strategy as shown in Box 4,[26] the main lines of which will be developed in the remainder of this section.

Attempts at reform so far

While measures have been taken in recent years to address the causes of deficits in the social security funds, these were insufficient to arrest their rapid

Figure 20. **Demographic projections**



1. Ratio of elderly population (65 years and over) to working age population (15-64 years).
Source: World Bank.

growth. In 1995, the government implemented legislation to increase compliance in the payment of contribution premia,[27] but this resulted in only a modest improvement in SSK collection rates. A 1997 amnesty on penalty charges to encourage workers to pay outstanding contribution debts produced little extra revenues.

Awareness of the need for more serious policy action has grown as the unsustainability of the present pension system becomes more apparent. In 1995, the ILO made a study of the social security system, financed by the World Bank, proposing a variety of pension reforms, including alternative mixtures of a reformed and expanded pay-as-you-go and new private funding schemes, and institutional reforms (Annex III). This exercise led to an intensive domestic policy dialogue with international participation. The government presented draft laws for the reform of SSK and BK in 1997 and 1998, respectively, as part of its medium run fiscal stabilisation programme. The main reform proposals included: an increase in the minimum retirement age to 50/55 (female/male) for existing workers, with a 15-year transition, and to 57/60 for new entrants; an increase in the contribution ceiling to 3 times the minimum wage; a 10 per cent premium contribution to be paid by the state on behalf of all contributors; re-indexing of the

Box 4. Recommendations for reform of the pension system

Stabilising the pay-as-you-go system

- *Raise the retirement age* for existing workers to 60 and gradually raise it to 65 for future workers, and lengthen the contribution period required to obtain a full pension to 30 and 35 years, respectively. The transition to these new parameters should be rapid.
- *Revise the benefit formula* to reduce distortions that encourage both early retirement and income underreporting. This will require maintaining a constant benefit accrual rate and switching to lifetime average income as the reference wage for benefits. CPI indexing may also help to control pension costs.
- *Improve contribution collection rates through institutional reform*. This should eventually allow the statutory contribution rates to be reduced, in turn improving the incentives to comply.

Partial pre-funding of public pensions

- *Build a fund in anticipation of the future benefits associated with expanded coverage of the system.* All contribution revenues resulting from expanded coverage – whether due to increases in registration, female participation, or contribution ceiling – should be used for this purpose. This will prevent any future growth in the unfunded pension debt, hence assuring a more sustainable system.
- *Ensure that management of such a fund is free from all political interference.*

Adding a private funded pillar

- *Establish a second private funded pillar*, in order to provide a broader base for retirement income and a better diversification of risks.
- *Allow occupational or individual plans*. A mandatory, defined contribution individual savings plan, with opt-outs for occupational plans, may provide the greatest flexibility and the widest coverage.
- *Provide subsidies to top up the savings of poorer investors in individual plans*, while making contributions to occupational plans tax-deductible.
- *Reduce inflation on a durable basis, and establish a modern and effective regulatory framework for financial markets*. These are key preconditions for an expanded private sector arrangement.

Expanding social protection

- *Develop a social assistance pillar* providing, on a means-tested basis, perhaps half of the minimum wage as a minimum guarantee. This will pick up those who have never worked or have not worked long enough, while providing cover for market risk on defined contribution plans.
- *Inaugurate a minimal unemployment insurance scheme* in order to compensate workers for the existing *de facto* protection against unemployment risk in the form of early retirement.
- *Strengthen the administrative apparatus* in order to minimise the moral hazard inherent in this added social protection.

flat-rate assistance element (SYZ); and "discretionary" pension indexing, taking into account both the financial position of the fund and the CPI. Given the unsettled political climate of late 1998 and early 1999, these laws were never acted upon by Parliament, however. It is expected that the new government will redraft these laws, taking into account a broader array of reforms including institutional restructuring and unemployment insurance.

Making social security solvent

As noted above, the system support ratio is low and is out of line with both Turkey's youthful demographic structure and the relatively high ratio of benefit to contribution rates. The generous benefit formulas and supplementary allowances, poor collections of contributions, and perverse incentives which lie behind these features all need to be brought back into balance in order to realign these ratios.

The system support ratio can be raised by increasing the average retirement age to closer to the OECD norm, *i.e.* to the 60-65 range (Table 17). This is not unreasonable, as adult life expectancy is virtually the same in Turkey as elsewhere in the OECD.[28] Since women live longer than do men, as elsewhere, the gap between female and male retirement ages should ideally be eliminated, as it has been in most OECD countries. Thus, the minimum retirement age could be set at 60 for those already in the workforce, while gradually raising it to 65 for future adherents. Coherence of the system will require that the minimum contribution period required for a full pension be correspondingly lengthened as well. This could be set to be 30 years for current workers and 35 years for future entrants, the latter being about average for the OECD (Table 17). With a minimum retirement age of 65 for future entrants, those entering as late as 30 could still qualify for a full pension.

Benefit formulas contain costly hidden features and often distort work and income reporting incentives. To correct these problems, benefits should be based on indexed lifetime average earnings and post-retirement benefits indexed to growth to the CPI, which have been common reforms in other OECD countries (Box 5 below). Lifetime income averaging would not only reduce benefit costs but also boost contributions by eliminating the incentive to under-report income in early contribution years and over-report once the reference period for benefits starts (in SSK and BK). Linking the indexation of post-retirement benefits automatically to the CPI index would make the benefit formula more transparent than it has been, an important consideration for maintaining credibility of the system.[29] Transparency and predictability would also be greatly assisted by abolishing the SYZ flat benefit payment and replacing it within the context of an explicit social assistance pillar (below).

Table 17. **Main parameters of public pension schemes in selected countries**

	Statutory retirement ages (men/women)	Contribution period for full pension	Benefit accrual factor[1]	Reference period for benefits	Maximum replacement ratio (per cent)	Indexation of benefits
Turkey	None (average age 49)	14 years	4	Last 5 or 10 years	85	Discretionary, but linked to civil service wage
United States	65/65	35 years	..[2]	Career	44	Prices
Japan	60/55	40 years	0.75	Career	30	Net wages
Germany	65/65	40 years	1.5	Career	60	Net wages
France	60/60	38 years	1.75	Best 12 years	50	Prices (basic scheme) Wages (earning-related portion)
Italy	62/67	40 years	2	Last 5 years	80	Prices
United Kingdom	65/60	50 years	0.4	Career	20	Prices
Sweden	65/65	30 years	..[3]	Best 15 years	60	Prices
Argentina[4]	63/58 rising to 65/60 by 2001	30 years	Flat 27.5% of average wage plus 0.85	Last 10 years	52.5	Discretionary; previously linked to wages
Mexico (old system)	65/65	10 years	Flat 80% of average wage plus 0.56-2.4	Last 5 years	100	By evolution of minimum wage

1. Benefit accrual factor per year of contributions, as a per cent of reference period earnings. Note that pensionable contributions are capped in Turkey (see text).
2. Accrual factor rises as assessed earnings decline.
3. Accrual factor declines as number of contribution years increases.
4. Reformed public system. Most younger workers are now in a Chilean-style private system.

Source: Chand, S.K. and A. Jaeger (1996), "Aging Populations and Public Pensions Schemes", IMF Occasional Paper No. 147, December.

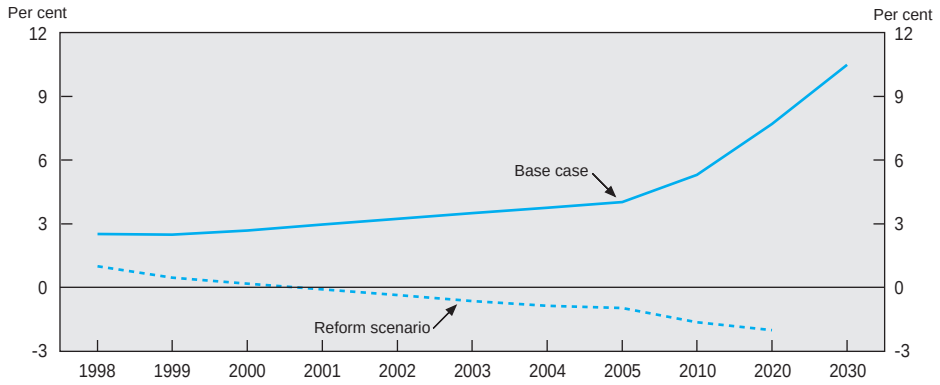
The structure of benefit accrual rates (*i.e.* yearly increases in the pension replacement rate of insurable wages) also needs to be reconsidered. Over a 30-year minimum contribution period, an accrual rate of 2 – half the current level but still quite generous in OECD terms – would allow maintenance of the current full-pension replacement rate of 60 per cent of the insurable wage. Beyond the minimum contribution period, the accrual rate should not be reduced, as it is presently; otherwise, there will be a strong incentive to retire early, or to stay in the informal economy up until the age of thirty. The cap on the replacement rate could also be raised to 100 per cent to further reward working beyond the minimum retirement age. Pensions awarded after less than the minimum contribution period, conditional on the minimum retirement age having been reached, should likewise earn the same yearly accrual rate. By contrast, in the present set-up, shorter than minimum contribution periods actually earn *higher* accrual rates (Box 2 above), which encourages late entry and effectively rewards participation in the informal labour market.

The current low ceiling on contributions (in SSK and BK) means that, even with a generous benefit formula, corresponding benefits are meagre.[30] The switch to lifetime income averaging will further reduce pension income in cases where lifetime average earnings fall short of the contribution wage threshold. Benefit levels may need to be made more realistic by raising the contribution ceiling. On the other hand, the social security tax is highly regressive, with the lowest-earning workers, who pay at the full statutory rate of 35 per cent (including health), being the most vulnerable to falling into informal employment. The effect of a ceiling increase would be to increase the range of workers subject to this maximum contribution rate. Thus, the contribution ceiling should preferably be raised by no more than proposed in the government's earlier draft law, that is to a threshold 3 times the minimum wage. This would imply that a full pension, earning a 60 per cent replacement rate on the contribution wage, would, on average, provide 35 per cent replacement of previous wages (the average wage at present being 5 times the minimum wage), falling in the mid-range of the OECD spectrum (Table 17). The actual replacement rate for each individual would still vary inversely around this average with his/her wage level, although over a whole lifetime, the effect of the ceiling would be actuarially neutral due to the corresponding regressivity of the contribution structure.[31]

Figure 21 shows the estimated effects on system balances of establishing a minimum retirement age of 60 (58 for women)[32] and lengthening of the contribution period to 30 years (all else being equal). It is seen that, if implemented immediately, these two reforms alone could ensure balanced accounts by the year 2005, and even generate small surpluses thereafter. However, it is probably unrealistic to expect immediate transition, as it will take time for workers near retirement under the old system to readjust their behaviour in response to as much as a 20-year increase in the minimum retirement age, and such workers

Figure 21. **Impact on pension deficits of increased retirement age and longer contribution period**[1]

Pension deficits as per cent of GNP



1. Retirement age raised to 58/60 for women/men and contribution period lengthened to 30 years. Reforms are assumed to take place in 1998. Both base case and reform scenarios assume that collection rates improve from 85 to 90 per cent for SSK, and from 51 to 72 per cent for BK, over 10 years.

Source: Undersecretariat of Treasury.

might be allowed to make less than the full adjustment. On the other hand, the longer the transition takes, the more costly it will be, since the stock of young retirees that have to be supported for the rest of their lives will grow substantially during the phase-in. Furthermore, a rush to retire before the rules change could increase these costs more than anticipated. Thus, the transition period should be kept short, preferably to 4 years or less.[33] A lengthier transition, such as that proposed by the ILO (see Annex III) or in the government's draft legislation (15 years), would be too long to address the short-to-medium term fiscal problem, while substantially increasing the unfunded debt that is left to future generations of workers.

At the same time, reforms to benefit formulas, such as CPI indexing, lifetime income averaging, and abolition of SYZ, could help to offset transition costs and such savings could grow over time. A ceiling increase would, in the short run, raise contributions faster than benefits much as in an "immature" system, also helping to cover transition costs, but remain neutral in its impact over the longer run. While revenues clearly associated with immaturity of the system should be funded (see below), eventual surpluses resulting from benefit formula revisions might be used to lower contribution rates.

Table 18. **Pension rates of return**[1]

Pension regime	Parameter assumptions					Implicit rate of return
	Contribution rate	Benefit replacement rate	Years of contribution	Years of retirement[2]	Reference period for benefits	
1. Current pay-as-you–go system[3]						
Minimum: retirement age 43, contribution period 14 years	.20	.60	14	37	Last	.123
Average: retirement age 50, contribution period 20 years	.20	.69	20	30	Last	.095
2. Raise minimum retirement age to 60, contribution period to 30 years	.20	.60	30	20	Last	.054
3. Regime 2 with lifetime income averaging	.20	.60	30	20	Life	.030
4. Regime 3 with contribution rate set to maintain real return of regime 2	.115	.60	30	20	Life	.054
5. Private scheme: contribution rate implied by a real return of 7%[4]	.042	.60	30	20	Life	.070

1. The implied real pension rate of return is calculated as that interest rate which equates the cumulated value of contributions at retirement date with the present value of lifetime benefits (assuming constant real pension) at retirement date (see Annex IV for more detail).
2. Assumes life span of 80 years.
3. Current SSK parameters.
4. By comparison, a rate of return of 6 per cent would imply a contribution rate of 5.3 per cent.

Source: OECD.

The reform impact can also be seen in a declining rate of return on pension contributions. The *implicit pension rate of return* may be calculated as that interest rate which equates the cumulated value of contributions with the present value of benefits at retirement (Table 18 and Annex IV). Although calculation of the overall rate of return is complicated by the rules of the system (see Box 2), estimates using simplifying assumptions show that the rate of return implied by current pension arrangements is 9½ per cent at the average retirement age and contribution period, rising to 12¼ per cent for the minimum retirement age and contribution period. This is clearly unsustainable for a pay as you go system, as this is well above the country's long run growth rate.[34] The reforms suggested above to increase the retirement age and the contribution period would reduce pension returns to the 5 to 6 per cent range, which appears to be more sustainable. The switch to a lifetime average reference period would further reduce the rate of return to 3 per cent. The latter compares unfavourably with the potential rate of economic growth, not to mention the market rate of interest, in addition to exacerbating the problem of insufficiency of pension income while asking people to work longer. This in turn suggests that such a reform – desirable in its own right for its incentive effects – might justify a reduction in the contribution rate. To maintain the previous 5½ per cent real return, for example, the contribution rate would need to be cut from 20 to 11½ per cent. This would release resources which could eventually be placed in second tier savings arrangements.

It should be noted that these hypothetical rates of return are based on the assumption that premium contributions are fully and promptly paid on honestly reported incomes.[35] As long as current problems in collection and reporting persist, actual effective returns (and thus fund deficits) could be higher. Thus, reforms to improve contribution revenues – notably lifetime averaging and improved collections – will need to *precede* any reduction in statutory contribution rates. Only if everyone pays can the rate then come down. Some level of continuing budget support to the funds might be justified, in the meantime, as an offset to the "externality" of a large and implicitly tolerated informal sector. However, the subsidy of 10 per cent of the contribution wage as proposed in the government's draft legislation, seems too high.[36] It should also be transitory, with an automatic phase-out.

Administrative reform of the social security system

In the domestic policy debate, the social security funds have argued for greater autonomy in keeping with the system's founding principles. Autonomy implies the introduction of appropriate mechanisms to ensure political insulation of the funds in terms of determining pension rules, personnel practices, and investment strategies. It may also provide scope for the funds to produce detailed actuarial studies of proposed changes to system parameters, which is taken into account in the legislative process (as in the United States). In most

OECD countries, in fact, the social security funds are autonomous and, within limits, have the right to set contribution rates in line with legislated benefits.

Once autonomy is established, it should be conditional on a marked improvement in the social security funds' (SSK and BK) collection records. Fund managers should be obliged to meet certain performance targets which could involve monetary incentives (*e.g.* performance-related pay, demotions or dismissals for non-performance), but is above all likely to require significant new investments in information technology and associated training of staff. Better tools and skills will permit a more systematic tracking of individual work and contribution histories. Consolidation of the three social security funds into one fund may also be desirable in order reap administrative economies of scale, eliminating wasteful duplication of scarce skills and technology. A common set of pension rules for all workers would furthermore reduce existing distortions and non-transparency, making compliance and monitoring simpler.[37]

These measures will need to be complemented by better enforcement of system rules.[38] Tougher penalties for late payment or evasion will be needed (*e.g.*, making collusive evasion a criminal offence as under the general tax code), as will new approaches to dealing with the chronic problem of non-payment in the state economic enterprises, and effective judiciary and executive enforcement of sanctions. Ways might also be found to extend recently-strengthened enforcement in the general tax administration (see Chapter II) to the social security funds, so as to obtain economies of scale in tax administration and allow auditing expertise to be shared. For example, the new individual taxpayer numbers could double-serve as social security identification numbers, while crosschecks between social security and tax records, including third party information, should also be possible. These processes are likely to be long-term, implying that collection rates and membership rolls can probably increase only slowly over time.[39]

Pre-funding of social security

Once the social security system is put on a sound financial footing, the next stage of reforms should be towards greater pre-funding of future pension liabilities in order to smooth the intergenerational tax burden and assure sustainability of the system. Most OECD countries now face large unfunded pension liabilities due to the coincidence of programme maturation and population ageing.[40] In Turkey's case, by contrast, the initial conditions for reform will be those of a quite immature system, with the demographic transition still some years away. Less than half of the work force is covered directly (as active insured) by social security; the level of benefits is low, and female participation is very low (see Chapter IV). The potential for increased coverage is thus very large. Previous policies, nonetheless, have created an artificial early "ageing" of the system, which the above reforms, if implemented, should reverse. The task would then be

to prevent the growth of unfunded debt within the extended pension system. This means that all contribution revenues arising from increased coverage should be used to build a fund *in anticipation* of the implicit liability linked to future benefits of the expanded system. Such funds should not be used to avoid reforms in contribution and benefit rules or to finance general government spending. Safeguards will be needed to prevent short-term surpluses from being spent. OECD-wide experience suggests that tight restrictions need to be imposed.

Adding a private funded pillar

A well-designed public pay-as-you-go (PAYG) system needs to be buttressed by a privatised second tier system to provide a broader basis for retirement income. PAYG systems, for their part, provide intra- and intergenerational risk-sharing that are absent in private schemes. But they remain subject to political risks and pay, at best, a return which rarely exceeds the economy's long run growth rate. On the other hand, private retirement savings provide scope for individuals to obtain higher returns on investment, albeit while facing greater risks associated with financial market fluctuations, inflation, and higher administrative costs. Thus, a mixed system could help to diversify risks and benefits. Moreover, requiring workers to contribute to private funded schemes, as opposed to charging higher PAYG taxes,[41] could help to alleviate microeconomic distortions. Private funding improves incentives in the formal labour market (where innovation is higher) by substituting highly remunerated savings, with strengthened ownership rights and responsibility, for a tax with high dead-weight costs.[42] It also encourages greater depth and diversity of financial markets and increases the generation of savings, in the economy.[43, 44] These factors would increase economic growth, eventually permitting more resources to be released for old-age income protection. The experiences of other countries can provide guideposts for the balance between these two elements of retirement savings and income and for the reform strategies (Box 5).

A workable strategy for Turkey might be as follows. As shown above, the reformed PAYG system would provide, on average, 35 per cent replacement of lifetime real incomes. A more adequate replacement rate of 70 per cent (the international standard recommended by the ILO) could be targeted by instituting a private mandatory defined contribution scheme for new workers (and allowing other workers to opt in voluntarily), which is designed to yield another 35 per cent of lifetime real income after a full working life of contributions.[45] Assuming, over the long run, a real gross yield of 8 per cent on a prudently-managed bond portfolio in Turkey[46] and a 1 per cent of assets average annual management fee,[47] such an outcome would require setting aside 4 per cent of wages for investment in the privatised scheme (Table 18 above).[48] This compares with 11½ per cent required for the same outcome under reformed pay-as-you-go, reflecting the much better rate of return, even net of fees, on market based investments.

Box 5. Pension reforms in other countries

Countries such as Mexico and Chile have chosen to make compulsory private individual retirement accounts their primary pension insurance pillar while phasing out PAYG altogether. These "clean break" privatisations entail significant up-front fiscal costs because accrued rights under the old system have to be honoured by issuing explicit debt. Explicit debt then needs to be serviced at market rates (higher than the rate of return on old implicit debt) while private pension funds are being built up from scratch.[1] Such transition costs can be reduced, to some extent, by prior reform of the public scheme to reduce its implicit liability before turning it into explicit debt. Other countries, such as the United Kingdom, Argentina, and Hungary, have pursued partial privatisations allowing for a choice between public and private schemes, or cohort-specific transitions, which reduces or spreads the transition costs more widely.[2] The United States, Sweden, Canada, among others, have sought to pre-fund part of their public schemes, while providing fiscal incentives for voluntary individual retirement accounts, in effect creating a mixed system. Singapore, Malaysia, and Australia have had only funded schemes, in the former two countries as public provident funds and in the latter, compulsory occupational pension trusts. Although this group of countries has avoided the transition problem and is in this sense well prepared for the demographic transition, publicly-managed funds have provided poor returns for investors.[3]

In the OECD countries, the reforms may alternatively be grouped into three broad categories:

Reductions in generosity of public pension benefits

- Reductions in final benefit relative to number of years of work/contributions in Italy, Norway, Canada, Finland, United Kingdom, New Zealand, Sweden, and Portugal.
- Increases in the number of earnings years used to calculate final benefit in Spain, France, and Sweden.
- Less generous adjustment of benefits to changes in inflation in Japan and Finland.
- Adjusting benefit calculation to take account of increases in longevity of new retirees in Sweden and Italy.

Working longer and reducing distortions that encourage early retirement

- Increase in age of entitlement from 65 to 67 in the United States over the next 25 years. Increase in female/male retirement ages from 58/63 to 60/65 by the year 2000 in Italy.
- Increasing retirement age for women up to the same age as for men in Australia, Belgium, Germany, Hungary, Japan, Portugal, and the United Kingdom.
- Increasing the minimum age at which (Finland and Poland), or the number of years of work/contributions before (Belgium, Hungary, and Italy), early retirement benefits can be accessed.
- Reductions in benefit payments for early retirees in Australia, Hungary, Sweden, and Slovak Republic.

(continued on next page)

(continued)

– Increasing the rate of pension for deferred retirement in Canada, Luxembourg, Sweden, Finland, Hungary, Germany, Slovak Republic, and Australia (lump sum payment).

Increased advance-funding

– Mexico is moving to a fully-funded privately managed system of individual accounts.
– Canada and Sweden are increasing the funded element of their public earnings-related benefit schemes.
– Hungary, Poland, and the Czech Republic are looking to private pension arrangements to supplement retirement pension benefits for workers without further expanding large public schemes.
– Australia, Denmark, Japan, Korea, Canada, New Zealand and the United States are encouraging greater reliance on the private pension system to provide retirement benefits.

1. At the same time, workers will be able to pay the higher general tax rates needed to service this new debt insofar as their pension contributions have been lowered by the excess of the market rate of return over the PAYG rate. The financial effects on workers sould be neutral, therefore.
2. See Disney R. (1998), "Crises in Public Pensions Programmes in OECD: What are the Reform Options?", *University of Nottingham Discussion Papers in Economics*, No. 98/20.
3. See Bateman, H. and J. Piggott (1997), "Mandatory Retirement Saving: Australia and Malaysia Compared", in Valdès-Prieto (ed.), *The Economics of Pensions*, Cambridge University Press.

Investing part of the fund in equities could further lower the required minimum contribution, as expected return would be higher, but at the cost of higher risk.

However, such individual retirement schemes can leave people with low incomes poor in old age. Direct subsidies, in the form of refundable tax credits, to top up savings contributions of low-income workers may hence be justified.[49] On the other hand, the actual replacement rate on the PAYG component is significantly higher for lower incomes, so that a worker earning up to the contribution wage could receive 60 per cent of his lifetime wage in retirement, and up to 100 per cent for a long work history. Thus, any tax credit subsidy could be kept small. Higher income workers could compensate for their low PAYG replacement rates (*i.e.* less than 35 per cent) by voluntarily contributing more than the mandated minimum to their individual savings scheme. Voluntary, tax-advantaged occupational schemes could be allowed to substitute for, or top up, the mandatory individual schemes (much as they do now for social security), although they may limit portability if vesting periods are long.[50]

The Turkish government, finally, faces a critical challenge in establishing the appropriate macroeconomic conditions and institutional capacity for a viable private funded pillar. In particular, inflation must be conquered and capital markets must be made deep, diversified, and transparent enough to provide adequate and safe returns for investors. The following measures will be needed:

- While the initial level of equity market development has been judged to be adequate,[51] and there is a large domestic market for government securities, other important savings vehicles are undeveloped, notably private fixed-income and inflation-indexed securities. Issuance of both private and government inflation-indexed debt would boost private pension income security. Life insurance benefits could then also be more easily annuitised and indexed,[52] providing protection against longevity risk and avoiding the large losses sustained in the past by holders of such policies due to inflation. But most importantly, inflation must be brought down on a durable basis in order to ultimately reduce inflation risk and inspire confidence in the longer-run safety of private savings vehicles.
- Restrictions on domestic equity and foreign investments by private pension funds have been imposed by other countries. In Turkey's case, this may help develop the domestic capital market. However, it does increase the risk facing private investors particularly if inflation is not brought under control. Such restrictions should therefore be strictly limited in time in order to allow for the full benefits of diversification.
- Legislation to ensure effective regulation and supervision of pension fund management companies and occupational schemes must be immediately put into place in order to protect investors against excessive management fees, asset mismanagement, or fraud. Such protections are notoriously absent in Turkey, accounting for the bleak record on private pension fund safety and for widespread distrust of a private sector role for pension provision.

Complementary social protection

A social assistance pillar, financed out of general tax revenues, is needed to pick up those who have never worked or those who have not worked long enough. It is also an essential back-up to those whose pensions from social security or from the private funded pillar are inadequate. To this end, the currently low level of old age assistance benefits needs to be raised toward the subsistence level (to perhaps half the minimum wage).[53] Assistance benefits should preferably take the form of a strictly means-tested minimum income guarantee (*e.g.* as in Australia), but they will need to be carefully controlled to minimise moral hazard effects. This will in turn require increased vigilance in the administration of means and asset tests. At the same time, the current nominal

freeze on the social security flat-rate benefit payment (SYZ) should be maintained, while transferring any such redistributive element to the explicit assistance pillar, where it would be better targeted on those in need and therefore less costly to the budget.

In addition, unemployment insurance may be needed in order to compensate workers for the current *de facto* protection against unemployment risk that is provided by early retirement. However, such a scheme should be kept minimal and its conditionality adequately enforced in order to minimise well-known labour-market distortions that can arise from ill-designed benefit schemes (see Chapter IV).

Health-care reform

Most of the problems with the health-care system noted above were already identified before the end of the 1980s. In the 1990s, the government aimed at creating a national consensus on the need for a comprehensive reform, building on international expertise, national conferences and improved data. The authorities hoped that the national health policy, thus formulated, would be least susceptible to changes of government. The main framework of the current reform strategy has been suggested by the *Health Sector Master Plan Study* completed by the SPO (State Planning Organisation) in 1990, and the *Draft Health Reform Project* completed by the Ministry of Health in 1992. The proposals included universal health insurance coverage, improved efficiency, and more equal access to health care services. These goals can be achieved through reforms to financing, service delivery, and organisational structure. Ministry of Health polls have revealed strong popular support for these proposals, reflecting widespread frustration with the existing system, and this creates favourable political circumstances for bold reform.

Financing reform

The proposed financing mechanism is the General Health Insurance Scheme (GHIS), which extends coverage to those who do not have health insurance with the social security organisations and to those who, in practice, are rationed in their access to services.[54] It would be managed by a new autonomous government agency, the Health Insurance Administration (HIA). A further study of policy options concluded that a single social insurance scheme that operates on actuarial and full cost principles, such as the GHIS, was the best option for Turkey.[55] The GHIS was submitted as a draft law by the government to the Parliament in July 1998 but could not be discussed by that legislature.

The members of this insurance scheme would be entitled to a package of comprehensive services. Contributions would be related to ability to pay, and the difference between the actuarially calculated premium and the member's

contribution would be met by subsidies from the state on a sliding scale. For this purpose, households would be classified into five income categories, with the very poor paying no contributions. Indeed, as noted above, the very poor have already been given access to free care by means of the "green card". The HIA would transfer its revenues to the Provincial Health Directorates – the regional arms of the Ministry of Health – on the basis of population, age structure and health status. The Directorates would then make contracts with service providers on behalf of the insured. The scheme would initially operate in parallel with the existing social insurance schemes. Over the longer term, however, once health and pension plans of the social security organisations have been separated, all the health insurance schemes could be integrated under a single umbrella.

The time required to implement the new scheme, and fully cover the presently uninsured, could take four years, after its "pilot" initiation in different provinces which remain to be determined by the Government.

Service delivery reform

The basic idea behind this part of the reform is to create a service delivery system in which autonomous (and often private) providers of services operate in a competitive environment, patients exercise freedom of choice and the government controls costs and quality of services through contractual arrangements with service providers. To set up such a system, separation of financing and the provision of services is envisaged (purchaser-provider split), with reforms affecting ambulatory, general hospital, and specialised hospital services.

- Restructuring primary care

The reform in this area aims at improving the efficiency of primary care provision by setting up a system of family physicians, providing individual preventive and ambulatory care in private practice. Under current arrangements, patients have tended to go directly to hospitals for all their health care needs, leading to congestion at hospital ambulatory care units. This in turn necessitates a change of status for most primary care physicians who are currently salaried civil servants working in hospitals, in order to improve their incentives and responsibilities in referring patients to proper care units. Family physicians would be paid mainly on a capitation basis (*i.e.* paid a flat rate for patients who register with them) by the Provincial Health Directorates. Capitation payments could be adjusted taking into account such factors as the age of patients and morbidity in the region in which the physicians practice. Patients would have the right to change their family physician after six months of registration, and this possibility of "voting with one's feet" is expected to instil some competitive forces into the system. Family physicians would play a gatekeeper role for referrals to specialists or hospitals and would also be responsible for the follow-up of specialist and hospital services received by their patients.

Family physicians would be supported by a network of small health-care units (health posts and health centres) and public laboratories in playing their role of promoting public health services, such as vaccination, health education and community diagnosis studies, as well as in accessing laboratory and radiological services. The main responsibility for monitoring, evaluation and co-ordination of public health services, however, would reside in the Provincial Health Directorates. A draft law (Primary Care Services and Family Physicians Law) was submitted to the Parliament in 1995. Although this draft law did not pass the Parliament, a pilot project related to the family physician system has been in progress since 1997 in two provinces.[56]

- Hospital decentralisation: the creation of Health Enterprises

A key element of the reform proposal at the level of out-patient and in-patient hospital care is to enable public hospitals to be decentralised through their transformation into autonomous Health Enterprises (HEs). This would apply to all public hospitals including those of the social security organisations and university hospitals. The HEs would have a separate legal status and full financial and managerial autonomy including the decisions to borrow, allocate resources and recruit and pay their own staff through contractual arrangements. They would be managed by professional managers, or doctors with management skills and the willingness to devote full time to management. A seven member Board of Directors, chaired by the Hospital General Manager and composed of four executive directors (including the General Manager) and three non-executive directors from community and local organisations would be the proposed governing body.

The main revenue source of the Health Enterprises (HEs) would be block contracts with Provincial Health Directorates as well as with private insurance companies. The HEs and private hospitals would have to compete for such contracts. The decentralised decision making and competitive pressure are expected to encourage the providers of hospital services to operate efficiently.

Organisational reform

The planned reforms will require a major reorganisation and cultural change at the Ministry of Health. It is presently organised around different kinds of services (such as health-care centres), target population groups (such as mothers and children) and specific diseases (such as tuberculosis), which make consistent policy development difficult. After reforms, most of the operational decisions, both concerning the management of the new scheme and the execution of Ministry of Health policies, would be devolved to the Provincial Health Directorates, with most services unified at the provincial level. The primary function of the Ministry of Health would thus be stripped down to that of setting national policies and standards, in particular in the areas of public health,

medical education and research, as well as monitoring and evaluating their implementation and outcome.

Assessment

The proposed comprehensive reform of the Turkish health-care system, if implemented, could go a long way towards ensuring access to quality care by all citizens at a reasonable overall cost to the nation (Box 6). Such comprehensive reform is to be preferred to piecemeal adjustments in the existing system, which is facing too many coverage and efficiency shortcomings. Much preparatory work has been done over the last decade in terms of studies, surveys, and preparation of necessary legislation so that it is now up to the politicians to act.

The single biggest issue is how reforms could be best introduced to improve overall performance. A major step concerns detaching the health plans from the pension plans in the SSOs and integrating the former into the GHIS once it is operational. This will allow the financing and provision of services to be fully separated, with social security and other hospitals negotiating contracts with the Provincial Health Directorates. Experience in other countries, however, has shown that contracting costs are high in the initial stages and information needed to monitor quantity and quality of care are often lacking. Learning from the experience in the United Kingdom, the writing of block contracts with both HEs and private hospitals is not a straightforward task, and it would be best to start a pilot project as soon as possible. Such contracts might have to adopt elements of case-mix funding as practised in Medicare in the United States and the state of Victoria in Australia,[57] which requires a period of experiments. Furthermore, given the disappointing impact of the current system on health outcomes – exemplified by the large regional differences in child and maternal death rates[58] – the Ministry of Health and, particularly, the Provincial Health Directorates will need to focus resources in areas where the impact on health outcomes is likely to be the highest.

Effective control of health-care expenditures will be an overriding priority in the new system, given the budgetary burden of providing universal health insurance coverage. While the introduction of *de facto* competition through the contracting process can be expected to hold down costs, this may not be enough. The method of paying doctors in primary care is another critical factor. The capitation payment system featuring in the reform proposals provides primary care doctors with a fixed payment, albeit with adjustments for special factors such as age and region, for each patient on their list. Unlike the alternative "fee for service" arrangement which gives doctors the incentive to expand the volumes and prices of services they provide, this payment method allows funders to control the overall level of primary health expenditures. However, under this approach there are risks of doctors registering too many patients and then

Box 6. Recommendations for reform of the health system

Provide universal health insurance coverage

- *Create a general health insurance scheme* and make it available to population groups not insured by social security organisations. Subsequently, integrate the health arm of the social security organisations into general health insurance.
- *Introduce means-tested subsidies for the subscription fees of lower income groups.* Divert general budget resources currently spent on the direct provision of services – apart from those dedicated to public health – to general health insurance.

Introduce competition in health-care provision

- *Develop the market supply of health services* by means of competitive contracts between the general health insurance and public and private service suppliers. Provincial Health Directorates of the Ministry of Health should be charged with negotiating these contracts on a regional basis, on behalf of the insurance fund.
- *Reorganise public hospitals as health enterprises* to compete with private suppliers. Health enterprises should be provided with access to state-of-the-art know-how in health sector management.
- *Separate the medical service provision of the SSOs from their financing activities,* allowing the integration of the latter into the general insurance system and subjecting SSO suppliers to the general contracting process.

Establish a system of primary and preventive care

- *Strengthen the "family physician" function* and finance it by capitation payments from general health insurance.
- *Family physicians should serve as gatekeepers for referrals to specialists and hospitals,* and monitor the health status of their patients by following-up services provided by specialists.
- Family physicians should be provided with *incentives to minimise referrals.*
- *Strengthen the focus of public health policy on preventive care,* including better pre-natal and infant care, and promotion of anti-smoking campaigns.

Control pharmaceutical costs

- Review the factors that have provoked a strong increase in public pharmaceutical costs.
- Remedy the cost-increasing trends in drug consumption, pricing and packaging, including through better bargaining with pharmaceutical companies.

under-serving them, selecting the better risks, and referring on too quickly to specialists or hospitals patients who could have been treated by the primary care physician directly. With no constraint (budgetary or otherwise) on the number of referrals, *overall* system health costs may thus prove difficult to control. And although a capitation system may enhance the incentive for doctors to practice "preventive medicine" so as to avoid larger problems later on, a greater focus on health outcomes and on preventive care by the public health system as a whole would still appear to be warranted. This could include raising the status of public

health nurses as the first point of contact for the rural population, infant immunisation and pre-natal health campaigns, and promotion of anti-smoking awareness starting at a young age.

The control of pharmaceutical expenditures should be another major objective, given the recent surge of drug costs. The factors underlying this cost increase need to be better understood, and this should help contain the pharmaceutical costs of the general health insurance system. Full utilisation of generic drugs, incentives to savings in physicians' prescriptions, better negotiation of prices with suppliers, and optimal packaging are possible avenues for cost reductions.

Finally, it would be worth putting further thought into the governance mechanism of the general health insurance scheme, and of health enterprises. A Health Insurance Administration is projected to manage the system; its design should be built on principles of efficient governance. Relevant approaches could include separation of chairman and general director roles and putting in place a strong and independent board of directors, alongside regional decentralisation and efficiency benchmarking of the Provincial Health Directorates by the Health Insurance Administration. The efficient management of Health Enterprises is equally important, and it is desirable that they gain access to state-of-the art health sector management know-how, including through co-operation and joint ventures with foreign health service organisations. It is likewise important to avoid too much bureaucracy in hospital governance; for example, one executive director, rather than the proposed four, should be enough and is more in line with international best practice.

IV. Structural reform for better-balanced growth

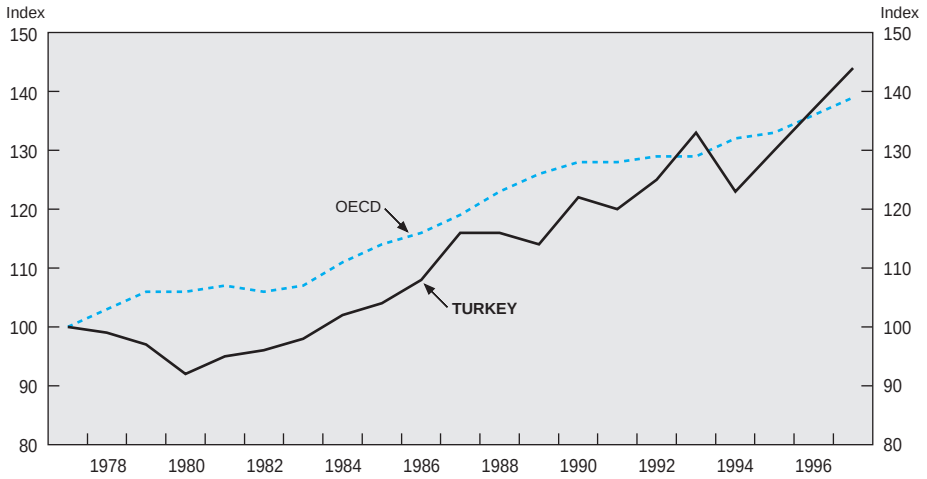
Structural policies that encourage flexibility and responsiveness to market signals can play a major role in improving long-run economic performance, although they also need to be accompanied by policies that address income distribution and other social issues. This Chapter reviews Turkey's longer-run economic performance and looks at the scope for structural reforms to enhance performance across all areas, apart from those already addressed above in relation to the budget and the social security system. For Turkey, key structural policy issues are: *i)* to reduce distortions in product markets, especially agriculture and state economic enterprises; *ii)* to develop better strategies for infrastructure development; *iii)* to strengthen the performance of the banking sector; and *iv)* to improve human capital. Policy proposals have been developed to address all these issues, but getting policy proposals adopted remains the major sticking point. Improving the policy-making process is therefore also a key policy challenge.

Turkey's longer-run economic performance

The Turkish economy has achieved an average annual growth rate of 4.1 per cent over the past twenty years. This growth rate has been considerably better than the OECD average of 2.5 per cent, as would be expected from a country that is "catching up". Strong population growth and urbanisation have played an important part. But real GDP per person has only just managed to grow at the same pace as the OECD average (Figure 22). Considering the problems that have beset the Turkish economy over that period, this could be construed as a surprisingly good economic performance, but this productivity record provides grounds to expect that Turkey could grow much faster if sounder economic policies were in place.

Turkey's economic performance has been underpinned by three major strengths. Its first strength is that it is internationally competitive and it trades

Figure 22. **Real gross domestic product per capita**
1977 = 100, at the price levels and exchange rates of 1990



Source: OECD, *National Accounts.*

extensively. It has benefited from unilateral dismantling of import barriers and from the customs union with the EU. In addition, it is well-placed geographically to take advantage of trade with emerging economies within the region, and "suitcase" trade has become an established feature. Its second strength is a young and rapidly growing work force. This has contributed significantly to Turkey's output potential. Its third strength is a well-established tradition of dynamic performance in many areas of the private sector. For example, one survey of the opinions of foreign business in Turkey found a consensus on the existence of entrepreneurial spirit in Turkey. Risk-taking, motivation, dedication and flexibility were also cited.[59] The most successful large Turkish businesses deliver world-class performance, while informal activity produces exceptional flexibility, responsiveness and dynamism, especially in small-scale operations.

Turkey's per capita GDP, at $ 6 463 in 1997, remains the lowest in the OECD and equal to only 30 per cent of the OECD average. Turkey also has one of the widest dispersions of income among the OECD countries, and income disparities actually widened between 1987 and 1994.[60] Basic measures of the standard of living, such as access to safe drinking water, also illustrate that Turkey has some way to go to catch up with the rest of the OECD.[61] Private consumption per person, often considered an overall measure of living standards, is not only the lowest in the OECD, but has also slipped further behind over the last twenty years.

Table 19. **OECD Economic Survey recommendations for structural reforms**[1]

Product markets and state enterprises

Competition framework

1995 Continue to lower border protection – particular concern for large coefficient of variation in protection which introduces large tax wedges between industries and products.

1996 Use opportunity of EU Customs Union to expose economy to greater competition through vigorous application of competition laws and deregulation.

Agriculture

1995 Shift to a more market-oriented system: from price support to specific subsidies for production inputs, provision of credit facilities and direct payments to targeted population.

1997 Minimise price support assistance, reorient agricultural subsidies towards direct income support payments.

State economic enterprises

1995 Replace non-economic incentives by rational cost and price structures, and commercial goals.

1996 Restructure or liquidate large loss-making SEEs to enforce hard budget constraints and to raise productivity as a move to promoting the stalled privatisation programme.

1997 Need to continue restructuring three largest loss-makers (Railways, Post Office and Hard Coal Company). Need to establish positive incentive structures and rational pricing practices more closely in line with market conditions.

Public infrastructure

1996 Address pressing public infrastructure needs, in particular unstable electricity supply.

Capital markets

1996 Strengthen prudential control of financial institutions.

1997 Tighten prudential standards.

Labour markets and human capital

Labour markets

1996 Exercise caution in any plans to introduce an unemployment insurance scheme to avoid adverse incentives and cost to the state.

1997 Encourage integration of unrecorded economy.

Human capital

1995 Reduce disparities between east/west and urban/rural in health and education standards. Provide more funding to education and health care.

1996 Raise compulsory schooling from 5 to 8 years (or beyond) as soon as feasible. Provide more and better financial and teaching resources.

1997 Earmark the efficiency savings from structural reforms to raising health and education standards, especially for the poorest elements of society. Reorientate spending priorities and raise the quality of human capital.

1. This table does not include previous structural recommendations on social security and tax reform, which have been dealt with in earlier chapters.

Source: OECD.

There is widespread agreement on what preconditions are important for generating high economic growth performance. These are an appropriate stock of physical capital, efficiently allocated; a well-qualified, increasingly productive and growing workforce; markets that work well; and well-functioning government. Judged against these preconditions, Turkish economic performance could be improved by implementing a wide range of structural reforms. Many required reforms have been recommended in previous *Economic Surveys* (Table 19). In particular, emphasis has been placed on the need for Turkey to adjust its policies and improve performance in five main areas: the further liberalisation of key product markets; developing appropriate infrastructure to support growth; improving the allocational efficiency of financial markets; upgrading human capital, and improving public policy making. The need for these reforms is widely accepted in Turkey, but progress towards implementation has so far been patchy. The overriding structural challenge is to get reforms put in place and the rest of this chapter concentrates on these issues.

Enhancing product market competition

A liberal import regime and strengthened framework for domestic competition

Turkey has achieved considerable progress in liberalising many of its product markets. Turkey is now reaping good returns from the liberalisation of imports undertaken since the early 1980s and the Customs Union with the European Union, which started in January 1996.[62] These have generated considerably more competition within Turkey. The average level of MFN tariffs on imports of manufactures has fallen from 27 per cent in 1993 to 12 per cent in 1998, and on average have fallen to 5.5 per cent for manufacturing other than food, beverages and tobacco manufacturing. Overall, restructuring is taking place, producing a more efficient manufacturing sector, to the benefit of consumers.

The new and independent Competition Authority, which previous *Economic Surveys* had recommended, started operating in November 1997. Its early work has concentrated on establishing regulations under the Competition Law and building up its expertise. The Authority received 250 applications for rulings on competition infringement in 1998, 130 of which did not in fact fall within the scope of the Authority. The remaining 120 applications ranged across many sectors, but food and beverages, printing and publishing, cement, transport and education were particularly high, relative to other sectors. The Authority has also received 220 applications for exemptions and negative clearances, again spread across many sectors. It also considered 72 applications for mergers and acquisitions. The Competition Authority also forwarded opinions to the Privatisation

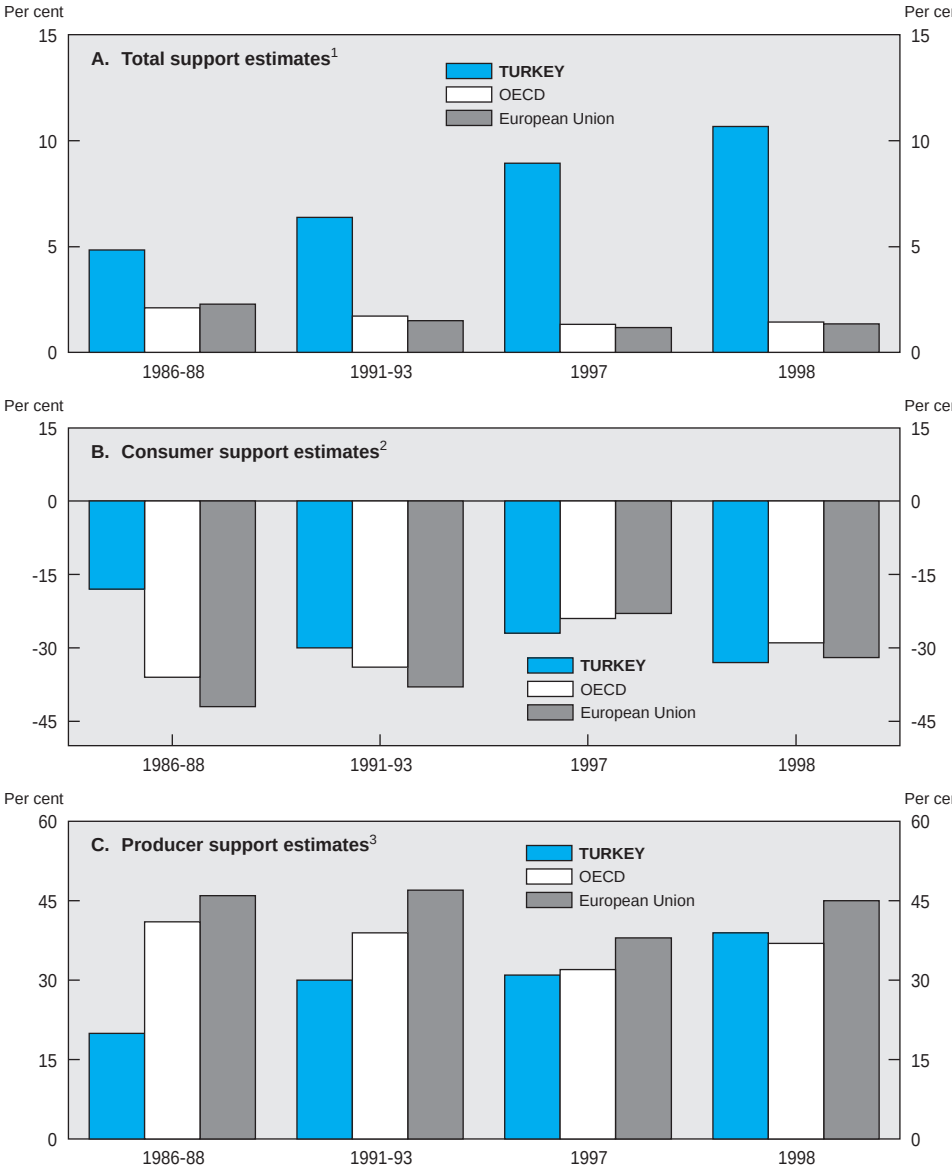
Administration, Ministries of Health, Environment, Education, Finance and Trade, and the Superior Board of Radio and Television, concerning possible clashes between the regulations of these agencies and the competition legislation. This is an encouraging sign as experience in other OECD countries, such as Spain and Australia, has shown how an independent and proactive Competition Authority can play a very important role in promoting economic efficiency more broadly.

A distorted agricultural sector

The agricultural sector is particularly important in Turkey. It still contributes around 15 per cent to GDP and accounts for around 42 per cent of all employment. OECD estimates of total support to agriculture were 10.7 per cent of GDP in 1998, compared with an OECD average of 1.4 per cent (Figure 23). Furthermore total support for agriculture has more than doubled over the past 10 years, while it has been reduced in other OECD countries. These OECD estimates include the financial costs associated with one-time payments of price premia for cotton, which have grown steadily since 1995 reaching 30 per cent of total support and 3.2 per cent of GDP by 1998. These financial costs were mainly the result of high inflation and interest rates on unpaid debt owed by the Treasury to the Ziraat Bank for premia payments to producers. Producer support estimates (PSEs), which measure the value of transfers from consumers and taxpayers to support agricultural producers, rose to 39 per cent in 1998 while the consumer support estimate (CSE), which measures the value of gross transfers from consumers of agricultural commodities, rose to –33 per cent (a negative CSE indicates a "tax" on consumers).[63] Producer receipts in 1998 were 65 per cent higher than they would have been at world prices and consumer prices were 50 per cent higher than if world prices prevailed. The OECD has extensively documented the damaging economic consequences of high levels of agricultural support and in 1998 OECD Agricultural Ministers again stressed the need to reduce market distortions. Previous *Economic Surveys* have also recommended reforms in agriculture.

Turkey has a complex web of agricultural support measures, delivered to producers through two main channels: output-based market support schemes, and the state-owned Ziraat Bank which delivers payments based on use of inputs (such as credit or fertilisers). Additional measures cover the cost of general services provided to agriculture and financed through ministerial budgets. Because of the way different support measures apply, the amount of support and the variation between different activities is quite unpredictable from year to year. This unpredictability makes it extremely difficult for producers to make reasonable and sound decisions about future production. (It also complicates fiscal stabilisation, as discussed in Chapter II). The largest share of benefits from price support programmes also goes to larger farmers, while close to 40 per cent of fertiliser subsidies are estimated to go to the largest 5 per cent of farms. At the

Figure 23. **Agricultural support measures**



1. Per cent of GDP.
2. Per cent of value of consumption (valued at farm gate).
3. Per cent of value of production (valued at farm gate).

Source: OECD, Directorate of Food, Agriculture and Fisheries.

same time, consumers have to pay higher prices for food (as indicated by CSEs), and this burden falls disproportionately on the low and middle-income consumers, including non-farmers in rural areas.

In 1998 OECD Agriculture Ministers agreed on a set of shared goals, policy principles and operational criteria for designing and implementing cost-effective policy measures (see Box 7). Turkey's current agricultural policies do not reflect these goals and principles and impose a heavy burden on the economy. Other OECD countries have already made progress towards refocusing the objectives of agricultural support and decoupling support from production, while at the same time reducing the three common measures of agricultural support: PSEs, CSEs, and total support estimates (TSEs). Turkey therefore needs to catch up with the OECD trend. The most effective long-term reform consistent with OECD goals and principles would be to completely replace Turkey's current agricultural support arrangements with a single transparent direct income scheme targeted on small farmers.

Current agricultural support arrangements could not be replaced overnight, and the Turkish authorities are working with the World Bank on projects to make possible a future shift to a direct income payments system. But there are a number of immediate measures that could be taken. One measure that could be taken immediately is to wind down the price support schemes that operate for cereals, tobacco, and sugar-beet. The government sets support buying prices for these by decree and they are well above world prices due to the high price levels of past years. The announced support buying price for sugar beet was increased by 50 per cent, tobacco by 60 per cent and cereals by 60 per cent compared with 1997, in line with the target inflation rate. In 1998, market price support alone amounted to around 5 per cent of GDP, as grains were purchased and later exported at a loss. Other OECD country experiences with price support schemes and the resulting stockpiles of over-produced and over-priced commodities have demonstrated clearly the shortcomings of such schemes. In addition, these schemes need large government or quasi-government agencies to manage them and import protection (especially tariffs) to maintain them. To get the full economic benefits of winding down the support schemes, tariffs should be reduced from the present very high levels at the same time, so that producers and consumers do actually receive and benefit from world price signals. If the Turkish authorities wish to maintain some type of price support on a few products in order to avoid extremely low prices (by world standards), this would best be achieved by setting up transparent and rule-based floor prices.

Another measure would be to "decouple" fertiliser subsidies from current use and prices by paying farmers a compensation payment based on past fertiliser use instead.[64] A record-keeping system already exists to make this possible. Another measure would be to phase out the credit subsidies made through

Box 7. OECD shared goals and operational criteria for agricultural policy measures

At their meeting in March 1998, OECD Agricultural Ministers outlined a set of *shared goals*, stressing that the goals should be viewed as an integrated and complementary whole. There was a broad consensus that OECD Member governments should provide the appropriate framework to ensure that the agro-food sector:

- is responsive to market signals;
- is efficient, sustainable, viable and innovative, so as to provide opportunities to improve standards of living for producers;
- is further integrated into the multilateral trading system;
- provides consumers with access to adequate and reliable supplies of food, which meets their concerns, in particular with regard to safety and quality;
- contributes to the sustainable management of natural resources and the quality of the environment;
- contributes to the socio-economic development of rural areas including the generation of employment opportunities through its multifunctional characteristics, the policies for which must be transparent;
- contributes to food security at the national and global levels.

Ministers stressed that agro-food policies should seek to strengthen the intrinsic complementarities between the shared goals, thereby allowing agriculture to manifest its multifunctional character in a transparent, targeted and efficient manner. The challenge in pursuing the shared goals is to use a range of well-targeted policy measures and approaches which can ensure that the growing concerns regarding food safety, food security, environmental protection and the viability of rural areas are met in ways that maximise benefits, are most cost-efficient, and avoid distortion of production and trade.

Ministers also agreed that policy measures should seek to meet a number of *operational criteria*, which would apply in both the domestic and the international context, and should be:

- transparent, having easily identifiable policy objectives, costs, benefits and beneficiaries;
- targeted, to specific outcomes and as far as possible decoupled;
- tailored, providing transfers no greater than necessary to achieve clearly identified outcomes;
- flexible, reflecting the diversity of agricultural situations, be able to respond to changing objectives and priorities, and applicable to the time period needed for the specific outcome to be achieved;
- equitable, taking into account the effects of the distribution of support between sectors, farmers and regions.

Ziraat Bank and pay farmers a compensation payment instead. Again the records already exist to make this possible. In both cases, these compensation payments should be seen as a transition measure that should be phased out quite rapidly, especially since as noted above, current payments are not generally going to smaller or poorer farmers. Another measure would be to immediately eliminate

preferential credit arrangements enjoyed by Agricultural Sales Co-operatives (ASCUs) through Ziraat Bank and eliminate the government's direct control over the ASCUs. Some amendments were proposed to the law governing ASCUs that would have eliminated direct government control, but these amendments have not yet been passed.

Implementing these reforms should bring significant benefits relatively quickly to the Turkish consumers and taxpayers. If consumers were able to obtain food at world prices they would immediately experience a rise in effective income. At the same time the budgetary savings could be used in priority spending areas such as health and education. Loss-making state economic enterprises managing the market support schemes of the agriculture sector would no longer be necessary and could be closed, while Ziraat Bank could concentrate on developing its financial intermediation skills rather than spending its efforts administering government subsidies. And ultimately the farming sector would gain because it would receive clearer signals about what to produce and the whole sector would become more efficient. However, the costs borne by some farmers during the adjustment process would be significant and these would need to be cushioned by compensation payments, despite the high fiscal cost. A move towards greater budgetary transparency would then allow the Turkish people to have an informed debate about what level of support would be appropriate over the long run.

Reforming the State Economic Enterprises

The Turkish authorities have embarked on an ambitious privatisation programme since 1984 with a target "to minimise state involvement in the industrial and commercial activities of the economy".[65] However, despite the completed sale of 130 companies, State Economic Enterprises (SEEs) still play a significant and even dominant role in some sectors of the economy. In 1997 Turkey's 500 *Major Industrial Enterprises Survey* included 44 state enterprises employing 210 000 workers, and the four largest companies in this survey were all state owned.[66] The manufacturing sector included 33 state enterprises with 140 000 workers, and in several sub-sectors, state enterprises played a particularly significant role (Table 20). Some of these enterprises are in the hands of the Privatisation Administration, while the rest remain under the direct control and responsibility of the Treasury.[67] There was considerable slippage in the 1998 privatisation programme and a number of planned sales were postponed until 1999, mostly because of administrative delays or concerns about the state of the market. In total, the privatisation agency currently has 42 companies with more than 20 per cent government ownership on its books. It plans to fully or partially privatise 26 companies in 1999, but there is considerable risk of further slippage. While competition issues need to be resolved before monopolies or dominant

Table 20. **Weight and relative productivity of state enterprises in manufacturing**

	Number of state enterprises	Sales	Gross value added	Employees	Relative productivity[1]
		As per cent of sector totals			
Food, beverages and tobacco	4	36	47	59	0.61
Textiles, apparel, leather and footwear	3	2	2	3	0.70
Timber products and furniture	0	0	0	0	n.a.
Paper, paper products and printing	3	9	6	30	0.30
Chemicals, petroleum products, rubber and plastics	9	60	70	44	1.39
Earthern and stoneware	1	2	1	1	1.49
Basic metal	5	29	35	44	0.22
Metal goods, machinery and equipment, and professional devices	5	3	5	12	0.22
Automotive	3	2	2	16	0.12
Other manufacturing	0	0	0	0	n.a.
Total manufacturing sector	33	27	40	28	0.96

1. Productivity of State enterprises relative to private-sector enterprises, measured as gross value added per employee.

Source: *Journal of the Istanbul Chamber of Industry,* No. 390, September 1998.

players are privatised, it is also important for Turkey's future economic performance that these enterprises be privatised as rapidly as possible once a sound regulatory framework has been established. And the new Competition Authority will need to play an ongoing monitoring role, in line with its statutory responsibilities.

A key policy challenge will be to address the SEEs that remain in the Treasury's portfolio. In total across all sectors these SEEs employed around 384 000 workers in 1998 at a wage bill of around $ 4.5 billion. Included here are Tekel (tobacco and alcoholic spirits and wines manufacturing) with 37 000 employees, Turkiye Seker (sugar processing) with 20 000 employees, and TKK (hard coal company) with 20 000 employees. The chronic inefficiencies of state economic enterprises overall were analysed in the past two OECD *Economic Surveys* of Turkey, and despite the efforts of the authorities, relatively little overall improvement can be detected since then.[68] The position of TTK is of particular concern as its operating losses amount to almost 0.2 per cent of GNP, and wages are growing considerably faster than sales revenue.[69] These operating losses are higher than the wages and salaries bill for the company.[70] If this company cannot be sold, which looks highly unlikely, or made profitable, then shutting it down completely should be seriously considered. Belgium and Portugal have already shut down their hard coal production and Germany, France and Spain all have

restructuring plans that will involve further mine closures.[71] As the experience of these countries has shown, restructuring this sector can be very difficult politically and any workable restructuring, privatisation or closure plan for TTK would probably involve a gradual withdrawal of state support and well-designed policies to deal with the workers involved. But given the size of the losses, maintaining the salaries and wages of the workers during some transition period after ceasing operations would still be cheaper for the taxpayer than keeping operations going.

Infrastructure improvement

Turkey has the potential to grow very fast if the obstacles to better performance are removed. So it will be especially important to develop efficient infrastructure. Otherwise bottlenecks may develop and hamper growth. Energy, transportation and communications are key sectors here. Each of these sectors faces challenges to overcome. But Turkey also has a "latecomer's advantage" of being able to develop highly efficient infrastructure using the latest technology and to benefit from lessons learned elsewhere.

Meeting electricity demands

The Ministry of Energy forecasts that total energy demand will double over the next ten years. Electricity consumption will increase even more rapidly. The electricity sector faces two major challenges, improving the efficiency of its existing generation and distribution network and expanding supply. In both cases, the government's strategy has been to involve the private sector in an industry traditionally dominated by the state-owned generation company (TEAS) and distribution company (TEDAS). In 1997, a law was passed providing for Transfer of Operating Rights schemes. But planned projects have been held up because of concerns that they were in conflict with Article 10 of the Turkish constitution, which prohibits the granting of "concessions" to private sector operators. The Turkish authorities have endeavoured to find ways around this constitutional hurdle and contracts covering five existing thermal power plants have now been approved by the Council of State (*Danistay*) with another three contracts awaiting approval.[72] In addition, long-term electricity sales agreements have been signed with several foreign investors, who wish to construct more than 5 000 MW of additional generating capacity. Progress has also been made in selling operating rights to the electricity distribution network. The network was divided into 29 regional sectors and operating rights to around half the distribution network had been sold by the end of 1998, to a range of Turkish and international investors.[73]

The other stumbling block to private investment in energy has been the constitutional barrier to international arbitration, which is usually a standard feature for dispute settlement in contracts involving foreign investment. The Turkish constitution requires that the *Danistay*, the highest Administrative Court should have the authority to make the final ruling in any disputes with foreign investors, which has discouraged foreign investment. The government sought a "gentleman's agreement" with the *Danistay* that it would not stand in the way of energy projects requiring international arbitration, pending Parliamentary approval of the necessary constitutional changes. However, this latest effort to overcome the constitutional barrier has been rejected by the *Danistay*. It remains to be seen whether in the circumstances foreign investors will have sufficient confidence for the energy projects to go ahead.

As the International Energy Agency has noted in its work on electricity reform policy, the three crucial challenges now facing OECD countries are: developing workable competition; getting consumers to share the benefits of liberalisation; and implementing regulatory mechanisms compatible with a market system. Furthermore, competition experts dealing with network industries have stressed that regulatory bodies should be financially and politically independent; that regulatory decision-making ought to be as predictable and transparent as possible; and that regulatory systems should encourage innovation and efficient operation. OECD country experiences show that it is important to address these regulatory issues before privatisation takes place. However, this has not been the case in Turkey and in its ruling on the transfer of operating rights of the distribution sector, the Competition Authority noted some of these shortcomings. These regulatory issues have not yet been fully settled in Turkey, and they need to be addressed as quickly as possible.

An inefficient rail system

Turkey is not alone in having an inefficient public railways company, and restructuring railways has often proved difficult in many OECD countries (see Box 8). The physical state of the rail network in Turkey is very poor and TCDD, the Railways Company, is the biggest loss maker of all state economic enterprises. In 1998, losses were around 0.5 per cent of GNP.[74] A 10-year restructuring plan is being implemented with independent business units being established for passenger, freight, workshops and infrastructure. A draft Railway law and a draft contract between the government and TCDD in which public service obligations could be redefined have been prepared. These drafts would have allowed the Railways to operate more in line with commercial and economic principles and determine its tariffs freely, according to market conditions, while remaining a State Economic Enterprise. The draft law would also have allowed the

Box 8. Reforming railways: lessons from OECD experience*

The following key points were drawn from an OECD review of 16 countries:

- The performance of the rail sector has been poor.
- This can, in many countries, be attributed to a "soft budget constraint".
- The soft budget constraint is partly a result of a lack of transparency in the costs of "public service" obligations.
- Action to improve the performance of railways and to introduce competition should seek to improve the commercial incentives on railways and address the public service obligations.
- Action to address public service obligations should subject them to a careful review and eliminate where possible and increase the transparency of the costs of such obligations.
- With a few exceptions, the rail industry faces significant inter-modal competition.
- As a result, regulatory intervention should be light-handed and focused on those markets in which the dominance of rail is most important.
- In some circumstances there may be opportunities for facilities-based intra-modal competition. These opportunities could be enhanced through attention to industry structure in the reform process.
- On-track competition in rail services may be achieved by regulation to ensure access to the rail infrastructure.
- A primary decision of rail policy makers is whether to go further and to separate the ownership of the rail infrastructure from the rolling stock.
- Where sector-specific regulation must be retained (*e.g.* due to public service obligations) price caps on a basket of services are preferred over traditional price controls.

* See OECD (1998), *Railways: Structure, regulation and competition policy*, Paris.

government to license new operators for particular routes. However, the draft law has not been yet passed, so the efficiency gains from restructuring will remain limited.

Turkey faces a greater challenge than most OECD countries because it also has to deal with the poor physical state of its rail network. Most rail routes are old, single track and circuitous and only around 10 per cent of the tracks are electrified. The authorities plan to upgrade the Istanbul-Ankara line to take high-speed trains. In other countries, the main challenge facing railway restructuring is to make more efficient use of the sunk costs of a highly developed network. However in Turkey the rail network is so poor that it becomes especially important to close down low-density lines, rather than invest in upgrading them. Experience from other OECD countries indicates clearly that low-density routes are a significant drain on profitability and an impediment to achieving static and dynamic efficiency. Given the low usage rates of rail in Turkey[75] and the

Table 21. **Railway statistics**

	Length of rail network (km)			Passenger-km (millions)			Freight net tonne-km (millions)			1994		Change 1980 to 1994	
	1980	1994	Percentage change	1980	1994	Percentage change	1980	1994	Percentage change	Passenger density[2]	Freight density[3]	Passenger density[2]	Freight density[3]
Turkey[1]	**10 144**	**10 466**	**3**	**6 011**	**5 797**	**–4**	**5 167**	**8 516**	**65**	**0.55**	**0.81**	**–0.04**	**0.30**
Austria	5 843	5 636	–4	7 380	9 202	25	11 002	12 424	13	1.63	2.20	0.37	0.32
Belgium	3 971	3 396	–14	6 963	6 638	–5	8 036	8 084	1	1.95	2.38	0.20	0.36
Denmark	2 944	2 349	–20	4 314	4 834	12	1 620	2 008	24	2.06	0.85	0.59	0.30
Finland	6 096	5 880	–4	3 215	3 037	–6	8 334	9 949	19	0.52	1.69	–0.01	0.32
France	34 362	32 275	–6	54 655	58 928	8	66 370	48 733	–27	1.83	1.51	0.24	–0.42
Greece	2 479	2 474	0	1 464	1 399	–4	814	324	–60	0.57	0.13	–0.03	–0.20
Hungary	7 826	7 715	–1	13 544	8 508	–37	24 339	7 707	–68	1.10	1.00	–0.63	–2.11
Ireland	1989	1 947	–2	1 032	1 260	22	637	569	–11	0.65	0.29	0.13	–0.03
Italy	16 138	16 002	–1	39 587	48 900	24	18 384	22 564	23	3.06	1.41	0.60	0.27
Netherlands	2 760	2 795	1	8 892	14 439	62	3 468	2 830	–18	5.17	1.01	1.94	–0.24
Norway	2 760	2 795	1	2 394	2 328	–3	3 081	2 678	–13	0.83	0.96	–0.03	–0.16
Poland	24 358	24 313	0	48 109	27 610	–43	134 734	65 788	–51	1.14	2.71	–0.84	–2.83
Portugal	3 588	3 070	–14	6 077	5 149	–15	1 001	1 826	82	1.68	0.59	–0.02	0.32
Spain	15 345	12 646	–18	14 826	14 853	0	11 232	9 048	–19	1.17	0.72	0.21	–0.02
Sweden	12 006	10 275	–14	6 998	5 906	–16	16 648	18 591	12	0.57	1.81	–0.01	0.42
United Kingdom	17 983	16 866	–6	30 300	28 866	–5	17 640	14 400	–18	1.71	0.85	0.03	–0.13
Average of above countries										1.61	1.51	0.05	–0.53

1. 1995.
2. Passenger-kilometres per km of track.
3. Freight net tonne-kilometres per km of track.

Source: ECMT (1998), *Statistical Trends in Transport*, Paris.

reductions in rail traffic that most countries have experienced, it is unlikely that many of these lines could ever be made profitable. A number of other OECD countries have reduced the number of lines, closing some of those that were uneconomic (Table 21).

Public service obligations should be reassessed within a broader transport framework to see if other ways of meeting them could be more economical.[76] However, the road network itself is under considerable pressure and needs upgrading. The main challenge is to upgrade the existing network rather than build new roads, since roads were built in 1980 to tolerate a load of 8 tonnes per axle, whereas modern trucks are carrying loads of up to 13 tonnes per axle. Freight transport by road has also tripled since 1980 adding to the stress on the road network.

Communications

The communications sector has already undergone some restructuring, most notably with the separation in 1994 of the government owned postal service and telecommunications. Telecommunications were incorporated as Turk Telekom Company (TTAS), and its privatisation is underway. Already Turk Telecom has demonstrated dramatic improvements in labour productivity with the greatest increase in access lines per employee of any OECD country between 1990 and 1997, with a 20 per cent reduction in staffing levels. Lines per employee are now significantly higher than the OECD average, although wages and salaries as a per cent of revenue remains lower than the OECD average. A new legal and regulatory framework for telecommunications is also being prepared, to replace the somewhat fragmented current arrangements. Although Turk Telecom currently has a monopoly over basic telephone services, its monopoly position will be abolished by 2005, in line with Turkey's commitments under the WTO 1997 Agreement on Telecommunication Services. In any case, the market for cellular phones has also grown dramatically after two new operating licences were issued to Turkcell and Telsim, already reaching around one tenth of the number of fixed line subscribers by the end of 1997. Two additional licences are planned; one will be awarded directly to Turk Telekom while the other one will be auctioned.

The Post Office remains under the direct control of the Treasury. The postal service does not play a major role in communication services in Turkey. Each Turkish inhabitant on average received just over 20 pieces of mail each year, compared with 380 pieces per year for EU citizens and 705 pieces for each American. The postal service has a monopoly over all sealed or unsealed letters and postcards and as a SEE it has been subject to political control and prevented from operating along commercial lines. It is over-staffed and loss-making although its true financial situation is difficult to establish. The majority of OECD countries now allow various forms of private ownership and operation of post offices,

although so far only the Netherlands has completely privatised the postal service. And many postal services have been restructured as corporate entities and given the freedom to operate on a fully commercial basis. Three countries have now removed all legal monopoly elements of the postal service (Sweden, Finland and New Zealand), while many other countries have reduced the range of monopoly services. A recent OECD study concluded that all regulatory restrictions on competition in the postal sector should be eliminated.[77] Compared with OECD experience, Turkey has not yet made much progress in reforming its Post Office. But as a chronic loss-maker and one of the larger state sector employees, corporatising the Post Office as a number of other countries have done would seem to be an important first step towards a more efficient service. However, OECD experience has also shown that this can be difficult and will require persistence, because it will generally involve removing public service employment privileges and a rationalisation of post office retail outlets, both of which are likely to be resisted.

Towards more efficient financial markets

Financial intermediation is under-developed...

Financial intermediation, the process of matching savings to the most productive and efficient investment projects, does not currently work well in Turkey. There is an unusually high degree of self-financing of business investment and few homes are mortgage-financed. At the same time, it is not clear whether all the investment that is taking place is going into the most productive sectors of the Turkish economy. There are three main reasons for this. The first is macroeconomic. With the budget deficit and interest rates on government paper so high, banks have concentrated their business on lending to the government rather than making loans for housing or business investment. As discussed earlier in the Survey, households and businesses are currently putting their savings into government bonds rather than spending or investing in expanding capacity. The second reason is that state-owned banks are making loans according to government policy and administering subsidy programmes. At the end of 1997, preferential lending accounted for 56 per cent of total state bank loans. There are six state-owned banks including the Ziraat (agricultural bank), Emlak Bankasi (housing bank) and Halk Bank (bank for small and medium sized enterprises). Etibank was privatised in 1998. The role of state-owned banks is significant; state banks held around 38 per cent of banking system assets, 34 per cent of loans, and 40 per cent of deposits in October 1998. These banks are diverting credit into particular sectors at the expense of others. The third reason is that many of the larger private sector banks are in fact part of industrial group holdings and are oriented towards financing the other activities of their groups. Because of the links with other parts of the conglomerates, lending may be being channelled into sectors

that could not successfully compete for funds on an open market basis and market signals are obscured.

... and banking supervision needs to be strengthened

As other policies are reoriented, as discussed elsewhere in the *Survey*, the banking sector will have to build up its financial intermediation skills. There will also probably need to be a certain amount of rationalisation and restructuring within the banking sector. Many other OECD countries have been through banking sector upheavals, although these were usually precipitated by financial market liberalisation. Nevertheless it is clear from these experiences that the banking sector can be particularly vulnerable during a period of adjustment to a new environment. This makes it particularly important that Turkey act to strengthen banking supervision. This is widely recognised and a draft banking law has been prepared. The key provisions of this law include strengthening banking supervision and establishing an independent Banking Regulation and Supervision Board.[78] However, although the Board will have the authority to take over, restructure and resell troubled banks, the authority to licence banks and revoke licences rests with the Council of Ministers. Nonetheless, this draft banking law would significantly strengthen supervision and accounting standards and passing it into law should be a top priority.

State banks must be restructured

Another major challenge will be reforming the state banks. These banks are overstaffed and subject to political inference. In some cases they have been required to make loans to other state institutions that are recorded on their balance sheets as assets, even though there is no real counterpart except future taxpayer liability.[79] Other OECD countries have had to face major restructuring of state owned banks, which have almost invariably been among the poorest performers in the banking sector.[80] There are several key lessons from OECD experience. First, if governments wish to follow policies that provide subsidised loans, these should be made transparent and open to all banks to administer rather than channelled only through state banks. These schemes should also be designed so that the bank itself shares the risk of default and therefore lends only to those likely to repay.[81] Second, the OECD has found that privatisation of state-owned banks has produced dramatic improvements in efficiency in the sector, either through internal restructuring or through rationalisation by mergers and acquisitions.[82] Third, sorting out state owned banks is costly. In a number of OECD countries, state banks have in the past been used to "hide" some forms of government intervention from scrutiny. There is no alternative to bringing these past shortcomings into the open and dealing with them, so that a viable and credible banking operation can be sold or a non-viable institution closed down.

Labour markets and human capital

Turkey faces major long-term challenges in labour markets and the development of human capital: to raise the participation rate; improve human capital; expand the formal sector without losing flexibility; and to address social safety net issues (see Chapter III). These issues were covered in detail in the 1996 *Economic Survey*, in the context of implementing the OECD's Jobs Strategy. Turkey is fortunate in having been able to avoid many of the labour market problems experienced in other OECD countries, such as problems of wage rigidities and excessively high labour costs, or at least these problems are confined to the relatively small formal sector. Nevertheless the challenges that remain are severe and require steady and integrated policy efforts to address them.

Increasing participation

Workers in Turkey work very long hours, 47.8 hours per week on average, and it would seem plausible to estimate annual working hours at somewhere between 2 300 and 2 400 hours per year.[83] This is at least 10 per cent higher than the next highest OECD country and up to 50 per cent higher than some OECD countries. Some 5.5 million of these Turkish workers are unpaid family members (excluding housewives), mostly female and mostly working in agriculture. But Turkey has the lowest participation rates of any OECD country (Table 22). The sharp decline in urban male participation rates from 45 years onwards (see Chapter III above), however, plays a small role in the overall participation rates, because the population still has a median age of less than 25 years (Figure 24).

Male participation rates show a pattern that is close to rest of the OECD. But women have an exceptionally low participation rate of only 26 per cent. It is significantly lower than in the rest of the OECD, where female participation rates range from 40 per cent in Mexico to 80 per cent for Iceland and are 59 per cent for the OECD as a whole. For urban women, participation is even lower at 15 per cent. This does not seem to be particularly linked to fertility rates, which in any case had fallen to 2.7 in 1993,[84] although rates remained higher in rural areas. Instead, these participation patterns indicate that working women are particularly concentrated in agriculture[85] and they may face considerable difficulties or even outright discrimination in finding work in manufacturing and services (Table 23). Although there is no discrimination embodied in labour legislation, the 1998 UNDP Human Development report on Turkey concludes that gender-based discrimination in the labour market is "structural and institutionalised". With the inevitable decline in agriculture as a source of employment, addressing this problem of female participation in industry and services will be important in raising output potential.

Table 22. **Labour force participation rates**[1]

Per cent

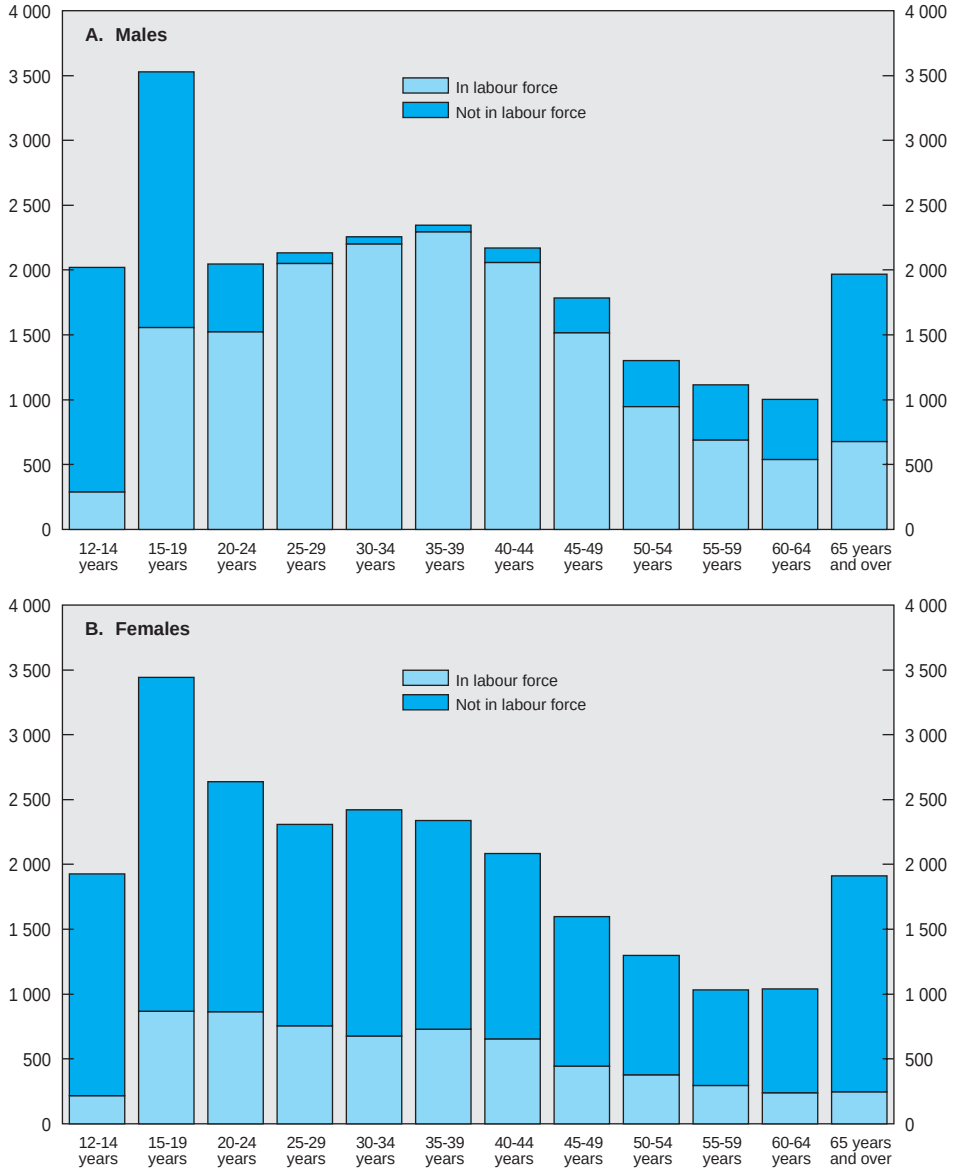
	Men	Women	Total
Turkey	**78.8**	**29.4**	**53.7**
Australia	81.9	63.0	72.5
Austria	80.0	61.8	70.9
Belgium	72.2	52.9	62.6
Canada	81.8	68.0	74.9
Czech Republic	80.5	63.7	72.1
Denmark	85.2	74.2	79.8
Finland	77.5	71.1	74.3
France	60.1	74.3	67.1
Germany	79.3	61.4	70.4
Greece	76.9	46.0	60.8
Hungary	66.6	49.2	57.8
Ireland	75.6	49.7	62.7
Italy	72.2	43.6	57.7
Japan	85.4	59.7	72.6
Korea	78.1	52.7	65.3
Mexico	87.0	41.4	63.1
Netherlands	81.4	61.3	71.5
New Zealand	78.5	62.6	70.5
Norway	85.4	75.6	80.6
Poland	73.2	59.9	66.4
Portugal	76.7	60.3	68.2
Spain	76.0	48.0	61.9
Sweden	79.1	74.5	76.8
Switzerland	89.8	72.7	81.5
United Kingdom	84.4	68.0	76.2
United States	84.2	70.7	77.4
European Union	77.6	57.7	67.7
Total OECD	81.0	58.5	69.7

1. In 1997.
Source: OECD (1998), *Employment Outlook*, Paris.

Enhancing human capital development

Labour productivity, measured by annual output per worker, is quite low in Turkey, at around $ 17 905 (calculated using PPPs) per worker. Since Turks work more hours than elsewhere, the gap in hourly productivity rates is even larger. Human capital development plays a major role here. Turkey currently devotes a smaller proportion of GDP to primary and secondary education than any other OECD country, despite its higher proportion of school age children (Table 24). And the amount spent actually fell by half a percentage point between 1990 and

Figure 24. **Work-force participation**
Thousands of persons, April 1998



Source: State Institute of Statistics (1998), *Household Labour Force Survey Results,* April.

Table 23. **Men and women in different sectors of activity**[1]

Thousands

	Men	Women	Total
Agriculture, forestry, hunting and fishing	4 896	4 135	9 030
Mining and quarrying	135	3	138
Manufacturing	2 772	610	3 382
Electricity, gas and water	116	7	123
Construction	1 215	21	1 235
Wholesale and retail trade, restaurants and hotels	2 640	238	2 878
Transportation, communication and storage	905	58	962
Finance, insurance, real estate and business services	361	146	507
Community, social and personal services	2 242	733	2 975

1. Population of 12 years and over.
Source: State Institute of Statistics (1998), *Household Labour Force Survey Results,* April.

Table 24. **Public investment in education**[1]

Percentage of GDP

	Primary and secondary	Tertiary	Total
Turkey	**2.1**	**0.7**	**2.8**
Australia	3.2	1.2	4.5
Austria	3.8	0.9	5.3
Canada	4.0	1.5	5.8
Czech Republic	3.4	0.7	4.8
Denmark	4.2	1.7	6.5
Finland	4.2	1.3	6.6
France	4.1	1.0	5.8
Germany	2.9	1.0	4.5
Greece	2.8	0.8	3.7
Hungary	3.3	0.8	4.9
Ireland	3.3	0.9	4.7
Italy	3.2	0.7	4.5
Japan	2.8	0.4	3.6
Korea	3.0	0.3	3.6
Mexico	3.4	0.8	4.6
Netherlands	3.0	1.1	4.6
New Zealand	3.8	1.1	5.3
Norway	4.1	1.5	6.8
Poland	3.3	0.8	5.2
Portugal	4.1	1.0	5.4
Spain	3.5	0.8	4.8
Sweden	4.4	1.6	6.6
Switzerland	4.1	1.1	5.5
United Kingdom	3.8	0.7	4.6
United States	3.5	1.1	5.0
Total OECD	3.4	0.9	4.7

1. In 1995. Public expenditure of educational institutions.
Source: OECD (1998), *Education at a Glance*, Paris; Turkish authorities.

1995. In absolute terms, spending per pupil is significantly lower at $ 710 per primary pupil and $ 510 per secondary pupil compared with OECD averages of $ 3 310 and $ 4 340, respectively. Turkey still has relatively high illiteracy rates, even among young adults, making improving basic education the key priority in efforts to raise human capital.

A key decision was taken in 1997 to increase compulsory schooling from 5 years to 8 years. Progress towards achieving this goal has been impressive. The strategy has been to start by increasing school attendance, followed immediately by a rapid programme of constructing the necessary additional school buildings and providing sufficient additional teachers. Inevitably, bottlenecks have occurred and are most evident in the current high teacher/pupil ratio of 1 teacher to around 34 pupils. The authorities aim to reduce this ratio to 1 to 30 by 2001, which will be a major improvement, although even this is still considerably higher than the OECD average of 1 teacher per 18.2 primary pupils. And it will be important to raise the quality of compulsory education, especially to develop reading recovery programmes to identify and provide remedial assistance to those students who fall behind. OECD country experience indicates that greatest attention has to be focused on the early years of education, ensuring that all children get a solid grounding in basic skills. However, even in within the compulsory education age bracket, enrolment rates are less than 90 per cent.[86] Some one million children in the age group 6-14 years are employed, and one-third of these are estimated to be less than 12 years old, as was discussed in the 1996 *Economic Survey*. It will be important to ensure that the children of lowest income families attend school, since for their families the immediate opportunity cost of sending their children to school is particularly high, even though education offers the greatest long-term opportunity for improvement in their living standards.

Table 25. **Adult illiteracy rates in selected OECD countries**

Per cent of population in 1995

	Total	Male	Female
Turkey (1998)	**15.4**	**6.9**	**23.9**
Greece	3.3	1.7	4.7
Hungary	0.8	0.7	1.0
Italy	1.9	1.4	2.4
Korea	2.0	0.7	3.3
Mexico	10.4	8.2	12.6
Portugal	10.4	7.5	13.0
Spain	2.9	1.8	3.9

Source: UN, *United Nations Statistical Yearbook*, 42nd edition; Turkish authorities.

Adult retraining is also important, including dealing with the high illiteracy rate even among relatively young cohorts of working age (Table 25). Around 18 per cent of Turkish adults (aged 15 years or over) are illiterate, which is much higher than other OECD countries, despite the generally younger population. And as recently as 1990, almost 10 per cent of those aged 15-19 years were illiterate. Tackling illiteracy amongst this group is an especially high priority, because providing them with reading skills will have a pay-off throughout the rest of their working lives, especially as the demands created by rapidly-changing technologies means that those adults with the weakest literacy skills will struggle most to keep pace.[87]

Integrating the informal sector

As already discussed above, the Turkish labour market is sharply divided into a formal and informal sector. As noted in the 1996 *Economic Survey*, it seems reasonable to assume that wages and employment conditions in the informal sector are very flexible and these conditions are acceptable to both workers and employers. Informal employment offers obvious advantages to employers (lower labour costs by avoiding tax and social security charges) and employees (access to a job that matches their productivity level). As noted above, the advantages of informal work exist not only because of high contribution and tax rates, but also because enforcement seems to be very weak. Indeed in Turkey, there is not only informal employment, but also probably a significant under-reporting of wage income. Notwithstanding these advantages, the informal sector has two major disadvantages. First, it places an unbearable strain on the public sector revenues, which must be redressed (see above). Second, it probably means that there is an under-investment in human capital within the firm and a lack of basic social protections and work-place (safety) regulations for those outside the social security net. Nevertheless, while it is essential for the long-term health of the Turkish economy to bring this informal sector into the tax and contributions net, it also needs to be recognised that this will have some adjustment costs. Either the workers will face a reduction in take home pay or employers will find that their competitive advantage is eroded. At the same time, the dynamism and flexibility that characterises the informal sector should not be lost in the process.

In the formal sector, wage setting is dominated by the state sector, with more than 60 per cent of the workers covered by collective agreements between 1989 and 1994 being employed in SEEs. In addition, the State sector would seem to be more diligent in adhering to labour market regulations. As noted in the 1996 *Economic Survey*, the main feature of formal labour markets is that it is dominated by the Turkish government, which does not face a binding budget constraint. Another feature of formal labour markets that was discussed in the same survey is high severance payments (see Chapter III above). At present Turkey has no

unemployment insurance although it has been argued that high severance payments are a substitute for this. There have been a number of calls to set up an unemployment insurance system, and a draft law has been proposed. Unemployment insurance may contribute to the efficient allocation of labour if it promotes better job match, rather than inducing people to take the first job that comes along. However the evidence from OECD countries strongly suggests that it may also blunt job search activity if badly formulated. Any scheme should be carefully designed to minimise the adverse work and job search incentives that can easily get embodied into these schemes.

The draft law, as it stands, contains features similar to those in a number of other OECD unemployment insurance schemes. The proposed 7-day waiting period is similar to a number of other countries, while the proposed replacement rate of 45-55 per cent is lower than a number of countries. The maximum duration of 300 days is also reasonably in line with OECD practice (where duration ranges from 6 months to 84 months). Perhaps the biggest challenges that Turkey would face in implementing an unemployment insurance scheme would be ensuring compliance with active job search requirements and avoiding benefit fraud, two problems that many other OECD countries have also had to deal with.

Improving the policy-making process

Perhaps the biggest structural challenge is to get reform proposals through the political process. Turkey has competent high level bureaucrats, who are able to give good policy advice and prepare sensible reform proposals. But even when proposals seem to have widespread support, they tend to languish at the political level. One constructive measure would be to establish an independent body with a wide-ranging mandate to undertake research into the costs and benefits of different economic policies, not only at the aggregate level, but also at the level of who wins and who loses from different policy options. Several OECD countries have agencies that perform this role – the Australian Productivity Commission, the Spanish competition authority or the Economics Institutes in Germany, for example. This analysis would help to strengthen policy making by identifying more clearly whose interests are at stake and how resistance to reform could be overcome. It would also help policy advisors to target and market policy changes more effectively. This is especially important because those who benefit from current policies most are those with the strongest political voice, while the losers under present policies may have little voice at all. Conversely those with the most to lose from policy reforms tend to be well organised and vocal, and are therefore able to resist change quite effectively.

A strengthened Economic and Social Council might be a useful forum for building a consensus for change. Care would have to be taken to ensure that all

Box 9. Recommendations for further structural reform

Making product markets function better

Agriculture

In the long term:

- Replace agricultural support programmes with a transparent direct income scheme targeted on small farmers.
- Reduce overall support for agriculture and direct the savings towards improving human capital.

In the short term:

- Decouple fertiliser and credit subsidies from current production and instead pay farmers a compensation payment based on previous usage.
- Eliminate preferential credit arrangements for Agricultural Sales Co-operatives.
- Wind down price support schemes.

State Economic Enterprises

- Maintain momentum on privatisation of those enterprises already in the hands of the Privatisation Administration.
- Re-examine the State Economic Enterprises that remain in the hands of the Treasury and either:
  - Include them in the privatisation programme, or
  - Clarify the commercial and social objectives of these enterprises and give managers the necessary incentives and scope to operate along commercial lines, including a hard budget constraint. Evaluate whether keeping operations going is the most cost-effective and efficient way of meeting social objectives. Consider closing down TTK, if it cannot be made economic at world price levels.

Improving infrastructure

Energy

- Remove constitutional obstacles to private sector participation in the energy sector so that:
  - private companies can obtain concessions to operate power generation plants and the distribution network, and
  - contracts with foreign private companies can be taken to international arbitration.
- Ensure that regulatory issues are properly settled before transfer of operating rights takes place and that the regulatory framework to be applied is transparent and predictable.

Railways

- Close down uneconomic and low-density rail lines.
- Critically evaluate "public service" elements of the railways service and examine whether other forms of transport could deliver these services more efficiently.

(continued on next page)

(continued)

– Pass legislation that specifies commercial and social objectives for the railways and ensure that managers are given both the scope and the incentive to perform efficiently.

Communications

– Get the new legal and regulatory framework for telecommunications established.
– Eliminate all regulatory restrictions on competition in postal services.
– Corporatise the Post Office, ensuring that managers are given both the scope and the incentive to improve efficiency.

Making financial markets function better

– Pass legislation that would provide for a strong and independent Banking Regulation and Supervision Board.
– Restructure and privatise the state-owned banks, after making transparent the cost of having used these banks to serve political goals and transferring these costs explicitly to the taxpayer.

Improving labour markets and human capital

– Identify and remove sources of gender discrimination in the work place and encourage greater workforce participation by urban women.
– Target programmes on the rural poor to ensure full compliance with compulsory schooling, especially for girls.
– Develop programmes to improve the quality of compulsory education, especially in developing remedial reading programmes to reduce illiteracy rates among school leavers.
– Allocate more public resources to elementary education.
– Develop adult education programmes to reduce illiteracy, especially among young adults.
– Integrate the informal sector into the formal economy, without sacrificing labour-market flexibility.

Improving the policy-making process

– Establish an independent body, with a wide ranging mandate, to undertake research into the costs and benefits of different economic policies, including analysis of winners and losers under different policy options.
– Strengthen the Economic and Social Council to provide a forum for building consensus for change.
– Build evaluation and adjustment mechanisms into the policy reform process.

stakeholders in the economy were properly represented and that the strategy of reform took income distribution aims into account. New Zealand, for example, found this consensus-building approach helpful when it undertook comprehensive reform in the mid-1980s. And comprehensive reform packages, where everyone shares the pain of the adjustment process, can be more effective than piecemeal reform, which can place a particularly heavy burden on some people,

especially as they may have to go through more than one adjustment as each reform is undertaken. In addition, building the scope for evaluation and adjustment into the policy reform process provides people with more reassurance and makes them more willing to try alternatives. Whatever approach is adopted, Turkey needs to move from proposals to action. A summary of OECD recommendations provided in this Chapter is given in Box 9.

Notes

1. A parent bank which is established in Turkey is required to consolidate the statements of the financial institutions that are members of the financial institutions group no matter whether they are institutions providing different financial services or are operating in different fields of expertise or have been established abroad.
2. An SMP is an informal and flexible instrument for promoting economic dialogue which is not supported by the use of the Fund's financial resources.
3. Inflation, combined with lags in tax collection, implies that the real value of tax collection declines (*Oliveira-Tanzi* effect), though this may be partially offset by lags in government payments for goods and services.
4. The sectoral breakdown of public-sector employment is the following: education 28.9 per cent, health 8.7 per cent, defence 7.4 per cent, police 6.7 per cent, other (including SEEs and civil servants) 48.2 per cent.
5. This interest rate was equal to the commercial interest rate plus a 20 per cent bonus since 1993. The extra-bonus of 20 per cent was abolished in 1998.
6. See OECD (1998), *Agricultural Policies in OECD Countries*, Paris. These calculations only refer to direct public support to agriculture (see Chapter IV below).
7. Under a special arrangement with the Treasury, Ziraat is also entitled to receive interest payments at the market commercial rate for an unsettled reimbursement owed since 1993 to cover cotton-related losses.
8. This measure of the real rate of interest refers to the average compounded yield across different maturities of Treasury bills.
9. If, as in the last few years, external interest payments were to remain close to 1.0 per cent of GNP, the corresponding conventionally-defined primary surplus would amount to 4.5 per cent of GNP, *i.e.* broadly the same level as in 1998.
10. For a wider discussion, see James H. Chan Lee (1998), "A decade of tax reform in Turkey", *Yapi Kredi Review*, November.
11. This commitment was announced in a protocol signed jointly by the Central Bank and the Treasury in July 1997. Until the eruption of the 1994 financial crises, the Treasury was entitled to overdraw from the Central Bank up to 15 per cent of the central government budget appropriation. After the crisis this amount was changed to 12 per cent of the increase in the central government budgetary expenditure relative to the previous year. This ratio was reduced gradually to 10 per cent in 1996, 6 per cent in 1997 and 3 per cent in 1998.
12. The tax is still, of course, insufficient to cover all such extra-budgetary lending activities.

13. According to the OECD, social expenditures cover public and mandatory private spending in the following thirteen policy areas: old age cash benefits; disability cash benefits; occupational injury and disease; sickness benefits; services for the elderly and disabled people; survivors' benefits; family cash benefits; family services; active labour-market programmes; unemployment; public expenditure on health; housing; other categories.

14. See World Bank (1994), *Averting the Old Age Crisis: Policies to Protect the Old and Promote Growth*, World Bank and Oxford University Press, which shows that for part of the 1980s Turkey had one of the lowest returns on public pension fund investments of a wide range of countries.

15. In SSK, these interest rates were raised in 1995 to 15 per cent monthly, with an immediate improvement in compliance rates, except for state economic enterprises, which are eligible for periodic transfers from Treasury to cover payments arrears (conditional on the requisite legislation being passed). Late payment interest-rate penalties are still too low in BK (5 per cent monthly), and explain its low contribution collection record (see Box 2).

16. The last contribution amnesty was legislated in May 1997. For private-sector employees (SSK) and the self-employed (BK), all of the unpaid contributions and arrears and 50 per cent of the late payment penalty up to the end of December 1996 were to be erased, together with 42 per cent of interest charges owing to these arrears up to May 1997. The contribution debt of the public sector (including state economic enterprises and municipalities) until June 1997 was paid by Treasury from the transfers of the budgetary appropriation for SSK.

17. See OECD (1998), *Maintaining Prosperity in an Ageing Society*, Paris, Figure 1.2.

18. They do not appear to be out of line with regressions which relate spending to GDP. See Oxley, H. and M. Macfarlan (1995), "Health Care Reform: Controlling Spending and Increasing Economic Efficiency", *OECD Economic Studies No. 24*.

19. These rights are awarded after a minimum period of employment but often rise steeply after 5 years, and can reach the amount of a month's earnings multiplied by years of employment. The average liability under collective agreements was estimated in 1998 at $ 8 100 (8.5 months of average 1998 earnings) per worker employed. See *Turkiye Isveren Sendikalari Konfederasyonu, Genel Kurul Calisma Raporu* (Yearly Activity Report of Turkish Employers' Confederation, 1998), Ankara, December.

20. Health Insurance Commission of Australia (1996), *Health Financing Policy Options Study, Turkey*, Ankara.

21. The weighted average of 1998 average benefits in each of the three funds shown in Box 2 is 39 per cent of the average wage, of which SYZ (the flat-benefit component) accounts for 6 percentage points. As SYZ is now frozen in nominal terms, its importance will decline in future years, implying a basic average benefit in the range of 35 per cent of the average wage.

22. Health Insurance Commission of Australia (1996), *op. cit.*

23. Undersecretariat of Treasury estimates, abstracting from the health accounts (see Figure 21 below). Given recent trends in health expenditures in Turkey and the experience of other countries in the area of health spending, these projections (like those of the ILO; see Annex III) probably understate the problem.

24. Pay-as-you-go insurance systems require a balance between revenues and expenditures for each given period/year: there is a probabilistic balance between yearly

contributions by the active insured and yearly expenses of retirees and sick people. In funded insurance schemes, there must be intertemporal (actuarial) balance between the lifetime contributions and lifetime benefits of each age cohort; between their times of activity and good health, and their times of retirement and bad health.

25. See OECD (1998), *op. cit.*

26. This strategy is also broadly consistent with a major World Bank report on ageing (World Bank, 1994, *op. cit.*), which advised separation of the three key functions of pension systems – insurance, redistribution and savings – into separate pillars each with its own financing and managerial arrangements.

27. These measures included: mandating increased inspection staff and actual work place inspection; eliminating the possibility for deferring interest on contribution debts and deducting unpaid premiums from the tax base; increasing fines on late payment and increasing interest on contribution debt to 12 per cent per month in SSK; introducing personal liability for employers and their accountants following an amnesty for private sector premium arrears to SSK and BK; and deduction of SSK contributions from municipal authorities' share of centrally-collected revenues. At the same time, an amnesty extended to agricultural members allowed them to deduct outstanding premiums by up to 1 per cent of sales revenue.

28. Life expectancy at age 60 in Turkey is 77 years for men and 79 years for women. At age 65, it rises to 79 and 80 years, respectively. Life expectancy at birth, however, is much less, reflecting high infant mortality rates. See Kenar, N., Teksöz, T., and H. Coskun (1996), "The Social Protection System and its Reform", Undersecretariat of Treasury, *Proceedings of the International Conference on Social Security and Health Insurance Reform*, Ankara.

29. However, CPI indexing would limit the scope for intergenerational risk sharing. For example, in a prolonged recession, declining per capital incomes would lag behind constant real pensions; conversely, a more likely outcome (if the policy environment is good) is that pensions lag increasingly behind wages. This might justify the proposals in the government's previous draft law allowing for discretion in setting the index.

30. The contribution base is used to calculate the benefit. The pension fund for civil servants (ES) has no contribution ceiling. However, wages are relatively low in the public sector and generous pension benefits may be a way of attracting qualified workers into the public service and assuring their loyalty.

31. Research in other countries has found that in a lifetime sense, social insurance tends to be regressive insofar as there is a strong positive correlation between life expectancy and income level.

32. While a common retirement age for men and women is the preferred recommendation (above), a more politically realistic reform is likely to include a small continuing advantage in favour of women (in recognition of a still-low female participation rate).

33. In a 4-year transition, the current minimum retirement age of 38/43 for women/men would be increased by 5 months for women and 4¼ months for men for each month remaining for them to qualify for retirement under previous arrangements. Those with more than 4 years to retirement would immediately jump to the new minimum retirement age of 58/60. Clearly, this does not leave much time for adjustment.

34. The steady state rate of return in a pay-as-you-go system equals the economy's potential growth rate. See Tirole J. (1985), "Asset Bubbles and Overlapping Generations", *Econometrica* 53, No. 6.

35. The calculations assume fixed real pension benefits. In principle, current formulae for indexing post-retirement benefits could allow for real benefit increases in line with the growth of civil servants' wages. However, in practice civil servants' wages, and hence post-retirement pension benefits, have both increased and decreased in real terms, with perhaps little real gain over time. It is also assumed that the current freeze on nominal SYZ payments (the flat benefit portion) is continued indefinitely.

36. In 1998, average collections in SSK were 11 per cent of the contribution wage, compared with the 20 per cent statutory rate. Collections could never be as high as 20 per cent, however, insofar as some people earn less than the contribution wage, so that the true collection gap was less than 9 per cent. (Nonetheless, with a normal distribution of income, contributions should be closer to the 20 per cent ceiling than to the 11 per cent outcome). The implied shortfall requiring a subsidy from the government would therefore be less than 9 per cent of the contribution wage.

37. In the 1981 Chilean pension reform, 32 social security funds were consolidated into one fund with common contribution and benefit rules.

38. It may also be worth considering reforms that increase the incentives to contribute by individuals. This would in part be already achieved by the above parametric reforms, for example, by equalising the pension benefit accrual rate across different working periods. Evasion might also be reduced by improving the transparency of the system, for example, by developing a system of "notional accounts", as in Sweden and Latvia. This mimics the structure of a funded retirement savings account by clearly calculating the pension rate of return and incremental accrual of pension benefits, and providing clear rules for indexing of post-retirement benefits.

39. A slow adjustment may be advisable, however, since many labour-intensive firms are profitable only because they are exempt from paying social security taxes, given lax enforcement. Bringing these under the tax umbrella will imply significant cost pressures in the non-registered sector.

40. The unfunded liability is the extent to which current benefit promises exceed the assets on hand to pay for them. The debt related to the maturing of the system reflects the lack of pre-funding in the early stages of the programme which often is related to paying earlier pensioners more than their accumulated pension rights, allowing them to receive markedly higher than market rates of return. (In the United States, for example, to service this debt, *all* successive generations will have to accept a lower than market rate of return, equivalent to a permanent 25 per cent tax on their pension contributions). While this debt is a sunk cost, there is still time to pre-fund much of the debt associated with the ageing of current contributors.

41. As noted above, another way of increasing retirement income would be to increase significantly the relatively low contribution ceiling on PAYG contributions.

42. It can be shown that a pension system paying financial returns below market interest rates implies very high marginal tax rates on earnings from formal or dependent employment which induces an increase in independent employment or in informal or illegal employment relationships. Assuming a real annual interest rate of 4 per cent and a pay-as-you-go return in a mature (non-redistributive) system of 1½ per cent, the marginal tax rate on pension contributions made at age 20 is 70 per cent and 20 per cent at age 60. See Valdès Prieto, S. (ed.), *The Economics of Pensions*, Cambridge University Press, 1997.

43. Empirical evidence for Chile suggests that development of a major private funded pillar under the 1981 pension reform accelerated financial market development and

thus efficiency of resource allocation, enabling an estimated 1 per cent annual gain in total factor productivity, while also affecting positively national saving and capital formation, raising the capital stock by some 5 per cent in the long term. See Holzmann, R. (1996), "Pension Reform, Financial Market Development, and Economic Growth", *IMF Working Paper* WP/96/94.

44. Pre-funding of public pensions would also increase national savings, as investing any pension fund surpluses rather than mixing them with the general government accounts makes the true public sector deficit more transparent, improving incentives in the public sector.

45. Although voluntary schemes might imply fewer behavioural distortions, in that they are not forced savings, they could increase the state's future assistance burden insofar as savers are either myopic or save little on the expectation of public assistance. But in addition, if subsidised by tax credits (as IRAs in the United States), they might leave poor people with little coverage while mainly benefiting the better-off.

46. A gross market real interest rate of this magnitude would reflect a relatively robust long-run rate growth which is needed in Turkey to complete the catch-up process and which would be unleashed by the structural reforms discussed in Chapter IV.

47. The average management fee charged by the Magellan Fund in the United States and that observed in Chile after the privatisation reforms is higher (1.7 per cent; see Valdès Prieto, 1997, *op. cit.*). However, these funds also include equities which generate higher returns.

48. A one percentage point drop in the net yield, to around 6 per cent, would require a contribution rate of 5 per cent, while an increase to 8 per cent would allow a contribution rate of 3 per cent.

49. The United States government has proposed a scheme of pension accounts which would provide subsidies to top up the savings contributions of lower income people. See OECD (1999), *Economic Survey of the United States*, Paris.

50. Such schemes can either be defined contributions where investment risks are borne by individuals, as above, or defined benefit, where risks are borne by employers. Employers' contributions to such schemes would constitute an added element of compensation, though they would be to some extent offset by reductions in the basic wage.

51. See Demirgüç-Kunt, A. (1996), "Turkish Security Markets", World Bank Policy and Research Department, mimeo.

52. A US Social Security Technical Panel found in 1995 that if the US Treasury were to issue indexed bonds, this would probably facilitate private sector issuance of indexed annuities. See Mitchell, O. and S.P. Zeldes (1996), "Social Security Privatization: A Structure for Analysis", *NBER Working Paper* No. 5512.

53. OECD calculations suggest that this increase could cost ½ per cent of GDP for those individuals currently in the scheme, and considerably more insofar as more people would fall under the higher threshold.

54. The actual number of people without effective SSO health coverage is difficult to estimate. The Ministry of Health estimates this number at around 18 million subsequent to the extension of BK health insurance to rural area members. This estimation assumes that the number of "dependants" of SSO members may in practice be smaller than that estimated by the SSOs themselves (see Figure 16 above).

55. Health Insurance Commission of Australia (1996), *op. cit.*

56. The $ 18.5 million Primary Health Care Services Project is co-financed by the World Bank (78 per cent) and the Turkish Government. The Project has three objectives: to develop a pilot primary health care system based on the family physician and an effective referral system supported by improved infrastructure; to evaluate the system thus developed for nation-wide replication; and to develop capacity for economic analysis within the Ministry of Health. The two provinces chosen are Eskisehir and Bilecik.

57. Otherwise known as Diagnostic-Related Group (DRG), case-mix is a way of measuring hospital output. Under this system, each treatment available in a hospital is classified to a DRG on the basis of its average expected cost of treatment.

58. In the latest period for which official regional data were available (1985/87), infant mortality rates were 44 per thousand in the west, and 96 and 103 per thousand in the southern and eastern regions, respectively.

59. M. Oktay (1996), *Turkish Business Life via the Eyes of Foreign Businessmen*, Istanbul Chamber of Industry Customs Union Information Office.

60. The poorest 20 per cent of households received less than 5 per cent of all income, while the richest 20 per cent of households received almost 55 per cent of all income. Reflecting this, the Gini coefficient rose from 0.44 to 0.49. See TESEV (Turkish Economic and Social Studies Foundation)/UNDP (1998), *Turkey Human Development Report*, Ankara.

61. Less than half of all households had access to safe drinking water in 1995. In urban areas, some 63 per cent now have access, while in rural areas it is only 28 per cent (*Turkey Multiple Indicator Cluster Survey*, 1995).

62. Turkey has until 2001 to fully implement the Customs Union Decision.

63. It should be noted that the PSEs are calculated for a standard set of 13 common commodities across OECD countries. These commodities form only 42 per cent of agricultural production in Turkey, the lowest of OECD countries. Market price support on commodities such as tea, tobacco, apples and tomatoes are omitted from these estimates.

64. The same approach could also be adopted for the subsidies currently paid on pesticide use.

65. The other stated targets of the privatisation programme are to: provide legal and structural environment for free enterprise to operate; to decrease the financial burden of State Economic Enterprises on the national budget; to transfer privatisation revenues to the major infrastructure projects; to expand and deepen the existing capital market by promoting wider share ownership; and to provide efficient allocation of resources.

66. As measured by sales from production. They were TUPRAS, TEAS, TEKEL and *Eregli Demir Celik Fabrikalari A.S.* See *Journal of the Istanbul Chamber of Industry*, No. 390, September 1998.

67. Once the government has decided to privatise a State Economic Enterprise, the enterprise is passed to the Privatisation Agency for management of the privatisation process. See *http://www.oib.gov.tr/3eng.htm* for further information on the Privatisation Agency, its responsibilities and its activities. Where SEEs remain under the direct control of the Treasury, there are either no intentions to sell them or no agreement reached yet.

68. The ratio of wages and salaries to sales of goods and services provides a crude measure of labour productivity. For the SEEs governed by code 233, labour productivity fell by 1.1 per cent in 1996, rose by 4 per cent in 1998 and is projected to rise by 0.3 per cent in 1999.
69. Much of the output of TKK goes to ISDEMIR, a state owned flat steel producer, which is in the Privatisation Agency's portfolio of companies. ISDEMIR also made significant losses in 1997 and the first half of 1998.
70. In 1996, the operating losses were 1.9 times higher than wages and salaries, while in 1997 and in 1998 they were 1.6 times higher. Projections indicate that losses may be again around 1.5 times higher than wages and salaries in 1999.
71. Less than 10 per cent of hard coal production in IEA countries receives any state assistance.
72. These sales complete the government's tender for the transfer of eight lignite-fired power plants. These plants have a total installed capacity of 4 910 MW and the sales are expected to generate $ 1.3 billion. The operating rights are transferred for 20 years.
73. These transactions were projected to raise $ 1.7 billion.
74. Including the amount covered by budgetary transfers, which amounted to 0.2 per cent of GNP.
75. In 1997, 94 per cent of inland freight traffic and 97 per cent of surface passenger traffic (*i.e.* excluding air) went by road.
76. In some cases, other countries have found that it would be cheaper to meet public service requirements by providing free taxi services than by continuing to operate manifestly uneconomic passenger lines.
77. See OECD (1999), *Promoting Competition in Postal Services*, Paris, forthcoming.
78. Banking supervision is currently shared between the Central bank and the Treasury.
79. However, this future taxpayer liability is not currently recorded anywhere in the government's financial statements.
80. These traditionally state-owned banks need to be viewed differently from weak private sector banks that had been taken over by the state as a result of financial crises. In the latter case, the state's role has been to restructure the banks and prepare them for return to the private sector or for closure.
81. A further alternative would be to consider administering such subsidies through non-bank channels, such as the tax system.
82. See OECD (1998), *Financial Market Trends* No. 70, July, "Privatisation: Recent Trends". Mergers and acquisitions within the banking sector are currently governed by both the Competition Authority and the present Banking Law which predates the Competition Authority. Once the new banking law is passed, sole responsibility will rest with the Competition Authority.
83. Assuming 48 to 50 working weeks per year. It is difficult to obtain reliable data on holidays, because so much of this work is informal.
84. In other words, each woman of childbearing age produces on average 2.7 children.
85. In 1955, 77 per cent of working age women were in the labour force, and by 1989, it had fallen to 35 per cent. Urbanisation seems to have played a large role in this, because of the high use of female labour in the agriculture sector.

86. These overall enrolment figures obscure significant regional and gender differences. The lowest levels were recorded in east Anatolia, where only 63 per cent of girls were enrolled.

87. See OECD (1999), *Overcoming Exclusion through Adult Learning*, Paris.

Annex I

Quantifying output gains of disinflation

The calculations have been obtained using the INTERLINK model. As a benchmark, it has been assumed that Turkish consumers and business investors behave in a way consistent with the average of the OECD countries. Typically, in the standard OECD economy, a 1 percentage point decline in the real interest rate relative to the central projection would lead to a rise in activity of 0.6 points within a period of two years, assuming unchanged real exchange-rate and government spending. Calculations for Turkey suggest that if both the real interest rate and the rate of inflation fell gradually to within one digit, total aggregate output would increase relative to the baseline by a cumulative 15 per cent over a period of six years (Annex Table A1). A particularly strong positive impact would be felt by total domestic demand whose level would be 20 per cent higher than the baseline. Stronger domestic demand would stem not only from investment but also from consumption, via the job creating consequences of surging investment. The pick-up in domestic demand would entail a strong import acceleration and a worse current account (5 percentage points of GDP higher than the baseline). In level terms the current account deficit would not exceed 3 per cent, however.

Table A1. **Illustrative scenario of possible output gains from disinflation**[1]

Percentage difference from central projections

	Years					
	1	2	3	4	5	6
Private consumption	1.5	2.5	4.9	8.3	14.0	21.7
Government consumption	0.0	0.0	0.0	0.0	0.0	0.0
Gross fixed investment	1.6	3.4	6.6	10.6	16.8	24.7
Final domestic demand	**1.4**	**2.7**	**5.1**	**8.4**	**13.8**	**21.1**
Stockbuilding	0.0	0.0	0.0	0.0	0.0	0.0
Total domestic demand	**1.4**	**2.7**	**5.1**	**8.4**	**13.8**	**21.1**
Exports of goods and services	0.0	0.0	0.0	0.0	0.0	0.0
Imports of goods and services	2.6	3.5	6.6	9.3	13.0	19.2
Foreign balance	**–0.9**	**–1.2**	**–2.4**	**–3.6**	**–5.2**	**–7.9**
Real GDP level	**0.7**	**1.6**	**2.9**	**5.3**	**9.5**	**14.4**

1. Relative to the central projections, the scenario assumes implementation of a disinflation policy, including support from structural reforms in a number of critical areas as programmed by the *Memorandum of Economic Policies* of July 1998. Income policies are also assumed to assist wage and price moderation. The exchange rate is managed in such a way as to grant an overall stable real effective exchange rate over the period. In such a context, the real interest rate is assumed to fall from around 35 per cent to about 7 per cent by the end of the period.

Source: OECD.

Annex II

The dynamics of government debt accumulation

This Annex presents the methodology used to assess the exposure of Turkey to adverse debt dynamics. Changes in total government liabilities can be decomposed into three elements, *primary deficit*, *debt dynamics* and *other factors*. Thus,

$$D_t - D_{t-1} = P_t + r_t D_{t-1} + O_t \qquad (1)$$

where:

D_t = central government domestic financial liabilities;
P_t = central government primary deficit, defined as the central government domestic cash deficit excluding net interest payments;
r_t = interest rate paid on central government financial liabilities;
O_t = central government net financial liabilities not arising from financial balances and for which no corresponding flows are therefore recorded, including statistical discrepancies.

It can be shown that:

$$\frac{D_t}{Y_t} - \frac{D_{t-1}}{Y_{t-1}} = \frac{P_t}{Y_t} + \left(\frac{r_t - g_t}{1+g_t}\right)\frac{D_{t-1}}{Y_{t-1}} + \frac{O_t}{Y_t} \qquad (2)$$

where:

Y_t = Nominal GNP;
g_t = Nominal GNP growth rate.

Otherwise, drawing on a slightly modified version of (1), it can be shown that:

$$\frac{D_t}{Y_t} - \frac{D_{t-1}}{Y_{t-1}} = \frac{P_t}{Y_t} - \left(\frac{g_t}{1+g_t}\right)\frac{D_{t-1}}{Y_{t-1}} + \frac{R_t}{Y_t} + \frac{O_t}{Y_t} \qquad (3)$$

where:

R_t = Central government interest paid on debt.

The second terms in the right hand side of equations (2) and (3) are alternative versions of the *debt dynamics* term, both suggesting that pressures on debt arising from high interest rates may be offset by strong nominal GNP growth. Tables 12 and 14 in the main text were prepared using equation (2) whereas Table 13 is based on equation (3).

Annex III

The ILO pension reform proposals

Main recommendations

The 1995 ILO (International Labour Office) pension reform proposals stress the need for action on a number of fronts in order to address the crisis in the Turkish pension system. The main recommendations are as follows:

- A substantial increase in the retirement age: to 53/55 for women/men immediately, rising to 58/60 over a 10-year period. During the transition, workers would receive the right to a pension equal to the higher of that under the old or the new system.
- A large increase in insurable earnings, which in turn define contribution and benefit ceilings, from around 2 times the minimum wage currently to 3 to 5 times the minimum wage.[1]
- A stronger link between benefits and contributions via a more realistic pension benefit formula, comprising: a much lower rate of accrual of pension entitlements, 1.3 to 1.5 per cent per year versus around 4 per cent currently; average (revalued) lifetime earnings as the reference wage; and elimination of the existing bias in favour of short contribution periods, though early retirement would be permitted once the pension entitlement reaches one-quarter of average lifetime earnings.
- Major improvements in the efficiency of social security institutions, particularly in the areas of administration and enforcing compliance, and greater harmonisation among the institutions in both their benefit and contribution structures (separation of pension and health insurance, and integration of social security collection and general tax administration, are not recommended).
- A possible role for a second tier private funded pension arrangement (see below). This would help to develop financial markets in Turkey, but would require careful preparatory work in terms of correcting deficiencies in capital market regulations, introducing private pension scheme regulations, establishing a Pensions Supervisory Board and a mandatory system of reinsurance to guard against the risk of pension scheme bankruptcy.
- A system of social assistance for the elderly and disabled (but not for the entire population) to complement the pension reform. The value of the benefit should be brought up to around 35 per cent of the 1995 minimum wage (and in the longer run to nearly 90 per cent of the 1995 minimum wage), and then indexed to the rate of growth in national average wages. The scheme would initially cost about 0.6 per cent of GDP, to be financed by resources released from pension reform.

– Greater efforts to bring about a social consensus for change, notably by bringing the social partners into the dialogue and giving them a role to play in the governance of the reformed system. Also, compulsion should be associated with the freedom to enhance mandatory schemes under collective agreements as well as for individuals, and reforms should be widely perceived as adding to equity and economic performance. The process of reform thus involves a major exercise in public relations.

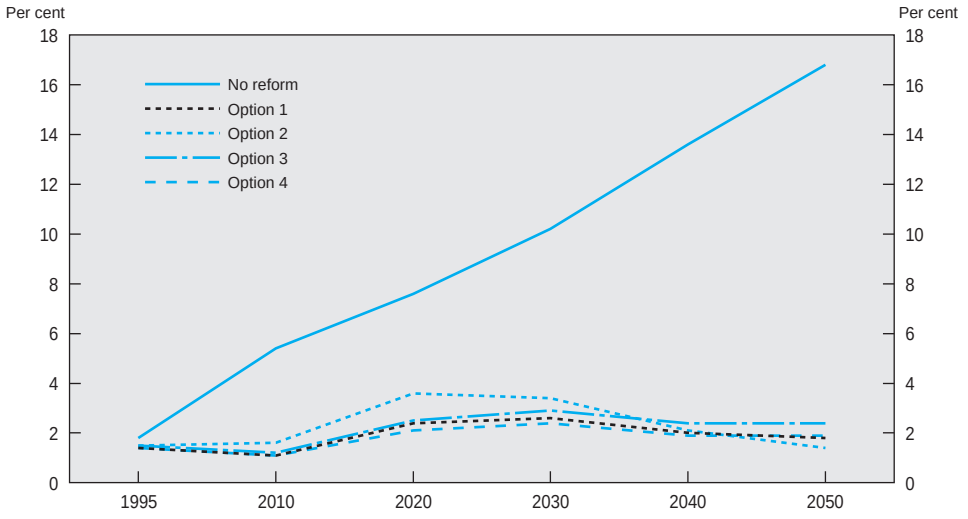
The reform options

In order to assess the possible benefits and costs of reforms, a quantitative analysis is carried out in terms of four different policy options: a pure restructured public PAYG system (Option 1); a pure system of mandatory individual savings accounts for new workers, with existing workers given the right to remain under the old PAYG system or switch to the new scheme (Option 2); a mixed system with a primary PAYG tier and a secondary tier of mandatory individual savings accounts (Option 3); and mixed system where the second tier consists of voluntary savings schemes, either occupational or individual (Option 4).[2]

The key assumptions are as follows. In all the options, the mandatory contribution rate is kept at the current statutory contribution rate of 21.5 per cent. In Option 2, this mandatory contribution rate represents 17 per cent for additions to individual savings

Figure A1. **Pension deficits under ILO reform options**

Per cent of GNP



Source: International Labour Office (1996), *Republic of Turkey, Social Security Final Report,* Geneva, March.

accounts, 4 per cent for invalidity and survivors benefit insurance, and 0.5 per cent for administrative costs. In Option 3, the rate of contribution to mandatory individual savings accounts is 5 per cent, leaving 16.5 per cent for the PAYG scheme. In Option 4, supplementary contributions to voluntary schemes are considered to be over and above the mandated 21.5 per cent for the public scheme. In Options 1, 3, and 4, moreover, the public pension contributions of self-employed farmers and agricultural labourers receive a matching contribution by the government. The PAYG pension benefit rate is set at 1.5 per cent (1.33 per cent in Option 3) of average lifetime (revalued) earnings per year of contributions. The real interest rate, determining the rate of return on private defined-contribution plans, declines from its 1995 level of 9.2 per cent to 2.6 per cent by the year 2000 and to a permanent level of 2.2 per cent by 2006. Post-retirement benefits are indexed to the national average wage for the public tier, and to the CPI for the private tier (for example, by the purchase of an indexed annuity upon retirement).[3] The insurable earnings base for public pensions rises to 5 times the minimum wage (3 times the minimum wage in Option 4).

Figure A1 shows the estimated impacts of each of the reform options over a long-run horizon. It is seen that all options are sufficient to prevent the blow-up of social security deficits projected under the no-reform scenario. All options produce declining deficits until 2010, when demographic factors cause a renewed rise until 2020-25, after which deficits start to fall again. However, none of the options is sufficient to bring the system back into balance even by the year 2050, at best keeping deficits in their original 1½ -2½ per cent of GDP range.[4] In the short to medium run, Option 2 is clearly the costliest according to these calculations. This largely reflects the need by the government to honour accrued rights under the old PAYG scheme, and the right for all existing workers to remain under that scheme, for which there will be a declining number of contributors. The ILO concludes that it would also impose a heavy burden on the next two generations which would, through contributions and taxes, have to finance both existing pensions of the preceding generation and their own future pensions (the transition problem).[5]

Notes

1. The civil servants' fund (ES) currently has no ceiling, but very few civil servants earn as much as 5 times the minimum wage, so that apart from top-level staff, they would be largely unaffected by the reform.
2. Two sub-options are furthermore considered under each of the two mixed schemes: in Option 3*a*, the first tier includes a flat-rate benefit element, while in option 4*a*, the second tier would provide defined benefits. The discussion here (and Figure A1 below) focuses on Option 3*b*, where first-tier benefits are wholly contribution-related, and Option 4*b*, where the second tier is of the defined contribution-variety.
3. A minimum public pension guarantee, equal to 20-25 per cent of the national average wage, is offered in Options 1, 3 and 4 on condition of a minimum 15-year contribution period, while in Option 2 (where there is no public pension), a state guarantee equal to 40 per cent of the personal reference wage is offered, on condition of a contribution history of 30 years, to guard against investment risks.
4. Moreover, the starting point for reform (the actual pension deficit) has deteriorated by 1 percentage point of GDP since the calculations were made in 1995. The number of pensioners grew by 3 times what the ILO baseline had assumed over 1996-98, while real wages grew much more strongly than assumed in 1996-97 (from their 1994 low point), pushing up real pensions in line with the indexing formula.
5. It may be noted that this conclusion also may be traced to the rather pessimistic interest rate assumption. By contrast, most analyses of the costs of pension privatisation assume that the rate of interest exceeds the rate of growth, which is sufficient to eliminate the extra burden on the transition generation (see Box 5 above). In the present analysis, the rate of GDP growth is assumed to be 5.9 per cent from 1995 to 2005, 4 per cent from 2006 to 2025, and 3 per cent from 2026 to 2050 – considerably in excess of the market rate of return and thus biasing the results in favour of a pay-as-you-go system (where the rate of return is essentially the rate of growth of the economy).

Annex IV

Calculation of implied real pension rates of return

The implied real pension rate of return can be calculated as that interest rate which equates the cumulated value of pension contributions at retirement date with the present value of lifetime benefits at retirement date.[1] To simplify the calculations, it is assumed that the flat-rate benefit (SYZ) is zero and that benefits are held real in constant terms after retirement (indexation to the CPI).

Variable definitions:

c = average contribution rate
b = income replacement ratio
T = number of years contributions are made
S = number of years in retirement (assuming average life expectancy at retirement)
g = real growth rate in earnings = .04
r = real rate of return

Value of contributions at retirement date:

$$= cw_0 \sum_{t=1}^{T} (1+g)^t (1+r)^{T-t}$$

$$= cw_0 (1+g)^T \left[\left\{ \left(\frac{1+r}{1+g} \right)^{T-1} - 1 \right\} \Big/ \left(\frac{r-g}{1+g} \right) \right] \qquad (1)$$

Present value of benefits at retirement date (assuming constant real pension):

a) Reference wage is lifetime average real wage:

$$= b\left[\frac{w_0}{T}\sum_{t=1}^{T}(1+g)^{T-t}\right]\left[\sum_{t=1}^{S}(1+r)^{-t}\right]$$

$$= b\left[\frac{w_0[(1+g)^{T-1}-1]}{Tg}\right]\left[\frac{1-(1+r)^{-S}}{r}\right] \qquad (2a)$$

b) Reference wage is final real wage:

$$= bw_0(1+g)^T\left[\sum_{t=1}^{S}(1+r)^{-t}\right]$$

$$= bw_0(1+g)^T\left[\frac{1-(1+r)^{-S}}{r}\right] \qquad (2b)$$

To calculate r, set (1) = (2*a*) or (2*b*), then solve, given selected prior values of c, b, T, S, and g. (It is seen that w_o drops out of the calculation.)

Note

1. See Blake, D. (1997), "Pension choices and pensions policy in the United Kingdom", in Valdès-Prieto (ed.), *The Economics of Pensions*, Cambridge University Press.

Annex V

Chronology of main economic events

1997

June

The government lead by Mr Erbakan falls. It had been in office since June 1996.

July

A new three-party minority coalition government headed by Mr Yilmaz is appointed.

August

Parliament approves the new education law extending compulsory state schooling from five to eight years.

November

Mr Yilmaz rules out a controversial one-year "shock therapy programme" announced earlier by the Economy Minister Taner and intended to cut wholesale price inflation to 15 per cent by the end of 1998.

The First National Agricultural Congress is held for negotiating problems in this sector.

December

At the Luxembourg summit, the EU decides not to include Turkey as a candidate for EU membership in the next round of enlargement.

The Yilmaz government announces a plan to reduce wholesale price inflation to 50 per cent by end-1998.

Parliament approves the 1998 budget.

The withholding tax rate on interest earnings on government securities is reduced from 12 to 6 per cent, while a 6 per cent withholding tax is imposed on revenues from repo transactions.

1998

January

Reserve requirements are imposed on banks' cash and non-cash credits extended.

A law is approved providing new incentives for business investment in poor Southeast regions.

The government announces the 1998-2000 stabilisation programme providing a framework for concerted anti-inflation actions on a number of fronts, including stronger co-ordination between macroeconomic policies and structural reforms. Talks with the IMF.

The Treasury introduces a new borrowing instrument on which nominal interest rates are determined by realised increases in CPI at the time of redemption (indexed bonds).

February

The Treasury issues $ 400 million in bonds in the Eurobond market, a move intended to reduce the domestic debt stock burden.

March

In a move to prepare the privatisation of the oil refinery company, the government announces that the price of oil products will be adjusted in line with international market prices.

Political uncertainties, combined with failure to introduce structural reforms more rapidly, induce the leading credit rating agencies to maintain Turkey's credit rating at low levels.

The government opens a tender to sell 51 per cent of its stake in the oil distributor company (*Petrol Ofisi*).

April

The government finalises a licence agreement with two cellular phone operators, *Türkcell* and *Telsim*, worth a total of $ 1 billion.

Parliament ratifies an agreement between Turkey and Russia on the pipeline transportation of Russian natural gas under the Black Sea. Legal arrangements for the use of Tax Identification Numbers are put into force.

May

The authorities sell 12.3 per cent of the public sector stake in *Turkiye Is Bankasi* through public placement while they also open a tender to sell 84.5 per cent of the public stake in *Türk Ticaret Bank*.

The grain board decides to increase grain purchase prices by 61 per cent, *i.e.* significantly in excess of the inflation target.

June

The government presents to Parliament draft legislation aimed at reforming the tax system.

The government and IMF agree on an 18 month Staff Monitored Programme (SMP). An SMP is an informal and flexible instrument for promoting economic dialogue which is not supported by the use of the Fund's financial resources. The terms of the agreement are recapitulated in the *Memorandum of Economic Policies*, consistent with the programme defined earlier in January.

The World Bank announces a $ 270 million loan destined to the energy sector.

A free-trade agreement is signed between Turkey and Lithuania.

July

The government announces an early general election for April 1999.

Petrol Ofisi sold to the oil employers union.

The operating rights of Antalya and Trabzon sea ports are transferred to the private sector.

The Central Bank announces its monetary targets for the second half of the year envisaging to replace reserve money with net domestic assets as the relevant monetary policy indicator.

Foreign exchange reserve holdings of the Central Bank reach a record high of $ 27 billion.

Parliament approves the new tax legislation.

August

In the wake of the financial crisis in Russia, the Yilmaz government announces measures to support the financial market, including the withdrawal of the withholding tax on government security interest earnings, a cut in the tax rate applied to bond tradings and the abolition of the withholding tax applied to interbank transactions.

September

Turkey and the World Bank agree on a $ 369 million loan to compensate for flood and earthquake damage.

The Yilmaz government announces tax exemptions on equity trading to help the stock market overcome difficulties induced by the Russian crisis.

A special programme created to encourage holders of undeclared money to bring their cash into the banking system (*Financial Millennium*) ends.

October

The Central Bank decides to transfer to the Savings Deposits Insurance Fund 98 per cent of *Bank Express* shares, a troubled private bank that came under strong pressure from a run on deposits.

Turkey and Turkmenistan sign a natural gas agreement, foreseeing the construction of a 1 700 km-long pipeline.

IMF quarterly monitoring mission. Discussions with IMF concentrate on measures to ensure an appropriate fiscal stance, as well as reforms designed to address structural weaknesses in the public finances and the economy.

November

Treasury launches a DM 600 million bond issue in the Euro DM market.

The Yilmaz government falls after losing outside support from the Republican People's Party.

December

Foreign exchange holdings of the Central Bank reach $ 20 billion by the end of the year.

The authorities announce measures to reinvigorate the economy, including special export credit funds made available to the state foreign trade bank *Eximbank*.

Turkey receives a package of ECU 113 million from the European Union for the financing of education and health programs.

Parliament approves a temporary budget for 1999 envisaging to contain spending in the first half of the year within a certain proportion of the 1998 allocations.

1999

January

A new caretaker government headed by Mr. Ecevit is appointed.

Interbank, a private medium sized bank, is transferred to the Savings Deposits Insurance Fund of the Central Bank. Under pressure from severe financial shortages, the business sector registers an increasing number of bankruptcies, most notably in the leather, textile and petrol sectors.

In a move to restrict imports, the authorities announce that foreign companies can only participate in international fairs in Turkey for the purpose of exhibition and promotion, but not for retail sale.

February

Standard & Poor's downgrades Turkey's foreign currency long-term sovereign credit rating to single B.

IMF quarterly monitoring mission. The IMF demands the reinvigoration of the macroeconomic and structural reform effort after the elections without delay.

An Economic and Social Council is held and its approved declaration is announced to the public.

March

The government announces that for 1999 subsidised tobacco prices will rise by 55.5 per cent.

It is allowed that banks can hold cash and convertible foreign reserves for their liquidity requirement based on their Turkish lira and foreign exchange deposit liabilities, as well as on non-deposit liabilities on their balance sheets.

April

The Ecevit government introduces measures to support the automotive and tourism sectors. The government promises to lift the quota on tobacco and sugar-beet purchases and to raise supports to textile producers.

Parliamentary elections are held, assigning the bulk of Parliamentary seats to the Democratic Left Party led by Mr. Ecevit and the extreme right National Action Party led by Mr. Bahceli.

June

Following the April election results, a new three-party coalition government headed by Mr. Ecevit receives a vote of confidence from Parliament.

STATISTICAL ANNEX AND STRUCTURAL INDICATORS

OECD 1999

Table A. **Gross domestic product, current prices**

TL billion

	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998
Private consumption	149 140	269 562	434 366	760 256	1 369 339	2 706 262	5 457 903	9 937 697	19 619 096	35 316 360
Government consumption	21 240	43 083	78 256	140 584	258 084	450 605	837 243	1 709 247	3 535 104	6 499 361
Gross fixed capital investment	51 837	89 892	149 272	258 406	525 506	952 322	1 850 225	3 706 404	7 618 372	12 794 101
Public sector	17 428	28 505	49 445	81 312	143 766	221 061	304 429	687 165	1 636 275	3 203 025
Machinery and equipment	5 917	8 809	14 523	25 839	46 734	55 774	85 418	172 001	392 721	832 417
Building construction	3 242	5 683	9 666	14 100	29 954	37 540	65 986	152 535	357 787	749 424
Other construction	8 269	14 013	25 256	41 373	67 078	127 747	153 025	362 629	885 767	1 621 184
Private sector	34 408	61 387	99 826	177 094	381 740	731 262	1 545 796	3 019 239	5 982 098	9 591 076
Machinery and equipment	9 817	23 695	37 436	70 831	173 159	286 019	754 172	1 600 071	3 284 866	4 968 421
Building construction	24 591	37 692	62 390	106 263	208 581	445 243	791 624	1 419 168	2 697 232	4 622 655
Increase in stocks	1 522	6 601	–6 160	2 454	21 619	–121 416	127 149	–79 656	–377 455	–103 257
Total domestic demand	223 739	409 138	655 734	1 161 700	2 174 548	3 987 773	8 272 520	15 273 692	30 395 117	54 506 565
Exports of goods and services	36 833	52 533	87 215	157 360	270 997	826 379	1 544 077	3 182 305	7 088 355	12 781 271
Imports of goods and services	40 420	69 092	104 819	189 646	383 358	788 530	1 890 238	4 110 584	8 762 826	14 572 641
Statistical discrepancy	7 172	480	–8 013	–36 045	–80 320	–157 193	–163 903	426 698	115 234	–1 090 052
Gross domestic product	227 324	393 059	630 117	1 093 369	1 981 867	3 868 429	7 762 456	14 772 111	28 835 880	51 625 143

Source: State Institute of Statistics.

Table B. **Gross domestic product, constant prices**

1987 prices, TL billion

	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998
Private consumption	51 105	57 803	58 913	61 282	66 545	62 962	66 011	71 614	77 620	77 661
Government consumption	5 831	6 297	6 579	6 765	7 344	6 938	7 411	8 047	8 379	8 798
Gross fixed capital investment	18 701	21 670	21 935	23 147	29 247	24 577	26 823	30 598	35 137	34 292
Public sector	6 348	6 916	7 043	7 201	7 721	5 346	4 340	5 400	6 933	7 979
Machinery and equipment	2 358	2 450	2 569	2 679	3 155	1 491	1 282	1 436	1 956	2 410
Building construction	1 132	1 283	1 266	1 180	1 496	880	935	1 218	1 496	1 870
Other construction	2 858	3 183	3 208	3 342	3 070	2 975	2 123	2 746	3 481	3 699
Private sector	12 352	14 755	14 892	15 945	21 527	19 231	22 482	25 198	28 204	26 312
Machinery and equipment	4 012	6 678	7 077	7 454	11 872	8 691	11 495	14 393	17 438	15 790
Building construction	8 340	8 077	7 815	8 491	9 655	10 540	10 987	10 805	10 766	10 522
Increase in stocks	520	1 513	–939	402	1 486	–2 925	1 772	–463	–1 420	–169
Total domestic demand	76 157	87 283	86 488	91 596	104 622	91 552	102 017	109 796	119 716	120 582
Exports of goods and services	13 751	14 184	14 627	16 236	17 484	20 138	21 746	26 521	31 593	34 898
Imports of goods and services	13 543	18 027	17 074	18 938	25 715	20 090	26 033	31 376	38 417	39 274
Statistical discrepancy	134	138	312	508	200	–279	159	–158	–261	–438
Gross domestic product	76 498	83 578	84 353	89 401	96 591	91 321	97 889	104 745	112 631	115 768

Source: State Institute of Statistics.

Table C. **Gross domestic product by kind of activity**

	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998
	TL billion, current prices									
Agriculture, forestry, fishing	37 704	68 692	96 074	163 827	305 525	598 169	1 218 178	2 489 774	4 170 001	8 947 885
Mining	4 625	6 269	9 762	14 946	21 814	54 099	98 545	183 080	336 889	551 772
Manufacturing	52 561	86 307	139 709	236 573	412 408	853 488	1 751 681	3 123 034	6 218 627	10 128 256
Electricity, gas and water	4 408	7 745	13 839	28 602	51 450	112 181	192 169	410 414	737 669	1 278 773
Construction	15 904	24 746	44 139	74 509	145 834	263 720	426 215	857 762	1 743 240	3 057 577
Wholesale and retail trade, restaurants and hotels	42 653	74 912	117 433	202 369	368 385	760 919	1 587 691	3 022 315	5 985 402	10 352 369
Transport, storage and communication	26 317	46 490	74 561	133 338	236 860	514 110	981 070	1 941 574	4 018 613	7 181 691
Financial institutions	6 499	12 660	26 060	43 492	84 453	115 011	322 590	732 340	1 474 426	3 191 373
Dwellings	8 390	13 218	23 228	41 216	68 178	127 918	249 169	442 935	850 332	1 670 981
Community, social and personal services	7 544	14 415	23 006	39 577	71 449	142 795	287 000	554 080	1 067 451	1 939 228
Producers of government services	15 535	32 525	61 051	111 841	203 922	344 530	619 785	1 238 527	2 579 910	4 782 332
Private non-profit institutions	891	1 521	2 332	3 553	6 015	9 070	14 577	26 922	53 021	98 742
(–) Imputed bank service charge	2 562	9 838	23 282	41 188	75 654	163 471	270 344	709 235	1 371 710	3 101 211
Import duties	6 854	13 397	22 206	40 713	81 229	135 892	284 130	458 588	972 011	1 545 375
Gross domestic product at factor cost	227 324	393 060	630 117	1 093 368	1 981 867	3 868 429	7 762 456	14 722 110	28 835 883	51 625 143
	1987 prices, TL billion									
Agriculture, forestry, fishing	13 272	14 177	14 049	14 651	14 463	14 358	14 640	15 284	14 927	16 069
Mining	1 590	1 550	1 620	1 624	1 521	1 642	1 529	1 565	1 639	1 736
Manufacturing	17 076	18 729	19 175	20 281	22 166	20 473	23 321	24 980	27 839	28 166
Electricity, gas and water	1 863	2 023	2 114	2 363	2 573	2 660	2 916	3 199	3 358	3 536
Construction	5 472	5 411	5 473	5 814	6 272	6 144	5 857	6 200	6 511	6 489
Wholesale and retail trade, restaurants and hotels	15 017	16 868	16 748	17 902	19 975	18 445	20 587	22 413	25 024	25 332
Transport, storage and communication	9 045	10 123	10 085	10 899	12 081	11 835	12 511	13 458	14 485	15 350
Financial institutions	2 447	2 496	2 515	2 463	2 453	2 417	2 424	2 476	2 573	2 751
Dwellings	4 504	4 616	4 725	4 841	4 976	5 117	5 224	5 352	5 475	5 601
Community, social and personal services	1 781	1 926	1 944	2 051	2 192	2 098	2 252	2 399	2 564	2 650
Producers of government services	3 906	4 019	4 117	4 259	4 336	4 371	4 481	4 469	4 473	4 501
Private non-profit institutions	331	365	377	386	397	388	383	386	390	397
(–) Imputed bank service charge	2 122	2 161	2 166	2 116	2 103	2 066	2 059	2 073	2 113	2 240
Import duties	2 317	3 436	3 576	3 982	5 289	3 429	3 822	4 637	5 486	5 430
Gross domestic product at factor cost	76 498	83 578	84 353	89 401	96 590	91 321	97 888	104 745	112 631	115 768

Source: State Institute of Statistics.

Table D. **Industrial production**

Index 1992 = 100

	1994	1995	1996	1997	1998	1998			
						I	II	III	IV
Total industry	**101.5**	**114.3**	**122.9**	**137.0**	**139.4**	**134.3**	**135.6**	**143.7**	**144.0**
Mining	**98.3**	**101.3**	**102.4**	**108.1**	**115.4**	**98.8**	**101.4**	**144.2**	**117.2**
Energy	**116.4**	**128.4**	**141.9**	**153.5**	**165.1**	**168.7**	**151.2**	**169.6**	**171.0**
Manufacturing	**99.9**	**113.4**	**121.3**	**136.7**	**137.7**	**132.2**	**135.7**	**140.6**	**142.4**
By manufacturing sector (weighted by Value Added)									
Foods, beverages, tobacco	111.4	117.1	124.2	134.0	138.4	126.1	119.4	145.2	163.0
Textile, clothing, leather	95.4	110.7	119.7	127.8	120.2	123.6	121.8	119.5	115.9
Paper and printing	105.0	116.7	112.0	110.3	103.9	107.0	101.3	102.2	105.2
Chemicals, petroleum	102.7	119.3	122.8	136.3	137.2	134.6	135.7	142.1	136.4
Non-metallic mineral products	101.5	113.6	119.6	132.8	134.0	113.1	137.2	151.4	134.3
Basic metal industry	108.9	113.7	123.2	134.8	134.0	126.4	137.1	140.4	132.1
Machinery and transport equipment	86.8	104.6	122.6	151.3	158.2	153.8	163.5	152.1	163.2

Source: State Institute of Statistics.

Table E. **Prices**[1]

Annual percentage changes

		Wholesale prices							Consumer prices	
		General index			Agriculture	Mining	Manufacturing industry	Energy	General	Food
		Total	Public	Private						
		Base year 1987 = 100								
Old series (1987 weights)		(100)	(27.74)	(72.26)	(23.03)	(2.54)	(69.80)	(4.62)	(100)	(32.10)
1988		70.5	70.3	70.5	44.1	64.1	81.5	40.9	73.7	83.2
1989		64.0	64.2	63.9	71.7	65.0	61.6	69.2	63.2	69.3
1990		52.3	56.7	50.6	70.6	48.7	46.9	56.5	60.3	64.3
1991		55.3	61.3	53.0	50.8	63.2	55.3	75.1	66.0	67.1
1992		62.1	65.2	60.8	62.7	61.6	55.3	97.7	70.1	71.3
1993		58.4	54.5	60.1	62.2	56.7	56.6	67.8	66.1	63.5
1994		120.7	122.5	119.9	97.8	132.8	129.4	102.3	106.3	110.0
1995		88.5	77.0	93.4	118.7	76.6	83.4	47.8	93.6	97.3
		Base year 1994 = 100								
New series (1994 weights)		(100)	(23.17)	(76.83)	(22.22)	(2.47)	(71.12)	(4.19)	(100)	(31.09)
1995		86.0	76.8	88.8	107.8	85.6	81.0	56.4	88.0	92.3
1996		75.9	81.9	74.3	86.5	89.3	70.4	101.7	80.4	72.2
1997	I	77.9	84.3	76.0	81.4	89.2	75.1	93.4	76.9	74.9
	II	74.4	67.7	76.3	85.5	70.4	70.2	71.0	77.6	82.4
	III	83.2	91.6	80.6	84.7	76.7	84.1	63.7	87.6	94.9
	IV	88.9	94.7	87.2	93.3	72.0	89.3	67.0	96.0	110.5
1998	I	89.4	83.3	91.2	106.6	58.5	85.1	66.8	99.4	111.4
	II	80.0	73.2	81.9	87.1	58.5	78.1	72.5	91.8	87.6
	III	68.5	50.7	74.1	84.8	45.7	62.5	78.3	82.4	78.0
	IV	58.3	40.7	63.8	76.2	38.2	51.3	69.1	73.0	66.0
1999	I	48.8	40.8	51.2	53.8	38.2	46.2	63.1	64.5	51.1

1. In January 1996, the State Institute of Statistics introduced new weights for both wholesale and consumer prices indices, and shifted the base year from 1987 to 1994.

Source: State Institute of Statistics.

Table F. **Imports by commodities**

US$ million

	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998
SITC classification										
0. Food and live animals	697	1 179	466	594	839	460	1 482	1 776	1 426	1 165
1. Beverages and tobacco	225	346	381	330	336	153	173	296	394	319
2. Crude materials, inedible, except fuels	1 602	1 680	1 614	1 872	2 499	2 284	3 536	3 636	3 928	3 501
3. Mineral fuels, lubricants and related materials	3 258	4 641	3 773	3 782	3 964	3 817	4 620	5 914	6 141	4 505
4. Animal and vegetable oils, fats and waxes	307	297	393	375	403	454	631	509	570	521
5. Chemicals and related products, n.e.s.	2 321	2 830	2 852	3 092	3 560	3 216	5 350	5 777	6 475	6 576
6. Manufactured goods classified chiefly by material	2 795	3 372	3 295	3 606	4 987	4 036	6 675	7 409	8 145	8 015
7. Machinery and transport equipment	4 095	7 030	7 168	8 043	11 236	7 663	11 492	15 538	18 663	18 220
8. Miscellanous manufactured articles	485	905	990	1 105	1 604	1 186	1 750	2 416	2 916	3 097
9. Commodities and transactions not classified elsewhere	4	22	114	73	0	0	0	357	0	0
Total	15 789	22 302	21 047	22 871	29 428	23 269	35 709	43 628	48 658	45 919

Source: State Institute of Statistics.

Table G. **Exports by commodities**

US$ million

	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998
SITC classification										
0. Food and live animals	2 015	2 302	2 728	2 760	2 794	3 300	3 424	3 559	4 079	3 756
1. Beverages and tobacco	493	455	593	356	466	471	469	742	754	642
2. Crude materials, inedible, except fuels	837	750	681	503	555	605	777	692	829	805
3. Mineral fuels, lubricants and related materials	260	298	292	235	177	244	290	276	193	259
4. Animal and vegetable oils, fats and waxes	153	139	197	215	187	182	328	232	272	238
5. Chemicals and related products, n.e.s.	936	745	603	629	611	733	890	886	1 169	1 149
6. Manufactured goods classified chiefly by material	3 391	3 841	3 717	4 155	4 464	5 661	6 231	6 550	7 796	7 767
7. Machinery and transport equipment	587	853	1 016	1 290	1 294	1 722	2 406	3 009	3 363	4 057
8. Miscellanous manufactured articles	2 949	3 567	3 758	4 561	4 798	5 187	6 821	7 052	7 791	8 210
9. Commodities and transactions not classified elsewhere	7	7	8	10	0	1	2	0	0	0
Total	11 627	12 957	13 593	14 715	15 345	18 106	21 638	22 998	26 246	26 883

Source: State Institute of Statistics.

Table H. **Geographical distribution of foreign trade**

Percentage share

	Imports					Exports				
	1994	1995	1996	1997	1998	1994	1995	1996	1997	1998
OECD countries	65.9	66.4	71.3	71.7	72.9	59.4	61.4	62.1	59.3	62.9
EU countries	46.9	47.2	53.0	51.2	52.5	47.7	51.2	49.7	46.6	50.0
EFTA countries	2.4	2.5	2.5	2.7	2.5	1.5	1.4	1.4	1.6	1.3
Other OECD countries	16.6	16.7	15.7	17.8	17.9	10.2	8.9	10.9	7.3	11.6
Non-OECD countries	34.1	33.6	28.7	28.3	27.1	40.6	38.6	37.9	40.7	37.1
European countries	11.2	12.0	9.4	9.6	10.2	13.5	16.1	15.7	17.8	14.8
African countries	3.7	3.9	4.6	4.5	3.8	4.7	4.9	5.0	4.7	6.8
American countries	1.4	1.7	1.5	1.6	1.6	0.6	0.6	0.6	0.8	0.9
Middle East countries	10.9	7.5	7.4	6.6	4.8	11.6	9.8	9.7	10.5	8.7
Other Asian countries	5.5	7.1	5.1	5.2	5.7	8.0	4.9	4.9	4.5	2.4
Other countries	1.4	0.9	0.7	0.7	1.0	2.2	2.2	2.0	2.3	3.6
Total (in million of US dollars)	23 270	35 709	43 627	48 559	45 921	18 106	21 636	23 225	26 261	26 881

Source: State Institute of Statistics.

Table I. **Balance of payments**

US$ million

	1990	1991	1992	1993	1994	1995	1996	1997	1998
Exports	13 026	13 667	14 891	15 611	18 390	21 975	32 446	32 647	31 128
Imports	22 581	21 007	23 081	29 771	22 606	35 187	43 028	48 005	45 540
Trade balance	–9 555	–7 340	–8 190	–14 160	–4 216	–13 212	–10 582	–15 358	–14 412
Services	2 437	2 499	3 157	3 959	3 755	6 377	3 698	7 854	11 377
Balance of goods and services	–7 118	–4 849	–5 033	–10 201	–461	–6 835	–6 884	–7 504	–3 035
Private transfers, net	3 349	2 854	3 147	3 035	2 709	3 425	3 892	4 552	5 568
Official transfers, net	1 144	2 245	912	733	383	1 071	555	314	159
Current balance	–2 625	250	–974	–6 433	2 631	–2 339	–2 437	–2 638	2 692
Long-term capital	1 037	623	2 252	5 909	933	930	2 818	6 976	–1 828
Basic balance	–1 588	873	1 278	–524	3 564	–1 409	381	4 338	864
Short-term capital	3 000	–3 020	1 396	3 054	–5 127	3 713	5 945	1 761	2 601
Assets	–409	–2 563	–2 438	–3 291	2 423	–383	331	–1 750	–1 464
Liabilities	3 409	–457	3 834	6 345	–7 550	4 096	5 614	3 511	4 065
Errors and omissions	–468	948	–1 190	–2 222	1 769	2 354	–1 781	–2 755	–3 018
Net transactions of monetary authorities	944	–1 199	1 484	308	206	4 658	4 545	3 344	447
Use of IMF credit	–48	0	0	0	340	347	0	–28	–231
Miscellaneous official accounts	–1	2	0	0	0	0	0	0	0
Allocation of SDR's	0	0	0	0	0	0	0	0	0
Change in reserves	895	–1 197	1 484	308	546	5 005	4 545	3 316	216

Source: State Institute of Statistics, Central Bank of Turkey.

Table J. **Money and banking**

TL billion, end of period

	1991	1992	1993	1994	1995	1996	1997	1998
Money supply								
Notes and coins	18 547	31 181	52 517	104 370	189 542	354 901	641 707	1 106 023
Sight deposits	29 344	47 952	79 792	134 611	206 505	476 514	736 897	1 178 151
M1	42 116	70 521	132 309	238 981	396 047	831 415	1 378 604	2 284 174
Time deposits	70 325	112 395	159 667	403 509	874 376	1 970 260	3 885 925	8 572 589
M2	113 563	182 989	291 976	642 490	1 270 423	2 801 675	5 264 529	10 856 763
Foreign exchange deposits	46 866	106 524	252 957	628 711	1 355 094	2 702 722	4 864 307	8 568 813
M2Y	160 429	289 513	544 933	1 271 201	2 625 517	5 504 397	10 128 836	19 425 576
Central Bank								
Total assets	96 789	177 244	285 801	693 005	1 421 112	2 681 841	4 700 989	6 881 008
Foreign assets	38 736	75 758	128 126	368 701	911 817	2 041 524	4 336 722	7 168 740
Domestic assets	58 053	101 486	157 675	324 304	509 295	640 317	364 267	–287 732
Cash credits	28 199	72 044	127 344	226 488	478 750	828 954	896 786	776 351
Credit to public sector	22 904	62 602	108 482	214 161	466 496	821 261	889 105	768 807
Credit to private sector	5 295	9 442	18 862	12 327	12 254	7 693	7 681	7 544
Other items	–2 575	–5 293	–1 593	–35 989	–14 467	–185 063	–342 438	–683 690
Devaluation account	32 429	34 735	31 924	133 805	45 012	–3 574	–190 081	–380 393
Total liabilities	96 789	177 244	285 801	693 005	1 421 112	2 681 841	4 700 989	6 881 008
Central bank money	43 993	87 837	128 113	193 140	372 794	706 094	647 966	528 032
Reserve money	37 244	61 195	101 721	185 738	343 484	621 483	1 186 386	2 145 691
Other Central bank money	6 749	26 642	26 392	7 402	29 310	84 611	–538 420	–1 617 659
Foreign currency liabilities	52 796	89 407	157 688	499 865	1 048 318	1 975 747	4 053 023	6 352 976
Deposit money banks								
Deposits[1]	164 669	296 151	522 500	1 203 149	2 543 787	5 876 127	11 964 592	..
Credits	106 491	195 182	363 026	630 583	1 433 121	3 261 926	7 549 472	..
Total bank credits								
(net of Central Bank advances to banks)	130 796	244 420	453 771	714 596	1 567 991	4 300 455	7 453 181	..
Central bank	18 454	42 584	82 936	148 208	193 877	351 308	738 789	..
Deposit money banks	101 452	185 419	342 182	518 908	1 297 004	3 775 503	6 384 251	10 389 202
Investment banks	10 890	16 418	28 653	47 480	77 110	173 644	330 141	597 941

1. Including interbank and foreign currency deposits.

Source: Central Bank of Turkey.

Table K. **Consolidated budget**

TL trillion

	1992	1993	1994	1995	1996	1997	1998
Revenue	174	351	745	1 394	2 702	5 750	11 671
Tax revenue	142	264	588	1 084	2 244	4 745	9 233
Non-tax revenue	33	87	157	310	458	1 005	2 438
Expenditure	222	485	897	1 711	3 940	7 990	15 397
Current personnel expenditure	94	186	295	503	974	2 073	3 870
Other current expenditure	20	35	73	142	309	706	1 271
Investment	29	37	50	92	238	590	886
Transfers	78	227	478	975	2 419	4 621	9 371
Budget balance	–47	–134	–152	–317	–1 238	–2 240	–3 276
Deferred and advance payments	–12	8	0	22	–30	20	–112
Cash balance	–59	–126	–152	–294	–1 268	–2 220	–3 838
Foreign borrowing (net)	4	21	–67	–81	–134	–447	–1 036
Domestic borrowing (net)	57	105	226	378	1 295	2 506	4 590
Other borrowing[1]	–1	0	–7	–2	107	162	284

1. Including errors and omissions and change in cash holdings in banks.

Source: Undersecretariat of Treasury.

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Czech Republic	Denmark	Finland	France	Germany	Greece
Population												
Total	Thousands	1996	18 289	8 060	10 157	29 964	10 316	5 262	5 125	58 380	81 877	10 465
Inhabitants per sq. km	Number	1996	2	96	333	3	131	122	15	106	229	79
Net average annual increase over previous 10 years	%	1996	1.3	0.6	0.3	1.3	0	0.3	0.4	0.5	3	0.5
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	8 344	3 737 (94)	3 675 (95)	13 676	4 918	2 593	2 087	21 951	35 360	3 824 (95)
of which: Agriculture	% of TCE	1996	5.1	7.2 (94)	2.5 (95)	4.1	6.3	4	7.1	4.6	3.3	20.4 (95)
Industry	% of TCE	1996	22.5	33.2 (94)	26.7 (95)	22.8	42	27	27.6	25.9	37.5	23.2 (95)
Services	% of TCE	1996	72.4	59.6 (94)	71.4 (95)	73.1	51.7	69	65.3	69.5	59.1	56.4 (95)
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	398.9	228.7	268.2	579.2	56.2	174.9	125.1	1 536.6	2 353.5	91.2 (95)
Per capita	US$	1996	21 812	28 384	26 409	19 330	5 445	33 230	24 420	26 323	28 738	8 722 (95)
At current prices using current PPPs[3]	Bill. US$	1996	372.7	172.4	222	645.1	..	118	96.7	1 198.6	1 736.1	133.5
Per capita	US$	1996	20 376	21 395	21 856	21 529	..	22 418	18 871	20 533	21 200	12 743
Average annual volume growth over previous 5 years	%	1996	3.9	1.6	1.2	2.2	2	2.2	1.6	1.2	1.4	1.3 (95)
Gross fixed capital formation (GFCF)	% of GDP	1996	20.3	23.8	17.3	17.7	33	16.7	16.1	17.4	20.6	17 (95)
of which: Machinery and equipment	% of GDP	1996	10.2 (95)	8.8 (95)	7.5 (95)	6.6	..	7.9 (95)	6.4 (95)	7.8	7.6	7.7 (95)
Residential construction	% of GDP	1996	4.6 (95)	5.9 (95)	4.6 (95)	5.4	..	3.2 (95)	3.5 (95)	4.4	7.3	3.3 (95)
Average annual volume growth over previous 5 years	%	1996	5.6	2.1	0.3	2.2	9.4	2	–4.1	–1.5	0.2	0.5 (95)
Gross saving ratio[4]	% of GDP	1996	18	21.9	22.2	17.8	..	17.6	19.6	18.7	20	16 (95)
General government												
Current expenditure on goods and services	% of GDP	1996	17	19.8	14.5	18.7	21.5	25.2	21.9	19.4	19.8	20.8 (95)
Current disbursements[5]	% of GDP	1995	35.6	48.6	52.2	45.8	..	59.6	55.9	50.9	46.6	52.1
Current receipts	% of GDP	1995	34.9	47.4	49.9	42.7	..	58.1	52.8	46.9	45.9	45
Net official development assistance	% of GNP	1995	0.36	0.33	0.38	0.38	..	0.96	0.32	0.55	0.31	0.13
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	12 596	12 152	13 793	12 959	..	12 027	10 282	12 506	12 244	9 473
Passenger cars, per 1 000 inhabitants	Number	1994	460	433	416	466	282	312	368	430	488	199
Telephones, per 1 000 inhabitants	Number	1994	496	466	449	576	209	604	551	547	483[8]	478
Television sets, per 1 000 inhabitants	Number	1993	489	479	453	618	476	538	504	412	559	202
Doctors, per 1 000 inhabitants	Number	1995	2.2 (91)	2.7	3.7 (94)	2.2	2.9	2.9 (94)	2.8	2.9	3.4	3.9 (94)
Infant mortality per 1 000 live births	Number	1995	5.7	5.4	7.6 (94)	6.3 (94)	7.7	5.5	4	5.8 (94)	5.3	8.1
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	1.7	5.2	2.7	2.4	..	3.2	3.8	2.6	4.2	11.8
Consumer prices	%	1996	2.4	2.9	2.2	1.4	11.9	1.9	1.5	2	3.1	11.6
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	60 288	57 870	170 223[7]	202 320	21 910	51 030	40 576	288 450	521 263	11 501
As % of GDP	%	1996	15.1	25.3	63.5	34.9	39	29.2	32.4	18.8	22.1	12.9 (95)
Average annual increase over previous 5 years	%	1996	7.5	7.1	7.6	9.7	..	6.2	12.1	6.3	5.4	5.8
Imports of goods, cif*	Mill. US$	1996	61 374	67 376	160 917[7]	170 931	27 721	44 987	30 911	271 348	455 741	27 402
As % of GDP	%	1996	15.4	29.5	60	29.5	49.3	25.7	24.7	17.7	19.4	30.4 (95)
Average annual increase over previous 5 years	%	1996	9.7	5.9	5.9	7.7	..	5.6	7.3	3.9	3.3	6.6
Total official reserves[6]	Mill. SDRs	1996	10 107	15 901	11 789[7]	14 202	8 590	9 834	4 810	18 635	57 844	12 171
As ratio of average monthly imports of goods	Ratio	1996	2	2.8	0.9	1	..	2.6	1.9	0.8	1.5	5.3

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Hungary	Iceland	Ireland	Italy	Japan	Korea	Luxembourg	Mexico	Netherlands	New Zealand
Population												
Total	Thousands	1996	10 193	270	3 621	57 473	125 864	45 545	418	96 582	15 494	3 640
Inhabitants per sq. km	Number	1996	77	3	52	191	333	458	161	48	380	14
Net average annual increase over previous 10 years	%	1996	–0.3	1.1	0.2	0	0.4	1	1.3	2	0.6	1.1
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	3 605	142	1 307	20 036	64 860	20 764	212 (95)	32 385 (95)	6 983	1 688
of which: Agriculture	% of TCE	1996	8.4	9.2	10.7	7	5.5	11.6	2.8 (95)	23.5 (95)	3.9	9.5
Industry	% of TCE	1996	33	23.9	27.2	32.1	33.3	32.5	30.7 (90)	21.7 (95)	22.4	24.6
Services	% of TCE	1996	58.6	66.2	62.3	60.9	61.2	55.9	66.1 (90)	54.8 (95)	73.8	65.9
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	43.7 (95)	7.3	70.7	1 214.2	4 595.2	484.8	17	329.4	396	65.9
Per capita	US$	1996	4 273 (95)	27 076	19 525	21 127	36 509	10 644	40 791	3 411	25 511	18 093
At current prices using current PPPs[3]	Bill. US$	1996	..	6.3	68.8	1 148	2 924.5	618.5	13.5	751.1	324.5	63.6
Per capita	US$	1996	..	23 242	18 988	19 974	23 235	13 580	32 416	7 776	20 905	17 473
Average annual volume growth over previous 5 years	%	1996	–2.4 (95)	1.5	7.1	1	1.5	7.1	4.8	1.7	2.3	3.7
Gross fixed capital formation (GFCF)	% of GDP	1996	19.3 (95)	17.5	17.2	17	29.7	36.8	20.8	18	19.7	20.9
of which: Machinery and equipment	% of GDP	1996	..	6.7	5.5 (95)	8.8	10.1 (95)	13	..	8.8	9.4	10
Residential construction	% of GDP	1996	. .	3.9	4.9 (95)	4.5	5.3 (95)	7.6	..	4.7	5	5.6
Average annual volume growth over previous 5 years	%	1996	–0.9 (95)	–1.4	6	–1.4	1.3	6.9	0.2	–0.7	2.2	9.6
Gross saving ratio[4]	% of GDP	1996	..	15.6	21.7	20.5	31.4	34.2	37.5	22.7	25.7	16
General government												
Current expenditure on goods and services	% of GDP	1996	24.9 (95)	20.8	14.1	16.4	9.7	10.6	13.6	9.7[10]	14	14.4
Current disbursements[5]	% of GDP	1995	..	35.1	39.2 (94)	49.5	28.5	15.1	..	..	51.8	..
Current receipts	% of GDP	1995	..	36	39.3 (94)	44.5	32	25.1	..	..	50	..
Net official development assistance	% of GNP	1995	..	..	0.29	0.15	0.28	0.03	0.36	..	0.81	0.23
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	..	14 244	10 020	12 224	13 912	7 354	17 811	5 045	12 477	10 895
Passenger cars, per 1 000 inhabitants	Number	1994	212	434	264	517	342	115	544	91	383	457
Telephones, per 1 000 inhabitants	Number	1994	170	557	350	429	480	397	564	93	509	470
Television sets, per 1 000 inhabitants	Number	1993	427	335	301	429	618	215	261	150	491	
Doctors, per 1 000 inhabitants	Number	1995	3.4	3.9 (94)	3.4	3.0 (94)	1.7	1.6 (92)	1.8 (94)	1.1	2.2 (93)	1.6
Infant mortality per 1 000 live births	Number	1995	11	6.1	6.3	6.6 (94)	4.3	9	5.3 (94)	17 (94)	5.5	7.2 (94)
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	..	..	3.7	3.5	1.8	..	..	–1.6	2.4	1.5
Consumer prices	%	1996	23.2	2.6	2.2	4.5	0.7	5.3	2.4	19.7	2.5	2
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	15 674	1 891	48 416	250 842	411 067	129 715	..	96 000	203 953	14 316
As % of GDP	%	1996	35.9	26	68.5	20.7	8.9	26.8	..	29.1	51.5	21.7
Average annual increase over previous 5 years	%	1996	8.9	4	14.8	8.2	5.5	12.5	..	17.6	8.9	8.2
Imports of goods, cif*	Mill. US$	1996	18 105	2 032	35 763	206 904	349 149	150 340	..	89 469	184 389	14 682
As % of GDP	%	1996	41.4	27.9	50.6	17	7.6	31	..	27.2	46.6	22.3
Average annual increase over previous 5 years	%	1996	9.6	3.4	11.5	2.5	8	13.9	..	12.4	7.8	11.8
Total official reserves[6]	Mill. SDRs	1996	6 812	316	5 706	31 954	150 663	23 670	..	13 514	18 615	4 140
As ratio of average monthly imports of goods	Ratio	1996	..	1.9	1.9	1.9	5.2	..	..	1.8	1.2	3.4

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Poland	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population											
Total	Thousands	1996	4 370	38 618	9 935	39 270	8 901	7 085	62 695	58 782	265 557
Inhabitants per sq. km	Number	1996	13	123	108	78	20	172	80	240	28
Net average annual increase over previous 10 years	%	1996	0.5	0.3	–0.1	0.2	0.6	0.8	2	0.3	1
Employment											
Total civilian employment (TCE)[2]	Thousands	1996	2 110	14 977	4 475	12 394	3 963	3 803	20 895	26 088	126 708
of which: Agriculture	% of TCE	1996	5.2	22.1	12.2	8.7	2.9	4.5	44.9	2	2.8
Industry	% of TCE	1996	23.4 (95)	31.7	31.4	29.7	26.1	27.7	22	27.4	23.8
Services	% of TCE	1996	71.5 (95)	46.2	56.4	61.6	71	67.4	33.1	71	73.3
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1996	157.8	117.9 (95)	103.6	584.9	251.7	294.3	181.5	1 153.4	7 388.1
Per capita	US$	1996	36 020	3 057 (95)	10 425	14 894	28 283	41 411	2 894	19 621	27 821
At current prices using current PPPs[3]	Bill. US$	1996	106.7	..	130.1	587.2	171.4	180.6	383.3	1 095.5	7 388.1
Per capita	US$	1996	24 364	..	13 100	14 954	19 258	25 402	6 114	18 636	27 821
Average annual volume growth over previous 5 years	%	1996	4.1	2.2 (95)	1.5	1.3	1	0.1	4.4	2.2	2.8
Gross fixed capital formation (GFCF)	% of GDP	1996	20.5	17.1 (95)	24.1	20.1	14.8	20.2	25	15.5	17.6
of which: Machinery and equipment	% of GDP	1996	8.4	..	11.7 (93)	6.1 (95)	7.9	9.3	11.9	7.6	8.3 (95)
Residential construction	% of GDP	1996	2.6 (94)	..	5.2 (93)	4.3 (95)	1.9	11[9]	8.4 (95)	3	4.1 (95)
Average annual volume growth over previous 5 years	%	1996	2.8	5.4 (95)	2.2	–1	–2.6	–0.8	6.9	1.3	6.9
Gross saving ratio[4]	% of GDP	1996	29.9	..	21.6	20.7	16	27.1	22.3	14.6	16.6
General government											
Current expenditure on goods and services	% of GDP	1996	20.5	16.9 (95)	18.5	16.3	26.2	14.3	11.6	21.1	15.6
Current disbursements[5]	% of GDP	1995	45.8	..	42.5 (93)	41.2	63.8	47.7	..	42.3 (94)	34.3
Current receipts	% of GDP	1995	50.9	..	39.8 (93)	37.9	57.5	53.8	..	37.2 (94)	32.1
Net official development assistance	% of GNP	1995	0.87	..	0.27	0.24	0.77	0.34	0.07	0.28	0.1
Indicators of living standards											
Private consumption per capita using current PPPs[3]	US$	1996	11 593	..	8 522	9 339	10 096	15 632	4 130	11 865	18 908
Passenger cars, per 1 000 inhabitants	Number	1994	381	186	357	351	406 (93)	451	47	372	565
Telephones, per 1 000 inhabitants	Number	1994	554	131	350	371	683	597	201	489	602
Television sets, per 1 000 inhabitants	Number	1993	427	298	190	400	470	400	176	435	816
Doctors, per 1 000 inhabitants	Number	1995	2.8	2.3	3	4.1 (93)	3.1	3.1 (94)	1.2	1.6 (94)	2.6 (94)
Infant mortality per 1 000 live births	Number	1995	4	13.6	7.4	6 (94)	4	5	46.8 (94)	6.2 (94)	8 (94)
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1996	3.2	..	..	5.8	4.8	..	..	4.9	2.7
Consumer prices	%	1996	1.9	..	5.6	4.7	2.7	2.2	81.6	2.7	2.9
Foreign trade											
Exports of goods, fob*	Mill. US$	1996	49 576	24 417	24 614	102 067	84 836	79 581	23 301	259 941	625 075
As % of GDP	%	1996	31.4	20.7	23.8	17.5	33.7	27	12.8	22.5	8.5
Average annual increase over previous 5 years	%	1996	7.8	..	8.6	11.2	9	5.3	11.1	7	8.2
Imports of goods, cif*	Mill. US$	1996	35 575	37 185	35 192	121 838	66 825	78 052	43 094	287 033	795 289
As % of GDP	%	1996	22.5	31.5	34	20.8	26.5	26.5	23.7	24.9	10.8
Average annual increase over previous 5 years	%	1996	6.9	..	6.1	5.5	6	3.2	15.1	6.5	10.3
Total official reserves[6]	Mill. SDRs	1996	18 441	12 409	11 070	40 284	13 288	26 727	11 430	27 745	44 536
As ratio of average monthly imports of goods	Ratio	1996	6.2	..	3.8	4	2.4	4.1	3.2	1.2	0.7

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